Beyond Frozen Encoders: Adapting 3-D CT Foundation Models to Multi-Label Classification and Segmentation

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Abstract

Clinical CT datasets are rarely annotated for a single target: a classification cohort carries many findings and a segmentation dataset many structures. Foundation models for 3-D CT are meant to be adapted to such datasets, yet their published evaluations mostly keep the encoder frozen and train a light head on its features, since this is cheap and measures the pretrained representation rather than the fine-tuning procedure. However, adapting a model to a multi-label cohort or a multi-structure dataset requires two decisions that frozen evaluations do not inform: which parameters to train, and whether to train one model per label or one model for the whole dataset. In this work, we go beyond frozen encoders and compare 11 classification and 7 segmentation adaptation methods, from frozen probes to low-rank adapters and full fine-tuning, each trained per label and jointly, in an effort to find out what to train and when one model can serve all labels of a dataset. We consider four public 3-D CT encoders and use 39 labels from five public classification cohorts and 37 structures from five public segmentation datasets (1988 runs). In classification, adapting the attention weights of the encoder improved on frozen probes on three of the four encoders, and one joint model per cohort was on average level with per-label models. In segmentation, a multi-depth decoder on the frozen encoder gave most of the gain, while one model per structure remained more accurate than a joint model at our training budgets. We condense the findings into a practical guide and release code, splits and per-case predictions.

Keywords: Foundation models, computed tomography, parameter-efficient fine-tuning, low-rank adaptation, multi-task learning, segmentation

1. Introduction

Clinical CT datasets are usually annotated for many targets at once: a cohort collected for trauma or hemorrhage carries several findings per scan, and a segmentation dataset delineates several organs or lesions. Pretrained 3-D CT encoders are meant to be adapted to such datasets, and several are now public, among them TAP-CT (Veenboer et al., 2026), SPECTRE (Claessens et al., 2026), Pillar-0 (Agrawal et al., 2025), CT-FM (Pai et al., 2025), Merlin (Blankemeier et al., 2026), CT-CLIP (Hamamci et al., 2026), FM-HCT (Zhu et al., 2026) and VoCo (Wu et al., 2024, 2026); 2-D radiology models such as Curia process a volume slice by slice (Dancette et al., 2025). Their evaluations mostly rank frozen representations with a linear, attentive or nearest-neighbour head (Veenboer et al., 2026; Claessens et al., 2026), some next to end-to-end fine-tuning (Pai et al., 2025; Zhu et al., 2026). For natural images, comparisons on the same pretrained models showed that a frozen evaluation can mislead: linear-probe and fine-tuning accuracy of masked autoencoders were largely uncorrelated, and tuning one transformer block raised ImageNet accuracy from 73.5% to 81.0%

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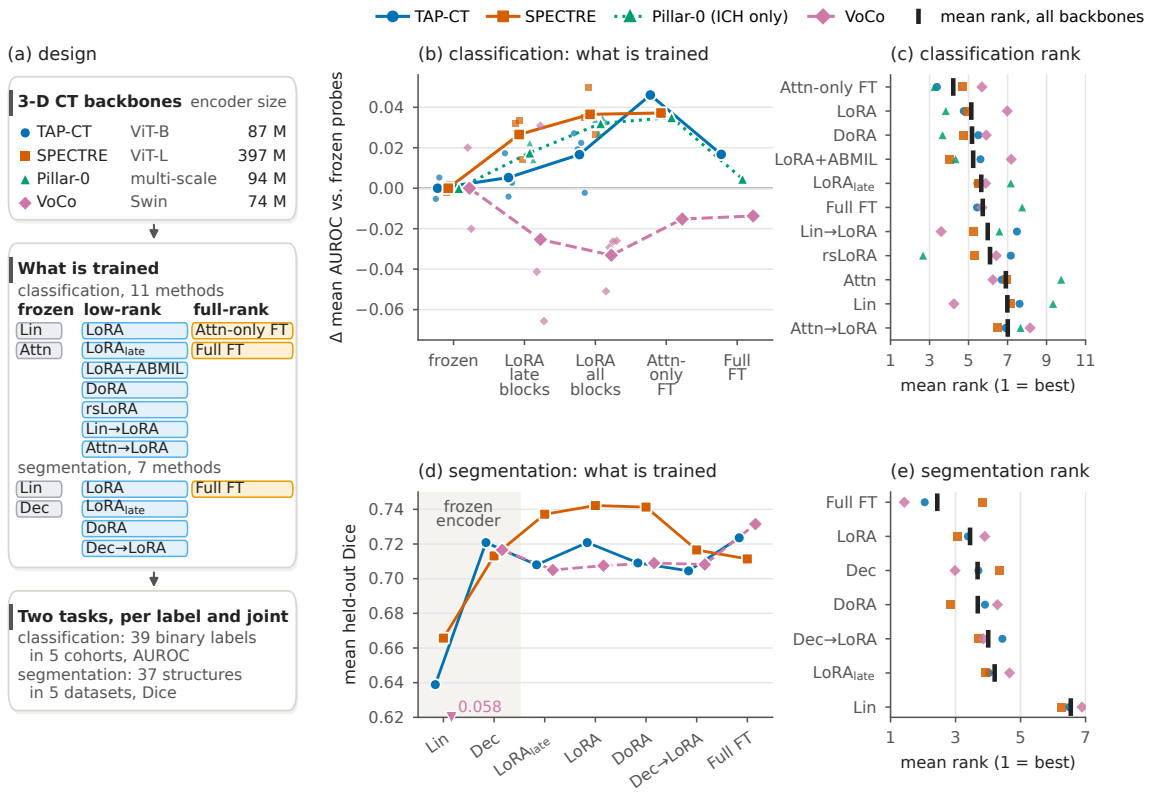


Figure 1: Study overview. (a) Backbones, methods and tasks. (b) Change in pooled held-out AUROC relative to the mean of the backbone's two frozen probes, by what is trained (small markers: methods; lines: group means; encoder parameters updated in Figure 5d); Pillar-0 covers the hemorrhage cohort only and VoCo three cohorts. (c) Mean classification rank per backbone (markers) and over all 188 label–mode pairs of the four backbones (bars). SPECTRE Full FT, which did not train, is left out of (b) and (c). (d) Mean held-out Dice per segmentation method; shaded methods keep the encoder frozen; VoCo's linear probe lies below the axis (value printed). (e) As (c), over 177 structure–mode pairs.

(He et al., 2022); prompt tuning with under 1% of the parameters beat full fine-tuning on 20 of 24 tasks (Jia et al., 2022); and full fine-tuning beat linear probing in distribution but not under distribution shift (Kumar et al., 2022). For 3-D medical images such as CT, such comparisons are few. However, a group that adapts an encoder to a multi-label cohort or a multi-structure dataset, often with a few hundred to a few thousand labeled scans, has to decide whether to keep the encoder frozen, adapt it with a low-rank update (Hu et al., 2022) or fine-tune part or all of it; whether to train one model per label or one for all labels of the dataset; and, for segmentation, which head turns tokens into a voxel map. A frozen probe answers none of these questions.

In medical imaging, the evidence for these choices comes mostly from 2-D backbones and single tasks. On 2-D medical classification, LoRA matches or surpasses full fine-tuning (Dutt et al., 2024; Zhu et al., 2024), and probing and fine-tuning of DINOv2 (Oquab et al., 2024) have been compared on radiology tasks (Baharoon et al., 2023). For 3-D classification, Liu et al. (2026b) found that frozen probing understates 2-D and 3-D foundation models, and on RSNA 2023 spleen injury, one of our labels, full fine-tuning of a convolutional and a Swin 3-D model surpassed LoRA. Parameter-efficient adaptation of 3-D medical encoders has otherwise been studied mostly for segmentation: adapters that lift the 2-D Segment Anything model (Kirillov et al., 2023) to volumes (Gong et al., 2024; Chen et al., 2024a; Wu et al., 2025), prompt tuning of a 3-D U-Net (Fischer et al., 2024), and LoRA, adapters and full fine-tuning in few-shot transfer (Silva-Rodríguez et al., 2025; Ly and Nguyen, 2026); for a 3-D CT vision–language model, full fine-tuning and parameter-efficient fine-tuning were close on one classification task (Kartika et al., 2026). Each study uses one backbone and one task, and LoRA variants (Liu et al., 2024; Kalajdzievski, 2023), probing before fine-tuning (Kumar et al., 2022) and attention-only fine-tuning (Touvron et al., 2022) were not part of them. Joint training on many labels can help (Kim et al., 2023) or hurt through conflicting gradients (Yu et al., 2020; Zhao et al., 2023), and task-specific LoRA weights from a hypernetwork have been proposed for chest CT (Liu et al., 2026a), but joint and per-task adaptation of a 3-D encoder have not been compared on the same labels (Veenboer et al., 2026; Liu et al., 2026b).

In this work, we run this comparison on five multi-label classification cohorts and five multi-structure segmentation datasets, under one protocol and on four encoders, two plain vision transformers (TAP-CT, SPECTRE), the convolutional-stem multi-scale transformer of Pillar-0 and the Swin encoder of VoCo: frozen probes, low-rank adapters and their variants, and partial and full fine-tuning, for classification on all four and segmentation on all but Pillar-0, each trained per task and jointly (Figures 1 and 2). Our contributions are:

- A controlled comparison of 11 classification and 7 segmentation adaptation methods on four 3-D CT encoders of three architectural families (1988 runs, plus 367 repeats and controls), with code, splits and per-case predictions released ([TODO: public repository URL]).
- What to train: for classification the attention weights, with LoRA ahead of both frozen probes on three of the four encoders and level with the matched probe on the fourth; for segmentation a multi-depth decoder, on which LoRA changes Dice little on average.
- One model per dataset or one per label: a joint model is on average level with per-label models in classification and costs Dice in segmentation at our training budgets.
- A practical guide (Section 5) that gives, for each decision, the setting we used and what it changed.

2. Methods

Backbones. **TAP-CT-B-3D** (Veenboer et al., 2026), pretrained by DINOv2-style self-supervision on in-house CT, is a plain 12-block ViT-B (Dosovitskiy et al., 2021) with $4\times8\times8$

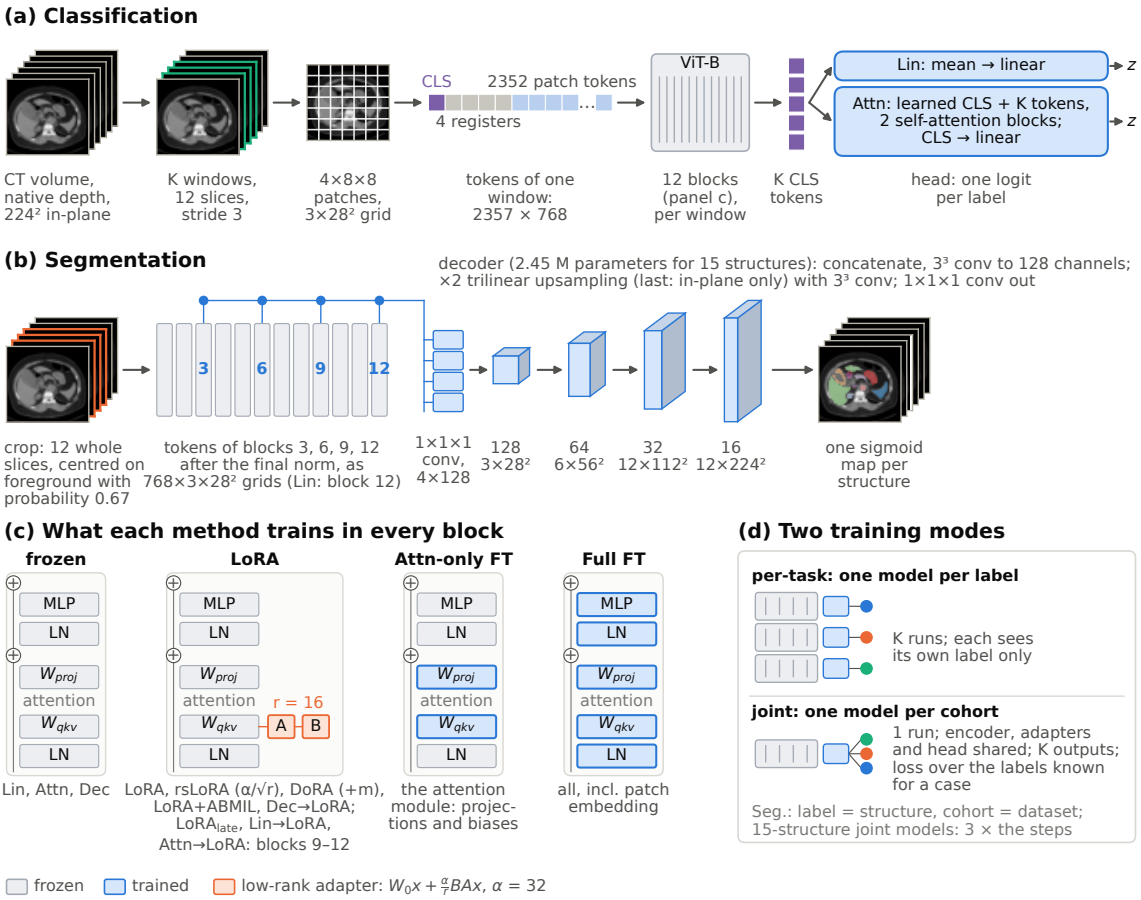


Figure 2: How each task is run. All panels show TAP-CT, drawn from the implementation; the other backbones are in Figures 5 and 6 (Appendix A). (a) Classification: 12-slice windows, $4 \times 8 \times 8$ patches; the CLS token of every window enters the head, a linear layer on their mean (Lin) or an attentive probe (Attn), with one logit per label. (b) Segmentation: the tokens of blocks 3, 6, 9 and 12 feed a multi-depth decoder (2.45 M parameters for 15 structures) with one sigmoid map per structure (Lin uses block 12 only). (c) What each method trains in every encoder block; the head is always trained. (d) Per-task and joint training.

voxel patches, a CLS token and four register tokens (Darcet et al., 2024), applied to windows of $12 \times 224 \times 224$ voxels at native slice spacing (Figure 2). **SPECTRE-Large** (Claessens et al., 2026), pretrained by self-distillation and vision–language alignment with radiology reports, is a plain 24-block ViT-L with $16 \times 16 \times 8$ patches on $128 \times 128 \times 64$ crops; for classification, 64 crops of a $512 \times 512 \times 256$ volume at $0.5 \times 0.5 \times 1$ mm are summarized by its pretrained 4-block feature combiner. **Pillar-0** (Agrawal et al., 2025), pretrained on CT and breast MRI of one academic center, is used through its head-CT encoder, a convolutional

stem followed by three stages of multi-scale window-attention blocks over the whole volume, which restricts it to the hemorrhage cohort. **VoCo-L** (Wu et al., 2024, 2026) is the self-supervised encoder of a SwinUNETR-v2 (Hatamizadeh et al., 2022a; He et al., 2023): four stages of shifted-window blocks (Liu et al., 2021) with patch merging and no class token; it encodes 96^3 windows at 1.5 mm, each summarized for classification by the concatenated spatial means of the stage-2 to stage-4 outputs (2688-d). Inputs follow each model's own orientation, spacing and intensity normalization (Figure 5 and Appendix A).

Classification methods. The methods differ only in what is trained (Figures 2c and 5d). *Frozen probes*: **Lin**, a linear layer on a pooled feature, and **Attn**, an attentive probe (Bardes et al., 2024; Chen et al., 2024b) in which a learnable CLS token joins the window, crop or scale tokens and two self-attention blocks run over the set. *Low-rank adapters* on the query/key/value projections, with the attentive head: **LoRA**, rank 16 ($\alpha = 32$) on every block (Hu et al., 2022); **LoRA$_{\text{late}}$**, on the last third of the blocks (the last stage for Pillar-0 and VoCo); **LoRA+ABMIL**, with tokens first pooled by gated attention (Ilse et al., 2018); **DoRA** (Liu et al., 2024); and **rsLoRA** (Kalajdzievski, 2023), whose $\alpha/\sqrt{r}$ scaling is four times the LoRA scale at $r = 16$. *Two-stage schedules*: **Lin→LoRA** and **Attn→LoRA** train the probe for half the budget and then add late-block LoRA, keeping the probe's head (Kumar et al., 2022). *Full-rank training*: **Attn-only FT** trains every parameter of the attention modules (Touvron et al., 2022) and **Full FT** every parameter of the encoder, both with the attentive head.

Segmentation methods. Segmentation uses TAP-CT, SPECTRE and VoCo. **Lin** is a $1 \times 1 \times 1$ convolution on the deepest token grid, upsampled trilinearly; for VoCo this grid has one token per 32^3 voxels (48 mm), far coarser than those of the two ViTs. **Dec** is a light multi-depth decoder in the spirit of UNETR (Hatamizadeh et al., 2022b): the token grids of four depths are projected, concatenated, fused and upsampled to the voxel grid by 3^3 convolutions (2.4 to 2.6 M parameters, no skip connections from the input; Figures 2b and 6). LoRA, LoRA$_{\text{late}}$ and DoRA add the adapters of classification under Dec, Full FT trains the encoder with it, and Dec→LoRA trains Dec for half the budget before adding all-block LoRA. Predictions are independent sigmoids per structure.

Per-task and joint training. Every cohort and dataset is trained in two modes with identical data and selection (Figure 2d): *per-task*, one model per label or structure, and *joint*, one model with shared encoder, adapters and head and one output per label or structure. A *label–mode pair* is one label (or structure) on one backbone in one mode. Joint losses sum over the labels known for a case, as in per-task training, and average over structures (Equations 1 and 2).

3. Experiments

Data. Classification uses five public CT cohorts with 39 binary labels (Table 21 and Figure 16 in Appendix F): the CVPR26 cohort, the release of the CVPR 2026 3-D CT foundation-model challenge, which groups AMOS (Ji et al., 2022) with 15 disease labels, COVID-CTset (Rahimzadeh et al., 2021) and LUNA25 nodule malignancy (Peeters et al., 2026); FDG-PET/CT (Gatidis et al., 2022), whose four diagnosis groups we predict from the CT alone; RSNA 2022 cervical-spine fractures (Lin et al., 2023); RSNA 2023 abdominal

trauma (Rudie et al., 2024); and RSNA 2019 intracranial hemorrhage (Flanders et al., 2020). Segmentation uses AMOS22 (15 organs; Ji et al., 2022), KiTS23 (Heller et al., 2023), LiTS (Bilic et al., 2023), MSD Pancreas (Antonelli et al., 2022) and TotalSegmentator (Wasserthal et al., 2023; Wasserthal, 2026) with its 117 structures merged into 15 (Table 22); 37 structures in all. For the CVPR26 cohort and AMOS22 the official validation cases are divided into a selection and a held-out half; the other sources use fixed seeded splits of the released labeled cases, and TotalSegmentator holds out its official test split. VoCo is run on FDG-PET/CT, RSNA 2022, RSNA 2023 and all segmentation datasets. Several evaluation sets appear, as images, in the pretraining data of SPECTRE and VoCo (Table 23).

Optimization. All runs use AdamW with 10% linear warm-up, cosine decay to 10^{-6} and bf16 precision. A classification run trains 8000 steps at batch size 1 on a masked, positive-weighted binary cross-entropy; a segmentation run trains 8000 steps of four crops on soft Dice plus binary cross-entropy; two-stage methods split the budget in halves. A joint model thus uses $1/K$ of the steps of the K per-task models it replaces, except the 15-structure joint segmentation models (AMOS22, TotalSegmentator), which train 24 000 steps. Each method has one default learning rate, set by what it trains: 10^{-3} for the classification linear probe and the segmentation decoder, 10^{-2} for the segmentation linear probe, 10^{-4} for the attentive probe and the adapters, and 2×10^{-5} for encoder weights; none was tuned on our data (Tables 2 and 3). The checkpoint with the best selection macro AUROC or mean Dice, evaluated every 1000 steps, is kept and its held-out metric reported.

Evaluation. AUROC per label and Dice per structure and case are the metrics (Maier-Hein et al., 2024). Methods are compared per label–mode pair with DeLong's test (DeLong et al., 1988) or a paired case bootstrap on Dice, with Benjamini–Hochberg correction (Benjamini and Hochberg, 1995), and ranked within each pair that has every method. The main comparisons are summarized at label level, each label counting once with its two modes averaged, with 95% intervals from a bootstrap over cohorts, labels and held-out cases (Appendix B); as four or five cohorts per backbone make the cohort stage optimistic, we also count the cohorts with a positive mean difference. VoCo, with three of the five classification cohorts, is reported per backbone and never pooled. A method whose held-out predictions barely vary between cases did not train: SPECTRE Full FT, with a median standard deviation of held-out probabilities of 0.004 against at least 0.114 for every other method; it stays in the tables and out of ranks and averages over methods. The joint models of Lin, LoRA, Attn-only FT and Full FT (classification) and of Dec, LoRA and Full FT (segmentation) were trained with two further seeds (Appendix D).

4. Results

Table 1 summarizes the main comparisons. Per-label and per-structure results are in Appendix E, paired tests and further analyses in Appendix C, seed repeats and controls in Appendix D, and success and failure cases in Appendix G.

4.1. Classification: train the attention weights

Updating the encoder's attention weights beats both frozen probes. On TAP-CT, SPECTRE and Pillar-0, LoRA raises held-out AUROC by +0.034 over the linear probe

Table 1: The main comparisons at label level: mean difference, 95% interval over cohorts, labels and held-out cases, and the number of cohorts (datasets) with a positive mean. Modes are averaged per label except in the LoRA − Dec sub-rows; LoRA − Attn compares methods with the same head. VoCo covers three classification cohorts; pooled estimates are in Table 6.

	TAP-CT		SPECTRE		Pillar-0		VoCo	
Comparison	Mean [95% interval]	> 0	Mean [95% interval]	> 0	Mean [95% interval]	> 0	Mean [95% interval]	> 0
Classification (AUROC)								
LoRA − Lin	+0.032 [+0.007, +0.053]	4/5	+0.037 [+0.009, +0.075]	4/4	+0.033 [+0.021, +0.044]	1/1	−0.071 [−0.168, −0.005]	0/3
LoRA − Attn	+0.022 [+0.006, +0.042]	5/5	+0.033 [+0.003, +0.078]	4/4	+0.034 [+0.021, +0.052]	1/1	−0.031 [−0.145, +0.043]	1/3
Attn-only FT − LoRA	+0.019 [−0.001, +0.054]	4/5	+0.002 [−0.037, +0.026]	2/4	+0.001 [−0.002, +0.004]	1/1	+0.036 [−0.024, +0.102]	2/3
Full FT − Lin	+0.022 [−0.061, +0.068]	4/5	−0.117 [−0.195, −0.059]	0/4	+0.004 [−0.013, +0.021]	1/1	−0.034 [−0.136, +0.027]	1/3
Joint − per-task	+0.012 [−0.014, +0.038]	3/5	−0.003 [−0.044, +0.051]	3/4	−0.008 [−0.013, −0.001]	0/1	−0.001 [−0.022, +0.018]	2/3
Segmentation (Dice)								
Dec − Lin	+0.085 [+0.055, +0.109]	5/5	+0.061 [+0.027, +0.088]	5/5	–		+0.680 [+0.433, +0.746]	5/5
LoRA − Dec	−0.005 [−0.021, +0.024]	4/5	+0.023 [−0.018, +0.076]	4/5	–		−0.018 [−0.047, +0.016]	2/5
per-structure models	−0.021 [−0.036, +0.002]	2/5	+0.006 [−0.003, +0.031]	5/5	–		−0.035 [−0.062, −0.007]	0/5
joint models	+0.011 [−0.021, +0.066]	4/5	+0.040 [−0.039, +0.147]	4/5	–		0.000 [−0.034, +0.047]	2/5
Full FT − Dec	0.000 [−0.024, +0.016]	4/5	−0.003 [−0.035, +0.009]	2/5	–		+0.015 [+0.007, +0.023]	4/5
Joint − per-task	−0.046 [−0.090, −0.008]	0/5	−0.058 [−0.093, −0.009]	0/5	–		−0.041 [−0.081, +0.004]	2/5

and by +0.027 over the frozen attentive probe, which has the same head; both intervals exclude zero, and the gain over the attentive probe is positive in all 10 cohort–backbone pairs (Table 1). Training the attention projections in full gains +0.045 and +0.038, again positive against the attentive probe in all 10 pairs. The gain comes from the encoder, not the head: the frozen attentive probe adds only +0.007 to the linear probe. It is largest where the frozen probe is weakest (+0.054 at a selection AUROC of at most 0.6, +0.025 above 0.8; Figure 14), it survives the removal of FDG-PET/CT and COVID-CTset, where we did not rule out shortcuts (+0.037), and LoRA is ahead of the linear probe in 27 of 30 cohort–seed pairs of the repeated joint models.

Attention-only fine-tuning ranks best; LoRA comes close at about a fiftieth of the parameters. Attention-only fine-tuning has the best mean rank over all slots (4.22; LoRA 5.14; Figure 1c), ranks above LoRA on every backbone, and is ahead of LoRA in 25 of 30 cohort–seed pairs, by +0.011 on average (interval −0.007 to +0.029), most on TAP-CT (+0.019). LoRA reaches this accuracy while updating 0.59 M instead of 28 M encoder parameters on TAP-CT and 2.4 M instead of 131 M on SPECTRE.

The head is chosen first: on VoCo the linear head wins. On VoCo the frozen attentive probe is behind the linear probe (−0.040), and every method that adapts the encoder under the attentive head inherits this deficit: against the attentive probe all eight are level (LoRA −0.031, attention-only fine-tuning +0.005, full fine-tuning +0.006), but against the linear probe they fall behind (LoRA −0.071, most on FDG-PET/CT). Lin→LoRA, which adapts the encoder under the linear head, is the best-ranked method on VoCo (3.59; Lin 4.25) and the one method with a positive mean difference to the linear probe on all four backbones (+0.001 to +0.035; VoCo +0.011). VoCo ran with the default settings of the other backbones.

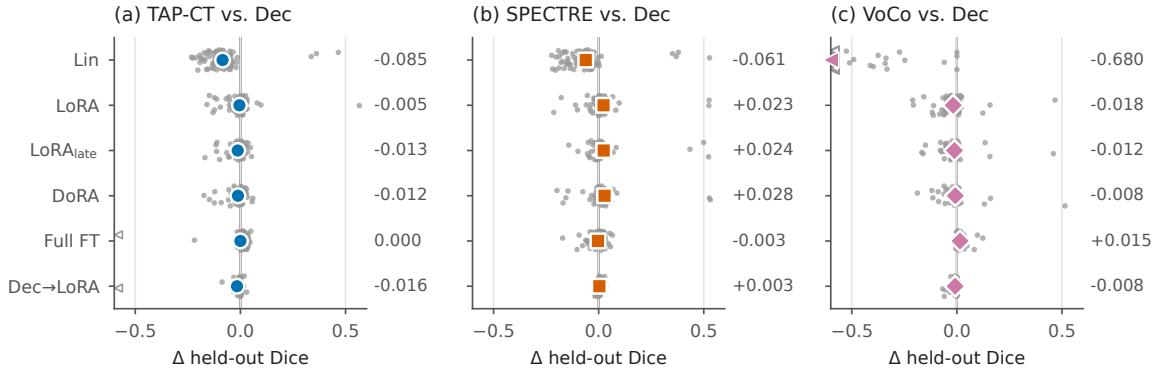


Figure 3: Segmentation: per-structure change in held-out Dice relative to the frozen decoder of the same backbone, dataset and mode, for (a) TAP-CT, (b) SPECTRE and (c) VoCo (grey: structure–mode pairs, triangles beyond the axis; colored: mean, printed on the right).

Full fine-tuning needs its own setting. With the default setting, Full FT is ahead of the linear probe on TAP-CT (+0.022, and best on RSNA 2023 trauma, as Liu et al. (2026b) found for spleen injury), level with it on Pillar-0 and VoCo, but does not train on SPECTRE: its median validation AUROC stays at 0.529, whereas attention-only fine-tuning at the same learning rates reaches 0.743 (Figure 15). With layer-wise learning-rate decay (0.75 per block), the best of three alternative settings on selection data, SPECTRE Full FT trains and is ahead of LoRA by +0.017 (Table 14), a comparison that favors Full FT, as LoRA was not re-tuned.

Plain LoRA is enough. No variant improves on LoRA on average: LoRA$_{\text{late}}$ changes held-out AUROC by −0.007, DoRA by −0.002, rsLoRA by −0.018, LoRA+ABMIL by +0.002 and the two-stage schedules by −0.017 and −0.022; rsLoRA at a quarter of the learning rate, which matches its update scale to LoRA's, is level with LoRA on SPECTRE (+0.003) and Pillar-0 (+0.008) and behind on TAP-CT (−0.020; Table 15).

4.2. Segmentation: a decoder on the encoder

A multi-depth decoder on the frozen encoder gives most of the gain. Replacing the linear probe by the frozen decoder raises Dice on every dataset and backbone, by +0.085 on TAP-CT, +0.061 on SPECTRE and +0.680 on VoCo, whose deepest grid has one token per 48 mm (Table 1). On TAP-CT this is all of LoRA's gain over the linear probe (+0.080) and on SPECTRE most of it (+0.085). The pretrained weights matter: the TAP-CT architecture trained from a random initialization is behind fine-tuning of the pretrained encoder on every dataset, by −0.103 to −0.236 Dice (Table 17).

On top of the decoder, full fine-tuning ranks best and LoRA changes Dice little. Full fine-tuning has the best mean rank (2.44; LoRA 3.45; Figure 1e), first on TAP-CT (2.05) and VoCo (1.42), where it is ahead of the decoder by +0.015 (interval +0.007 to +0.023); its

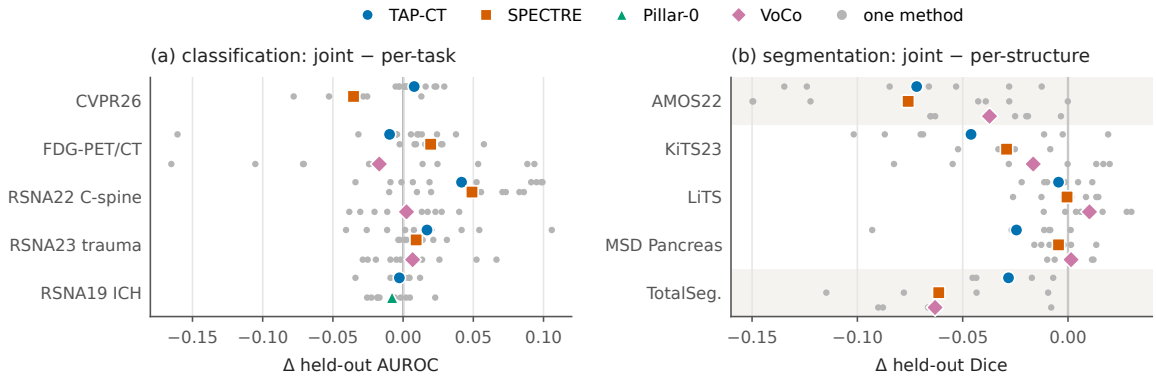


Figure 4: One model per dataset against one model per label. (a) Joint minus per-task held-out AUROC per classification cohort and (b) joint minus per-structure held-out Dice per segmentation dataset; large markers: mean over methods and labels (structures) per backbone; grey dots: individual methods. Per-label values are in Appendix E and Figure 12.

mean change on TAP-CT is 0.000 because 2 structure–mode pairs lose more than 0.1, among them the AMOS22 esophagus of the joint model (−0.713; Appendix G). LoRA changes Dice relative to the frozen decoder by −0.005, +0.023 and −0.018 on TAP-CT, SPECTRE and VoCo, with intervals that include zero (Figure 3); it gains most where the frozen decoder fails, in the SPECTRE AMOS22 joint model (0.658 to 0.784; Table 9). Tripling the steps of the 15-structure joint models raised their Dice by +0.121 and +0.147 (Table 10), and 75% of segmentation runs selected one of their last two validations (Figure 11), so these comparisons rank methods at a fixed budget, not at convergence.

4.3. One model per dataset or one per label

In classification, one model per cohort is on average level with one per label, at a fraction of the training. Joint minus per-task AUROC is +0.004 on TAP-CT, SPECTRE and Pillar-0 (interval −0.021 to +0.030) and −0.001 on VoCo (−0.022 to +0.018), while a joint model uses $1/K$ of the training steps of the K per-label models it replaces; per cohort, averaged over methods, the difference ranges from −0.035 to +0.049 (Figure 4a; per method in Figure 12).

In segmentation, the cost of one joint model lies in small structures and in the budget per structure. One joint model per dataset changes Dice by −0.046, −0.058 and −0.041 on TAP-CT, SPECTRE and VoCo, most on the 15-structure AMOS22 (Figure 4b). Much of this comes from 28 of 428 structure–method pairs on small structures (adrenal glands and esophagus of AMOS22, merged glands of TotalSegmentator), where the joint model scores below 0.01 Dice; without them the mean over structure–method pairs moves from −0.050 to −0.010 on TAP-CT and from −0.058 to −0.024 on SPECTRE. At matched total compute, the joint LoRA model is ahead of per-structure LoRA models on both 15-

structure datasets and both backbones (+0.042 to +0.136 Dice), whereas with the frozen decoder and full fine-tuning it is behind in three of the four pairs (Table 16).

5. A practical guide and discussion

Each step gives the setting we used and what it changed in our comparison (Table 1).

1. **Classification: choose the head on the frozen encoder first.** Train a linear and an attentive probe and keep the better one on selection data: the attentive head was slightly better on TAP-CT (+0.011) and SPECTRE (+0.004), the linear head clearly better on VoCo (−0.040 for the attentive head).
2. **Classification: then adapt the attention weights under that head.** Rank-16 LoRA ($\alpha = 32$, learning rate 10^{-4}) on every block added +0.034 AUROC over the linear probe and +0.027 over the attentive probe, positive in all 10 cohort–backbone pairs; on VoCo, adapt under the linear head (Lin→LoRA, +0.011). Fine-tuning the attention weights in full was ahead of LoRA in 25 of 30 seed pairs, by +0.011 on average, at about fifty times the trainable parameters. Use plain LoRA: no variant improved on it.
3. **Classification: train one model per cohort, not one per label.** It was on average level with per-label models (+0.004) at $1/K$ of the training.
4. **Segmentation: put a multi-depth decoder on the frozen encoder.** It added +0.061 to +0.680 Dice over a linear probe on every dataset. Adapting the encoder beneath it changed Dice by at most +0.023 on average; full fine-tuning ranked best but lost more than 0.1 Dice on some small structures, so check Dice per structure.
5. **Segmentation: train a joint model with LoRA and the steps of the per-structure models it replaces.** At matched compute it was ahead of per-structure LoRA by +0.042 to +0.136 Dice; at our shorter budgets one model per structure was more accurate.
6. **Full fine-tuning: lower the learning rate of early blocks.** With the encoder learning rate of the other methods it did not train on SPECTRE (near-constant predictions); layer-wise decay of 0.75 per block fixed it. Check that held-out predictions vary between cases.

Limitations. Each method has one untuned default setting, and only full fine-tuning was retrained after it failed, so the ranking among the stronger methods may change with tuning; seeds were repeated only for joint models. Segmentation had not converged and is scored on each backbone's working grid, so it ranks adaptation methods rather than measuring the distance to nnU-Net (Isensee et al., 2021) or to models pretrained on segmentation labels (Li et al., 2024; He et al., 2025). SPECTRE's and VoCo's pretraining images include several of our evaluation sets (Table 23), and one author co-developed TAP-CT.

References

Kumar Krishna Agrawal, Longchao Liu, Long Lian, Michael Nercessian, Natalia Harguindeguy, Yufu Wu, Peter Mikhael, Gigin Lin, Lecia V. Sequist, Florian Fintelmann, Trevor Darrell, Yutong Bai, Maggie Chung, and Adam Yala. Pillar-0: A new frontier for radiology foundation models. *arXiv preprint arXiv:2511.17803*, 2025.

Michela Antonelli, Annika Reinke, Spyridon Bakas, Keyvan Farahani, Annette Kopp-Schneider, Bennett A. Landman, et al. The medical segmentation decathlon. *Nature Communications*, 13(1):4128, 2022. doi: 10.1038/s41467-022-30695-9.

Mohammed Baharoon, Waseem Qureshi, Jiahong Ouyang, Yanwu Xu, Abdulrhman Aljouie, and Wei Peng. Evaluating general purpose vision foundation models for medical image analysis: An experimental study of DINOv2 on radiology benchmarks. *arXiv preprint arXiv:2312.02366*, 2023.

Adrien Bardes, Quentin Garrido, Jean Ponce, Xinlei Chen, Michael Rabbat, Yann LeCun, Mahmoud Assran, and Nicolas Ballas. Revisiting feature prediction for learning visual representations from video. *Transactions on Machine Learning Research*, 2024. URL https://openreview.net/forum?id=QaCCuDfBk2. arXiv:2404.08471.

Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.

Patrick Bilic, Patrick Christ, Hongwei Bran Li, Eugene Vorontsov, Avi Ben-Cohen, Georgios Kaissis, et al. The liver tumor segmentation benchmark (LiTS). *Medical Image Analysis*, 84:102680, 2023. doi: 10.1016/j.media.2022.102680.

Louis Blankemeier, Ashwin Kumar, Joseph Paul Cohen, Jiaming Liu, Longchao Liu, Dave Van Veen, et al. Merlin: a computed tomography vision–language foundation model and dataset. *Nature*, 652(8112):1318–1328, 2026. doi: 10.1038/s41586-026-10181-8.

Cheng Chen, Juzheng Miao, Dufan Wu, Aoxiao Zhong, Zhiling Yan, Sekeun Kim, Jiang Hu, Zhengliang Liu, Lichao Sun, Xiang Li, Tianming Liu, Pheng-Ann Heng, and Quanzheng Li. MA-SAM: Modality-agnostic SAM adaptation for 3D medical image segmentation. *Medical Image Analysis*, 98:103310, 2024a. doi: 10.1016/j.media.2024.103310.

Xiaokang Chen, Mingyu Ding, Xiaodi Wang, Ying Xin, Shentong Mo, Yunhao Wang, Shumin Han, Ping Luo, Gang Zeng, and Jingdong Wang. Context autoencoder for self-supervised representation learning. *International Journal of Computer Vision*, 132(1): 208–223, 2024b. doi: 10.1007/s11263-023-01852-4.

Cris Claessens, Christiaan Viviers, Giacomo D'Amicantonio, Egor Bondarev, and Fons van der Sommen. Scaling self-supervised and cross-modal pretraining for volumetric CT transformers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 13636–13647, 2026. arXiv:2511.17209.

Corentin Dancette, Julien Khlaut, Antoine Saporta, Helene Philippe, Elodie Ferreres, Baptiste Callard, et al. Curia: A multi-modal foundation model for radiology. *arXiv preprint arXiv:2509.06830*, 2025.

Timothée Darcet, Maxime Oquab, Julien Mairal, and Piotr Bojanowski. Vision transformers need registers. In *International Conference on Learning Representations (ICLR)*, 2024. URL https://openreview.net/forum?id=2dnO3LLiJ1.

Elizabeth R. DeLong, David M. DeLong, and Daniel L. Clarke-Pearson. Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach. *Biometrics*, 44(3):837, 1988. doi: 10.2307/2531595.

Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations (ICLR)*, 2021. arXiv:2010.11929.

Raman Dutt, Linus Ericsson, Pedro Sanchez, Sotirios A. Tsaftaris, and Timothy Hospedales. Parameter-efficient fine-tuning for medical image analysis: The missed opportunity. In *Proceedings of the 7th International Conference on Medical Imaging with Deep Learning*, volume 250 of *Proceedings of Machine Learning Research*, pages 406–425. PMLR, 2024. URL https://proceedings.mlr.press/v250/dutt24a.html.

Marc Fischer, Alexander Bartler, and Bin Yang. Prompt tuning for parameter-efficient medical image segmentation. *Medical Image Analysis*, 91:103024, 2024. doi: 10.1016/j.media.2023.103024.

Adam E. Flanders, Luciano M. Prevedello, George Shih, Safwan S. Halabi, Jayashree Kalpathy-Cramer, Robyn Ball, et al. Construction of a machine learning dataset through collaboration: The RSNA 2019 brain CT hemorrhage challenge. *Radiology: Artificial Intelligence*, 2(3):e190211, 2020. doi: 10.1148/ryai.2020190211.

Sergios Gatidis, Tobias Hepp, Marcel Früh, Christian La Fougère, Konstantin Nikolaou, Christina Pfannenberg, Bernhard Schölkopf, Thomas Küstner, Clemens Cyran, and Daniel Rubin. A whole-body FDG-PET/CT dataset with manually annotated tumor lesions. *Scientific Data*, 9(1):601, 2022. doi: 10.1038/s41597-022-01718-3.

Shizhan Gong, Yuan Zhong, Wenao Ma, Jinpeng Li, Zhao Wang, Jingyang Zhang, Pheng-Ann Heng, and Qi Dou. 3DSAM-adapter: Holistic adaptation of SAM from 2D to 3D for promptable tumor segmentation. *Medical Image Analysis*, 98:103324, 2024. doi: 10.1016/j.media.2024.103324.

Ibrahim Ethem Hamamci, Sezgin Er, Chenyu Wang, Furkan Almas, Ayse Gulnihan Simsek, Sevval Nil Esirgun, et al. Generalist foundation models from a multimodal dataset for 3D computed tomography. *Nature Biomedical Engineering*, 10(8):1610–1628, 2026. doi: 10.1038/s41551-025-01599-y.

Ali Hatamizadeh, Vishwesh Nath, Yucheng Tang, Dong Yang, Holger R. Roth, and Daguang Xu. Swin UNETR: Swin transformers for semantic segmentation of brain tumors in MRI images. In *Brainlesion: Glioma, Multiple Sclerosis, Stroke and Traumatic Brain Injuries (BrainLes 2021)*, Lecture Notes in Computer Science, pages 272–284. Springer International Publishing, 2022a. doi: 10.1007/978-3-031-08999-2_22.

Ali Hatamizadeh, Yucheng Tang, Vishwesh Nath, Dong Yang, Andriy Myronenko, Bennett Landman, Holger R. Roth, and Daguang Xu. UNETR: Transformers for 3D medical image segmentation. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 1748–1758, 2022b. doi: 10.1109/WACV51458.2022.00181.

Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15979–15988, 2022. doi: 10.1109/CVPR52688.2022.01553.

Yufan He, Vishwesh Nath, Dong Yang, Yucheng Tang, Andriy Myronenko, and Daguang Xu. SwinUNETR-V2: Stronger swin transformers with stagewise convolutions for 3D medical image segmentation. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2023*, Lecture Notes in Computer Science, pages 416–426. Springer Nature Switzerland, 2023. doi: 10.1007/978-3-031-43901-8_40.

Yufan He, Pengfei Guo, Yucheng Tang, Andriy Myronenko, Vishwesh Nath, Ziyue Xu, Dong Yang, Can Zhao, Benjamin Simon, Mason Belue, Stephanie Harmon, Baris Turkbey, Daguang Xu, and Wenqi Li. VISTA3D: A unified segmentation foundation model for 3D medical imaging. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 20863–20873, 2025. doi: 10.1109/CVPR52734.2025.01943.

Nicholas Heller, Fabian Isensee, Dasha Trofimova, Resha Tejpaul, Zhongchen Zhao, Huai Chen, Lisheng Wang, Alex Golts, Daniel Khapun, Daniel Shats, Yoel Shoshan, Flora Gilboa-Solomon, Yasmeen George, Xi Yang, Jianpeng Zhang, Jing Zhang, Yong Xia, Mengran Wu, Zhiyang Liu, Ed Walczak, Sean McSweeney, Ranveer Vasdev, Chris Hornung, Rafat Solaiman, Jamee Schoephoerster, Bailey Abernathy, David Wu, Safa Abdulkadir, Ben Byun, Justice Spriggs, Griffin Struyk, Alexandra Austin, Ben Simpson, Michael Hagstrom, Sierra Virnig, John French, Nitin Venkatesh, Sarah Chan, Keenan Moore, Anna Jacobsen, Susan Austin, Mark Austin, Subodh Regmi, Nikolaos Papanikolopoulos, and Christopher Weight. The KiTS21 challenge: Automatic segmentation of kidneys, renal tumors, and renal cysts in corticomedullary-phase CT. *arXiv preprint arXiv:2307.01984*, 2023.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022. URL https://openreview.net/forum?id=nZeVKeeFYf9. arXiv:2106.09685.

Maximilian Ilse, Jakub M. Tomczak, and Max Welling. Attention-based deep multiple instance learning. In *Proceedings of the 35th International Conference on Machine Learning*,

volume 80 of *Proceedings of Machine Learning Research*, pages 2127–2136. PMLR, 2018. URL https://proceedings.mlr.press/v80/ilse18a.html.

Fabian Isensee, Paul F. Jaeger, Simon A. A. Kohl, Jens Petersen, and Klaus H. Maier-Hein. nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 18(2):203–211, 2021. doi: 10.1038/s41592-020-01008-z.

Yuanfeng Ji, Haotian Bai, Chongjian Ge, Jie Yang, Ye Zhu, Ruimao Zhang, Zhen Li, Lingyan Zhang, Wanling Ma, Xiang Wan, and Ping Luo. AMOS: A large-scale abdominal multi-organ benchmark for versatile medical image segmentation. In *Advances in Neural Information Processing Systems*, volume 35, pages 36722–36732, 2022. doi: 10.52202/068431-2661.

Menglin Jia, Luming Tang, Bor-Chun Chen, Claire Cardie, Serge Belongie, Bharath Hariharan, and Ser-Nam Lim. Visual prompt tuning. In *Computer Vision – ECCV 2022*, Lecture Notes in Computer Science, pages 709–727. Springer Nature Switzerland, 2022. doi: 10.1007/978-3-031-19827-4_41.

Damjan Kalajdzievski. A rank stabilization scaling factor for fine-tuning with LoRA. *arXiv preprint arXiv:2312.03732*, 2023.

Febryan Putra Kartika, Cheng-Yu Ma, Ying-Jia Lin, Chi-Tung Cheng, and Kuan-Fu Chen. Efficient self-supervised adaptation of 3D abdominal vision-language model for institution-specific HCC classification via full fine-tuning and PEFT. In *Proceedings of The 9th International Conference on Medical Imaging with Deep Learning*, volume 315 of *Proceedings of Machine Learning Research*, pages 3245–3269. PMLR, 2026. URL https://proceedings.mlr.press/v315/kartika26a.html.

Sangwook Kim, Thomas G. Purdie, and Chris McIntosh. Cross-task attention network: Improving multi-task learning for medical imaging applications. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2023 Workshops*, Lecture Notes in Computer Science, pages 119–128. Springer Nature Switzerland, 2023. doi: 10.1007/978-3-031-47401-9_12.

Alexander Kirillov, Eric Mintun, Nikhila Ravi, Hanzi Mao, Chloe Rolland, Laura Gustafson, Tete Xiao, Spencer Whitehead, Alexander C. Berg, Wan-Yen Lo, Piotr Dollár, and Ross Girshick. Segment anything. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 3992–4003, 2023. doi: 10.1109/ICCV51070.2023.00371.

Ananya Kumar, Aditi Raghunathan, Robbie Jones, Tengyu Ma, and Percy Liang. Fine-tuning can distort pretrained features and underperform out-of-distribution. In *International Conference on Learning Representations (ICLR)*, 2022. arXiv:2202.10054.

Wenxuan Li, Alan Yuille, and Zongwei Zhou. How well do supervised 3D models transfer to medical imaging tasks? In *International Conference on Learning Representations (ICLR)*, 2024. URL https://openreview.net/forum?id=AhizIPytk4.

Hui Ming Lin, Errol Colak, Tyler Richards, Felipe C. Kitamura, Luciano M. Prevedello, Jason Talbott, et al. The RSNA cervical spine fracture CT dataset. *Radiology: Artificial Intelligence*, 5(5):e230034, 2023. doi: 10.1148/ryai.230034.

Fengbei Liu, Sunwoo Kwak, Hao Phung, Nusrat Binta Nizam, Ilan Richter, Nir Uriel, Hadar Averbuch-Elor, Deborah Estrin, and Mert R. Sabuncu. HyperCT: Low-rank hypernet for unified chest CT analysis. In *Proceedings of The 9th International Conference on Medical Imaging with Deep Learning*, volume 315 of *Proceedings of Machine Learning Research*, pages 1768–1801. PMLR, 2026a. URL https://proceedings.mlr.press/v315/liu26c.html.

Han Liu, Bogdan Georgescu, Yanbo Zhang, Youngjin Yoo, Michael Baumgartner, Riqiang Gao, Jianing Wang, Gengyan Zhao, Eli Gibson, Dorin Comaniciu, and Sasa Grbic. Revisiting 2D foundation models for scalable 3D medical image classification. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 30021–30031, 2026b. arXiv:2512.12887.

Shih-Yang Liu, Chien-Yi Wang, Hongxu Yin, Pavlo Molchanov, Yu-Chiang Frank Wang, Kwang-Ting Cheng, and Min-Hung Chen. DoRA: Weight-decomposed low-rank adaptation. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 32100–32121. PMLR, 2024. URL https://proceedings.mlr.press/v235/liu24bn.html.

Ze Liu, Yutong Lin, Yue Cao, Han Hu, Yixuan Wei, Zheng Zhang, Stephen Lin, and Baining Guo. Swin transformer: Hierarchical vision transformer using shifted windows. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 9992–10002, 2021. doi: 10.1109/ICCV48922.2021.00986.

Son Thai Ly and Hien V. Nguyen. Self-auditing parameter-efficient fine-tuning for few-shot 3D medical image segmentation. In *Medical Image Computing and Computer Assisted Intervention – MICCAI 2026*, Lecture Notes in Computer Science, pages 559–569. Springer Nature Switzerland, 2026. doi: 10.1007/978-3-032-38069-2_53. arXiv:2603.05822.

Lena Maier-Hein, Annika Reinke, Patrick Godau, Minu D. Tizabi, Florian Buettner, Evangelia Christodoulou, et al. Metrics reloaded: recommendations for image analysis validation. *Nature Methods*, 21(2):195–212, 2024. doi: 10.1038/s41592-023-02151-z.

Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jegou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024. URL https://openreview.net/forum?id=a68SUt6zFt. arXiv:2304.07193.

Suraj Pai, Ibrahim Hadzic, Dennis Bontempi, Keno Bressem, Benjamin H. Kann, Andriy Fedorov, Raymond H. Mak, and Hugo J. W. L. Aerts. Vision foundation models for computed tomography. *arXiv preprint arXiv:2501.09001*, 2025.

Dré Peeters, Bogdan Obreja, Noa Antonissen, and Colin Jacobs. The LUNA25 challenge: Public training and development set – imaging data, 2025. URL https://doi.org/10.5281/zenodo.14223624. Version 1.0.0.

Dré Peeters, Bogdan Obreja, Noa Antonissen, Zaigham Saghir, Ugo Pastorino, Mario Silva, et al. Benchmarking of AI and radiologists for indeterminate lung nodule malignancy risk estimation at screening CT: The LUNA25 challenge. *Radiology: Artificial Intelligence*, 8 (5):e260179, 2026. doi: 10.1148/ryai.260179.

Mohammad Rahimzadeh, Abolfazl Attar, and Seyed Mohammad Sakhaei. A fully automated deep learning-based network for detecting COVID-19 from a new and large lung CT scan dataset. *Biomedical Signal Processing and Control*, 68:102588, 2021. doi: 10.1016/j.bspc.2021.102588.

Jeffrey D. Rudie, Hui-Ming Lin, Robyn L. Ball, Sabeena Jalal, Luciano M. Prevedello, Savvas Nicolaou, et al. The RSNA abdominal traumatic injury CT (RATIC) dataset. *Radiology: Artificial Intelligence*, 6(6):e240101, 2024. doi: 10.1148/ryai.240101.

Julio Silva-Rodríguez, Jose Dolz, and Ismail Ben Ayed. Towards foundation models and few-shot parameter-efficient fine-tuning for volumetric organ segmentation. *Medical Image Analysis*, 103:103596, 2025. doi: 10.1016/j.media.2025.103596.

Hugo Touvron, Matthieu Cord, Alaaeldin El-Nouby, Jakob Verbeek, and Hervé Jégou. Three things everyone should know about vision transformers. In *Computer Vision – ECCV 2022*, Lecture Notes in Computer Science, pages 497–515. Springer Nature Switzerland, 2022. doi: 10.1007/978-3-031-20053-3_29.

Tim Veenboer, George Yiasemis, Eric Marcus, Vivien van Veldhuizen, Cees G. M. Snoek, Jonas Teuwen, and Kevin B. W. Groot Lipman. TAP-CT: 3D task-agnostic pretraining of computed tomography foundation models. In *Proceedings of The 9th International Conference on Medical Imaging with Deep Learning*, volume 315 of *Proceedings of Machine Learning Research*, pages 726–753. PMLR, 2026. URL https://proceedings.mlr.press/v315/veenboer26a.html. arXiv:2512.00872.

Jakob Wasserthal. Dataset with segmentations of 117 important anatomical structures in 1939 CT images, 2026. URL https://doi.org/10.5281/zenodo.22688904. Version 3.0.0.

Jakob Wasserthal, Hanns-Christian Breit, Manfred T. Meyer, Maurice Pradella, Daniel Hinck, Alexander W. Sauter, Tobias Heye, Daniel T. Boll, Joshy Cyriac, Shan Yang, Michael Bach, and Martin Segeroth. TotalSegmentator: Robust segmentation of 104 anatomic structures in CT images. *Radiology: Artificial Intelligence*, 5(5):e230024, 2023. doi: 10.1148/ryai.230024.

Junde Wu, Ziyue Wang, Mingxuan Hong, Wei Ji, Huazhu Fu, Yanwu Xu, Min Xu, and Yueming Jin. Medical SAM adapter: Adapting segment anything model for medical image segmentation. *Medical Image Analysis*, 102:103547, 2025. doi: 10.1016/j.media.2025.103547.

Linshan Wu, Jiaxin Zhuang, and Hao Chen. VoCo: A simple-yet-effective volume contrastive learning framework for 3D medical image analysis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 22873–22882, 2024. doi: 10.1109/CVPR52733.2024.02158.

Linshan Wu, Jiaxin Zhuang, and Hao Chen. Large-scale 3D medical image pre-training with geometric context priors. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 48(3):3801–3818, 2026. doi: 10.1109/TPAMI.2025.3639593.

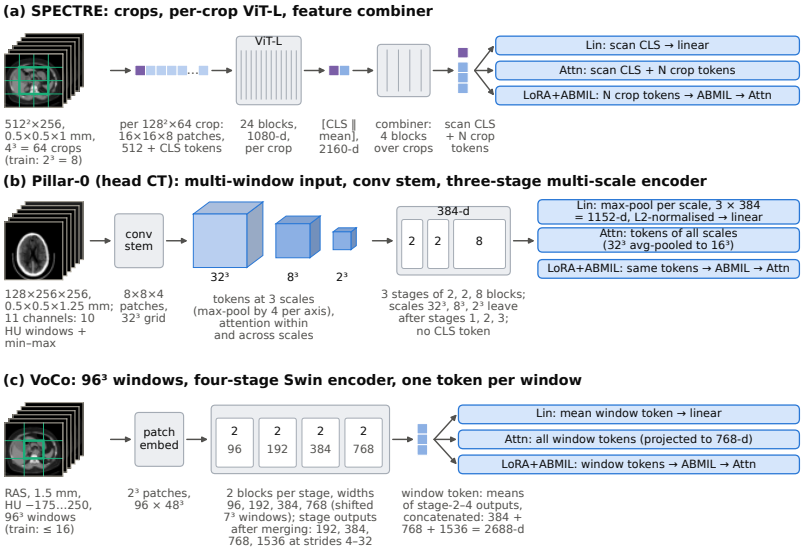
Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. Gradient surgery for multi-task learning. In *Advances in Neural Information Processing Systems*, volume 33, 2020. URL https://papers.nips.cc/paper_files/paper/2020/hash/3fe78a8acf5fda99de95303940a2420c-Abstract.html.

Yan Zhao, Xiuying Wang, Tongtong Che, Guoqing Bao, and Shuyu Li. Multi-task deep learning for medical image computing and analysis: A review. *Computers in Biology and Medicine*, 153:106496, 2023. doi: 10.1016/j.compbiomed.2022.106496.

Weicheng Zhu, Haoxu Huang, Huanze Tang, Rushabh Musthyala, Boyang Yu, Long Chen, Emilio Vega, Thomas O'Donnell, Reya Hayek, Lindsey Kuohn, Seena Dehkharghani, Jennifer A. Frontera, Arjun V. Masurkar, Kara Melmed, and Narges Razavian. 3D foundation model for generalizable disease detection in head computed tomography. *Nature Biomedical Engineering*, 2026. doi: 10.1038/s41551-026-01668-w.

Yitao Zhu, Zhenrong Shen, Zihao Zhao, Sheng Wang, Xin Wang, Xiangyu Zhao, Dinggang Shen, and Qian Wang. MeLo: Low-rank adaptation is better than fine-tuning for medical image diagnosis. In *IEEE International Symposium on Biomedical Imaging (ISBI)*, pages 1–5, 2024. doi: 10.1109/ISBI56570.2024.10635615.

Appendix A. Training and evaluation details



(d) What each classification method trains

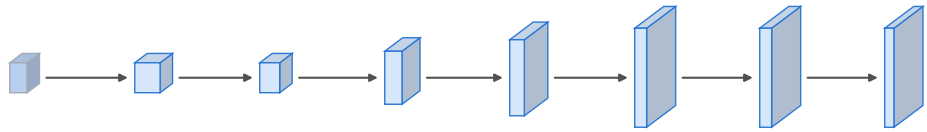
		early blocks				late blocks				encoder parameters updated			
		patch emb.	W_{qkv}	W_{proj}	MLP, norms	W_{qkv}	W_{proj}	MLP, norms	head	steps	TAP-CT	SPECTRE	Pillar-0	VoCo
Lin									pooled → linear	8000	0	0	0	0
Attn									attentive probe	8000	0	0	0	0
LoRA									attentive probe	8000	0.59 M	2.4 M	0.55 M*	0.18 M
LoRA$_{late}$									attentive probe	8000	0.20 M	1.0 M	0.25 M*	0.10 M
LoRA+ABMIL									ABMIL → Attn	8000	0.59 M	2.4 M	0.55 M*	0.18 M
DoRA	(+ m)		m			m			attentive probe	8000	0.62 M	2.5 M	0.57 M*	0.19 M
rsLoRA	(α/√r)								attentive probe	8000	0.59 M	2.4 M	0.55 M*	0.18 M
Lin→LoRA						2			linear	4000 + 4000	0.20 M	1.0 M	0.25 M*	0.10 M†
Attn→LoRA						2			attentive probe	4000 + 4000	0.20 M	1.0 M	0.25 M*	0.10 M†
Attn-only FT									attentive probe	8000	28 M	131 M	11 M*	6.5 M
Full FT									attentive probe	8000	87 M	397 M	45 M*	74 M

frozen rank-16 adapter (m: DoRA magnitude; 2: added in stage 2) trained in full

W_{qkv}: fused qkv (TAP-CT, VoCo) or q and kv (SPECTRE, Pillar-0); Attn-only FT trains the whole attention module (with biases; VoCo: also the relative position bias). Late blocks: TAP-CT 9–12, SPECTRE 17–24 and its combiner, the last stage of Pillar-0 (8 of 12 blocks) and of VoCo (2 of 8). CVPR26 joint models also ran 26 000 steps (13 000 + 13 000). Head parameters for one label (TAP-CT / SPECTRE / Pillar-0 / VoCo): linear 769 / 1081 / 1153 / 2689; attentive probe 14.2 / 28.0 / 31.9 / 16.2 M. †: same late blocks as LoRA$_{late}$. *Pillar-0: the per-scale blocks of scales that have already left the encoder, and the text projection, are also wrapped or unfrozen but never run and receive no gradient; counting them: LoRA 1.1 M, LoRA$_{late}$ 0.74 M, Attn-only FT 21 M, Full FT 94 M.

Figure 5: Classification with SPECTRE, Pillar-0 and VoCo, drawn from the implementation (TAP-CT: Figure 2). (a–c) How each backbone turns a volume into the tokens the heads read: SPECTRE encodes crops separately and summarizes them with its pretrained feature combiner; Pillar-0 encodes the whole head CT with a convolutional stem and a three-stage multi-scale transformer; VoCo encodes 96^3 windows with a four-stage Swin encoder and forms one 2688-d token per window. (d) The encoder components each of the eleven methods trains, its head, its number of steps and the number of encoder parameters that receive a gradient; for Pillar-0 this excludes per-scale blocks that never run, and the note gives the logged counts that include them.

(a) Segmentation decoder: tensor shapes (channels × depth × height × width)



	block tokens (4 taps)	1×1×1 conv, concatenate	3³ conv	upsample, 3³ conv	upsample, 3³ conv	upsample, 3³ conv	upsample, 3³ conv	1×1×1 conv
TAP-CT	768×3×28²	4·128×3×28²	128×3×28²	64×6×56²	32×12×112²	16×12×224²	–	C×12×224²
SPECTRE	1080×8³	4·128×8³	128×8³	64×16³	32×32³	16×64³	16×64×128²	C×64×128²
VoCo	192×24³… 1536×3³	4·128×24³	128×24³	64×48³	32×96³	–	–	C×96³

Taps: blocks 3, 6, 9, 12 (TAP-CT) or 6, 12, 18, 24 (SPECTRE) after the final norm; for VoCo the four stage outputs 192×24³, 384×12³, 768×6³ and 1536×3³ (strides 4 to 32, channel-normalised), resized to the stride-4 grid before concatenation. Each 3³ conv is followed by instance norm and LeakyReLU; trilinear upsampling along the axes whose stride is still above 1; no skips from the input. SPECTRE: per-crop ViT-L only. Decoder parameters: 2.45 / 2.62 / 2.42 M (TAP-CT / SPECTRE / VoCo, C = 15). Lin: a 1×1×1 conv on the deepest tap (block 12, block 24, or the VoCo stage-4 output at stride 32, i.e. 48 mm) with trilinear upsampling; 769 / 1081 / 1537 parameters per structure. C = 1 per structure, or the number of structures (joint).

(b) What each segmentation method trains

		early blocks				late blocks					encoder parameters updated		
		patch emb.	W_{qkv}	W_{proj}	MLP, norms	W_{qkv}	W_{proj}	MLP, norms	head	steps	TAP-CT	SPECTRE	VoCo
Lin									1×1×1 conv	8000	0	0	0
Dec									decoder	8000	0	0	0
LoRA									decoder	8000	0.59 M	2.1 M*	0.18 M
LoRA$_{\text{late}}$									decoder	8000	0.20 M	0.69 M	0.10 M
DoRA	(+ m)		m			m			decoder	8000	0.62 M	2.2 M*	0.19 M
Dec→LoRA			2			2			decoder	4000 + 4000	0.59 M	2.1 M*	0.18 M
Full FT									decoder	8000	87 M	339 M*	74 M

frozen rank-16 adapter (m: DoRA magnitude; 2: added in stage 2) trained in full

AMOS22 and TotalSegmentator joint models: 24 000 steps (12 000 + 12 000). *SPECTRE: LoRA, DoRA and Full FT also wrap or unfreeze the feature combiner, which never enters the segmentation features and receives no gradient; counting it, 2.4 M / 2.5 M / 397 M.

Figure 6: Segmentation decoder and methods, drawn from the implementation. (a) Tensor shapes through the decoder for TAP-CT, SPECTRE and VoCo, from the four encoder taps to the C output maps (one per structure). (b) The encoder components each of the seven methods trains and the number of encoder parameters that receive a gradient; the SPECTRE counts exclude the feature combiner, which the adapters wrap but which does not enter the segmentation features.

Input preprocessing. Every volume is reoriented to the frame of the backbone's pretraining and resampled to its grid. TAP-CT: LPS, native slice spacing, in-plane trilinear resize to 224×224, depth padded with air to at least 12 slices and to a multiple of 4; HU clipped to $[-1008, 822]$ and standardised with the model's mean and standard deviation; windows of 12 slices with a stride of 3 slices plus a final window ending at the last slice (at most 32 windows per training step, all windows at evaluation). SPECTRE: RAS, $0.5 \times 0.5 \times 1$ mm, centred $512 \times 512 \times 256$ field of view padded with the scan minimum, HU in $[-1000, 1000]$ scaled to $[0, 1]$; a $4 \times 4 \times 4$ grid of 64 crops of $128 \times 128 \times 64$ voxels at evaluation and a random contiguous $2 \times 2 \times 2$ sub-grid of 8 crops at training; each crop is summarised by its CLS token concatenated with its mean patch token before the feature combiner. Pillar-0:

LPI, $0.5 \times 0.5 \times 1.25$ mm, centred $128 \times 256 \times 256$ volume, eleven input channels (ten fixed centre/width windows and a per-volume min–max channel), one forward pass; the linear head reads the model's L2-normalised pooled embedding (three scales concatenated, 1152-d), the attentive head reads the tokens of the three scales, any scale with more than 4096 tokens being average-pooled on its grid. VoCo: RAS, 1.5 mm isotropic, HU in $[-175, 250]$ scaled to $[0, 1]$, padded to at least 96^3 and covered by non-overlapping 96^3 windows (the last window shifted to the border; at most 16 windows per training step); each window becomes one 2688-dimensional token, the concatenation of the spatial means of the stage-2, 3 and 4 outputs, standardised per channel with running statistics (momentum 0.01) that are stored with the checkpoint. For segmentation, TAP-CT keeps the classification grid (in-plane 224, native depth) and crops of $12 \times 224 \times 224$ voxels; SPECTRE resamples to 1 mm and uses crops of $64 \times 128 \times 128$ voxels; VoCo resamples to 1.5 mm and uses 96^3 crops. A training crop is centred on a random foreground voxel of the run's structures with probability 0.67 and placed uniformly otherwise; volumes smaller than the crop are padded with the image minimum. At evaluation the whole volume is covered by a sliding window with Gaussian weighting (overlap 0.5 for TAP-CT and VoCo, 0.25 for SPECTRE) and Dice is computed on the backbone's working grid, without post-processing.

Heads. The linear head is dropout (0.2) followed by one linear layer on the pooled feature: the mean of the window CLS tokens for TAP-CT (768-d), the feature combiner's CLS token for SPECTRE (1080-d), the pooled image embedding for Pillar-0 (1152-d) and the mean window token for VoCo (2688-d). The attentive probe prepends a learnable CLS token to the token set (the window CLS tokens for TAP-CT, the combiner tokens for SPECTRE, the multi-scale tokens for Pillar-0, the window tokens for VoCo; VoCo's tokens are first projected from 2688 to 768 dimensions), applies two pre-norm transformer blocks with eight-head self-attention and a GELU feed-forward layer of ratio 4, normalises the CLS output and maps it through dropout (0.2) and a linear layer. LoRA+ABMIL pools the window or crop tokens by gated attention (Ilse et al., 2018), $a_i = \mathrm{softmax}_i\big(\boldsymbol{w}^\top(\tanh(\boldsymbol{V}\boldsymbol{x}_i) \odot \sigma(\boldsymbol{U}\boldsymbol{x}_i))\big)$ with hidden size 128, and feeds the pooled token to the attentive probe. The segmentation linear probe is a $1 \times 1 \times 1$ convolution on the deepest grid (TAP-CT block 12, SPECTRE block 24, VoCo stage 4 at stride 32), upsampled trilinearly. The decoder projects each of the four grids to 128 channels with a $1 \times 1 \times 1$ convolution, resizes them to a common grid (the token grid for the two ViTs, the stride-4 grid for VoCo), concatenates them (512 channels), fuses them with a 3^3 convolution, instance normalisation and leaky ReLU to 128 channels, then applies stages of trilinear $\times 2$ upsampling followed by the same convolution block that halves the channels (to a minimum of 16) until the voxel grid is reached (three stages for TAP-CT, four for SPECTRE, two for VoCo), and ends with a $1 \times 1 \times 1$ convolution to one logit per structure. It has 2.45 M (TAP-CT), 2.62 M (SPECTRE) and 2.42 M (VoCo) parameters plus $17C$ or $33C$ for C structures, and no skip connections from the input.

Adapters. LoRA (Hu et al., 2022) replaces a frozen projection $\boldsymbol{W}_0\boldsymbol{x}$ by $\boldsymbol{W}_0\boldsymbol{x} + \frac{\alpha}{r}\boldsymbol{B}\boldsymbol{A}\,\mathrm{drop}(\boldsymbol{x})$ with $r = 16$, $\alpha = 32$ and dropout 0.1; $\boldsymbol{A}$ is initialised Kaiming-uniform and $\boldsymbol{B}$ at zero, so training starts from the pretrained model. It is applied to the fused query/key/value projection of every TAP-CT and VoCo block and to the separate query and key/value projections of every SPECTRE block, of SPECTRE's feature combiner, and of every Pillar-0 block. Late adapters cover TAP-CT blocks 9–12, SPECTRE blocks

17–24 and its feature combiner, the third stage of Pillar-0 (8 of 12 blocks) and the fourth stage of VoCo (2 of 8 blocks). rsLoRA scales the update by $\alpha/\sqrt{r}$ instead of α/r. DoRA (Liu et al., 2024) rescales each row of the adapted weight to a learned magnitude $\boldsymbol{m}$, $\boldsymbol{W} = \boldsymbol{m} \odot (\boldsymbol{W}_0 + \frac{\alpha}{r}\boldsymbol{B}\boldsymbol{A})/\|\boldsymbol{W}_0 + \frac{\alpha}{r}\boldsymbol{B}\boldsymbol{A}\|_{\text{row}}$, with $\boldsymbol{m}$ initialised to the row norms of $\boldsymbol{W}_0$; the magnitude is a two-dimensional parameter and receives weight decay like the adapter matrices. Attention-only fine-tuning unfreezes every parameter of the attention modules that LoRA would adapt (query, key, value and output projections with their biases, and for VoCo the relative position bias). In segmentation the SPECTRE adapters and full fine-tuning also cover the feature combiner, which the segmentation path does not evaluate; those parameters receive no gradient.

Two-stage methods. Lin→LoRA and Attn→LoRA take the probe weights (for VoCo also the running standardisation statistics of the window tokens) from the parent run's checkpoint at step 4000 of its 8000-step schedule, add late-block LoRA with $\boldsymbol{B} = 0$, and train 4000 further steps with a fresh optimiser and schedule. Dec→LoRA takes the decoder from the Dec run at the midpoint of its schedule (step 4000; step 12 000 for the 24 000-step joint models) and trains all-block LoRA with the decoder for the second half.

Optimisation. AdamW uses $\beta = (0.9, 0.999)$ and $\epsilon = 10^{-8}$ (the framework defaults; no run overrides them). The learning rate rises linearly from 1% of its peak over the first 10% of the steps and then follows a cosine to 10^{-6}. Weight decay is applied to parameters of dimension two or more, excluding biases and normalisation weights. Gradients are not clipped. Classification trains at batch size 1 without gradient accumulation; segmentation at a micro-batch of 2 crops with accumulation 2, so four crops per step. The classification loss for a case with logits $\boldsymbol{z}$, labels $\boldsymbol{y}$ and known-label mask $\boldsymbol{m}$ is

$$\mathcal{L}_{\text{cls}} = \frac{1}{B}\sum_{b=1}^{B}\sum_{t} m_{bt}\,\text{BCE}(z_{bt}, y_{bt}; w_t), \qquad w_t = \text{clip}\Big(\frac{n_t^-}{n_t^+}, 0.05, 20\Big), \tag{1}$$

where $\text{BCE}(\cdot; w_t)$ is the binary cross-entropy with positive weight w_t computed from the negative and positive counts of label t in the training split. The segmentation loss per case b and structure c, with $\boldsymbol{p} = \sigma(\boldsymbol{z})$ over the V voxels of a crop and reference $\boldsymbol{y}$, is

$$\mathcal{L}_{\text{seg}} = \frac{1}{BC}\sum_{b,c}\Big[1 - \frac{2\sum_v p_{bcv}y_{bcv} + 1}{\sum_v p_{bcv} + \sum_v y_{bcv} + 1} + \frac{1}{V}\sum_{v}\text{BCE}(z_{bcv}, y_{bcv})\Big]. \tag{2}$$

In the joint CVPR26 models the three sources are sampled in the ratio 10:8:8 (AMOS, COVID-CTset, LUNA25); every other run cycles one shuffled training set.

Table 2: Classification settings per method. LoRA-type adapters have rank 16, $\alpha = 32$ and dropout 0.1. Learning rate and weight decay apply to every trained parameter unless a backbone/head pair is given. All runs: AdamW, 8000 steps (two-stage: 4000 + 4000), batch size 1, warm-up 10%, cosine to 10^{-6}, bf16, head dropout 0.2.

Method	Trained in the encoder	Head	Learning rate / weight decay
Lin	nothing	linear	10^{-3} / 0
Attn	nothing	attentive	10^{-4} / 0.01
LoRA	LoRA on q, k, v of every block	attentive	10^{-4} / 0.01
$\text{LoRA}_{\text{late}}$	LoRA on q, k, v of the late blocks	attentive	10^{-4} / 0.01
LoRA+ABMIL	as LoRA	ABMIL then attentive	10^{-4} / 0.01
DoRA	DoRA on q, k, v of every block	attentive	10^{-4} / 0.01
rsLoRA	LoRA, scale $\alpha/\sqrt{r}$	attentive	10^{-4} / 0.01
Lin→LoRA	late LoRA from step 4000 of Lin	linear (inherited)	10^{-4} / 0.01
Attn→LoRA	late LoRA from step 4000 of Attn	attentive (inherited)	10^{-4} / 0.01
Attn-only FT	all attention parameters	attentive	encoder 2×10^{-5} / 0.05, head 10^{-4} / 0.01
Full FT	every parameter	attentive	encoder 2×10^{-5} / 0.05, head 10^{-4} / 0.01
Repeats and controls (Appendix D)			
Full FT, lower LR	every parameter	attentive	encoder 5×10^{-6} / 0.05, head 10^{-4} / 0.01
Full FT, layer decay	every parameter, LR ×0.75 per block from the top	attentive	encoder 2×10^{-5} / 0.05, head 10^{-4} / 0.01
Full FT, accumulation	every parameter; accumulation 8, clipping 1.0, 1000 steps	attentive	as Full FT
MLP-only FT	both MLP layers of every block	attentive	encoder 2×10^{-5} / 0.05, head 10^{-4} / 0.01
rsLoRA, matched LR	LoRA, scale $\alpha/\sqrt{r}$	attentive	2.5×10^{-5} / 0.01

Table 3: Segmentation settings per method. All runs: AdamW, 8000 steps (24 000 for the AMOS22 and TotalSegmentator joint models; two-stage methods split the budget in halves), four crops per step, warm-up 10%, cosine to 10^{-6}, bf16. LoRA-type adapters as in Table 2.

Method	Trained in the encoder	Head	Learning rate / weight decay
Lin	nothing	linear probe	10^{-2} / 0
Dec	nothing	decoder	10^{-3} / 10^{-4}
LoRA	LoRA on q, k, v of every block	decoder	10^{-4} / 10^{-4}
$\text{LoRA}_{\text{late}}$	LoRA on q, k, v of the late blocks	decoder	10^{-4} / 10^{-4}
DoRA	DoRA on q, k, v of every block	decoder	10^{-4} / 10^{-4}
Full FT	every parameter	decoder	encoder 2×10^{-5}, head 10^{-3} / 10^{-4}
Dec→LoRA	all-block LoRA from the midpoint of Dec	decoder (inherited)	10^{-4} / 10^{-4}
Controls (Appendix D)			
From scratch	every parameter, encoder re-initialised	decoder	as Full FT
Compute-matched per-structure	Dec, LoRA, Full FT at 1600 steps per structure	decoder	as above

Model selection and per-case outputs. Every 1000 steps the selection and held-out splits are evaluated and a checkpoint of the trained parameters is written; the checkpoint with the highest selection macro AUROC (mean over labels with both classes present) or mean Dice (mean over structures of the mean over cases in which the structure is present in the reference or the prediction) is kept, ties keeping the earlier checkpoint. For TotalSegmentator only the selection split is evaluated during training; the held-out split is evaluated afterwards with the selected checkpoint, and the recomputed selection Dice is required to match the recorded value within 0.005. Per case, the held-out probability of every known label (classification) and the Dice of every structure (segmentation) at the selected checkpoint are stored and are the input of every number in this paper. Dice thresholds the logits at 0; a structure enters a case's mean when the reference or the prediction contains it, so a false-positive structure scores 0 and an absent, unpredicted structure is skipped.

Appendix B. Statistical procedure

Every number in this paper is computed by one script from the stored per-case predictions of the selected checkpoint of every run (1988 main runs and 367 repeats and controls; the classification files hold the held-out probability of every known label of every case, the segmentation files the Dice of every structure of every case).

Per-run metrics. AUROC is the Mann–Whitney statistic with midranks (tied scores count one half); a label is skipped for a run when its selection or held-out split has fewer than two cases or a single class, which did not occur. Dice per structure is the mean over the held-out cases in which the structure is present in the reference or the prediction; an absent, unpredicted structure is skipped and a false-positive structure counts as 0. Cohort and dataset means (Tables 18 and 19) are means over the labels or structures of complete cells. The recomputed selection-split macro metric was compared with the value logged during training for every run: all 1354 classification runs agree to 10^{-7}, and all but one of the 1034 segmentation runs within 2×10^{-3} (108 segmentation runs had their per-case Dice re-inferred from the selected checkpoint after training; the one exception differs by 2.01×10^{-3}).

Did-not-train rule. For each backbone and method, the median over label–mode pairs of the standard deviation of the held-out probabilities is computed; a median below 0.010 flags a method that did not train. Only SPECTRE Full FT is flagged (0.004; every other method on TAP-CT, SPECTRE and Pillar-0 is at least 0.114). It stays in the per-cohort tables, the paired tests and the Full FT − Lin estimate, and leaves the ranks and every average over methods. VoCo spreads are reported (Table 11) but the rule is not applied to VoCo, whose methods have small spreads at held-out AUROCs well above chance.

Paired tests. Per backbone, cohort, mode (per-task, joint) and label, every method is compared with the baselines Lin, LoRA, Attn and Full FT (classification; two-sided DeLong test of the AUROC difference on the same cases (DeLong et al., 1988)) or Lin, Dec and LoRA (segmentation; paired case bootstrap of the mean Dice difference, 4000 resamples shared by every comparison of the same group, two-sided bootstrap p). Pairs require identical case sets; none had to be skipped. p-values are Benjamini–Hochberg adjusted (Benjamini and Hochberg, 1995) within each family of comparisons (task, baseline, VoCo or not), and "significant" means an adjusted p below 0.05; 7520 DeLong tests and 3591 bootstrap tests were made.

Label-level estimates. The unit is one label (structure) on one backbone. For a comparison A − B its value is the mean over the per-task and joint modes of the difference of the two methods' held-out metric (both modes must exist); for joint minus per-task it is the mean over the methods trained in both modes. The reported mean is over the labels of a backbone, or pooled over TAP-CT, SPECTRE and Pillar-0 (78 label–backbone units in classification, 74 in segmentation). Its 95% interval is a percentile bootstrap with 4000 resamples that draws cohort–backbone clusters with replacement, then labels within each drawn cluster, and then replaces each drawn label's value by one of 1000 case-level resamples of that label (positives and negatives resampled separately in classification, with the same case indices for every term of the difference; all cases in segmentation). Besides the

comparisons with the linear probe (Lin) and the frozen decoder (Dec), every method with the attentive head is compared with the frozen attentive probe (matched head), and LoRA − Lin is also computed without FDG-PET/CT and the COVID-CTset label. Table 6 also gives the interval without the case stage. The number of cohort–backbone clusters with a positive mean is reported next to every estimate.

Ranks. A slot is a label (structure) on one backbone in one mode with every method of the task present (minus methods that did not train); methods are ranked by held-out AUROC or Dice with ties sharing the average rank. Classification has 156 slots over the three main backbones and 32 on VoCo; segmentation 118 and 59 (TotalSegmentator per-structure slots lack three methods and are not ranked). Figure 1 pools all 188 and 177 slots; each slot is ranked within its backbone.

Seeds, controls and convergence. Seed repeats are compared through cohort means per seed and the number of cohort–seed pairs in which one method is ahead of another; full fine-tuning variants through the selection-split AUROC, the number of cohorts whose run trained (median spread at least 0.010) and, for the variant with the highest selection AUROC, its label-level difference to Lin and LoRA; compute-matched models through the mean over structures of joint minus matched Dice. Convergence is the share of runs whose selected checkpoint is the last or one of the last two validations. All random draws use fixed seeds, so the script is deterministic.

Appendix C. Paired comparisons and further analyses

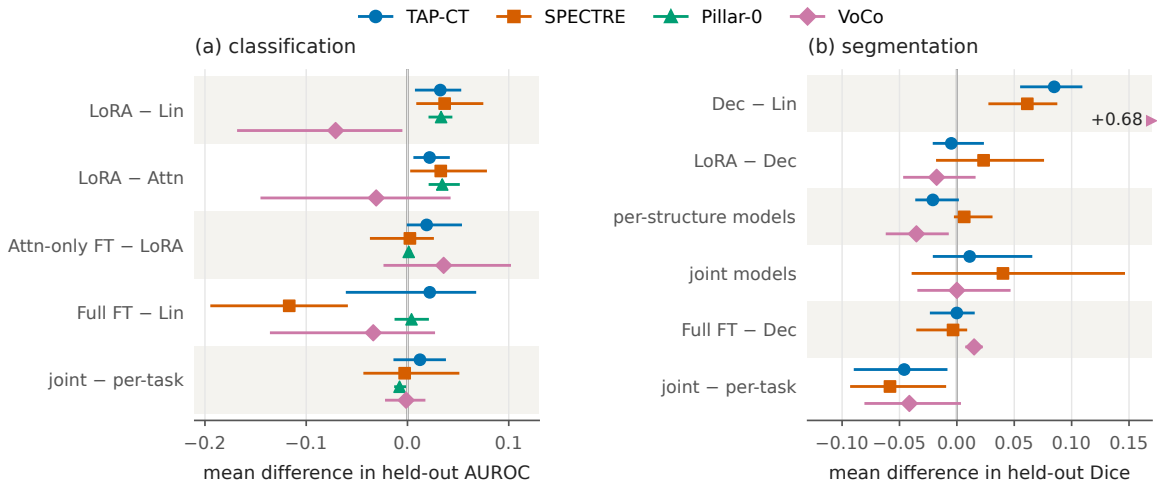


Figure 7: Label-level mean difference in held-out AUROC (a) or Dice (b) for the comparisons of Table 1, per backbone, with the 95% interval over cohorts, labels and held-out cases; 0 means no difference. VoCo was run on three classification cohorts. A triangle at the axis edge marks an estimate beyond the axis, with its value printed. The SPECTRE Full FT − Lin estimate includes runs that did not train.

Table 4: Classification headline: pooled held-out AUROC per backbone (mean over all label comparisons, n = labels × {per-task, joint}), and per-label paired comparisons of LoRA against the linear probe and against full fine-tuning (DeLong, Benjamini–Hochberg; "sig." = corrected $p < 0.05$ in the stated direction). The mean gain of LoRA over the probe is similar for small and large held-out splits (+0.034 below 100 cases, +0.031 above 400) but significant in 0% and 88% of these comparisons; every comparison above 400 cases is from the hemorrhage cohort.

		Pooled AUROC				LoRA vs. linear probe			LoRA vs. full fine-tuning		
Backbone	n	Lin	LoRA	Attn-only FT	Full FT	Mean Δ	LoRA better (%)	sig. (%)	Mean Δ	LoRA better (%)	FT sig. better (%)
TAP-CT	78	0.727	0.760	0.779	0.749	+0.032	78	17	+0.010	55	6
SPECTRE	66	0.718	0.755	0.757	0.601	+0.037	74	9	+0.153	94	2
Pillar-0	12	0.887	0.920	0.922	0.891	+0.033	92	83	+0.029	100	0

Table 5: Classification methods: trainable parameters of a single-label model as logged (head included; for Pillar-0 the count includes parameters of per-scale blocks that never run and receive no gradient, so the encoder parameters updated are about half, Figure 5); mean rank within each of the 156 label–mode pairs (↓; on SPECTRE among the ten methods that trained); share of pairs within 0.01 AUROC of the best method (↑); mean AUROC deficit to the best method (↓). **Bold**: best, underlined: second best of each rank and score column. Full FT's pooled values exclude SPECTRE, where it did not train, so they are not comparable with the other methods' and are not marked. Choosing the method per label on the selection split finds the held-out best in 33% of pairs and lowers the mean deficit from 0.036 (always attention-only fine-tuning) to 0.030, at eleven runs per label instead of one.

	Trainable parameters			Mean rank (1 = best)				Within 0.01	Δ to best
Method	TAP-CT	SPECTRE	Pillar-0	TAP-CT	SPECTRE	Pillar-0	All	of best (%)	(AUROC)
Lin	769	1081	1153	7.62	7.15	9.33	7.55	7	0.081
Attn	14.2 M	28.0 M	31.9 M	6.71	6.96	9.75	7.05	10	0.074
LoRA	14.8 M	30.4 M	33.0 M	4.76	4.94	3.83	4.76	23	0.046
$\text{LoRA}_{\text{late}}$	14.4 M	29.1 M	32.6 M	5.46	5.50	7.17	5.61	17	0.054
LoRA+ABMIL	15.0 M	30.7 M	33.3 M	5.61	**4.02**	4.33	4.84	26	0.045
DoRA	14.8 M	30.5 M	33.0 M	5.50	4.72	3.67	5.03	19	0.049
rsLoRA	14.8 M	30.4 M	33.0 M	7.17	5.31	**2.67**	6.04	18	0.065
Lin→LoRA	0.20 M	1.0 M	0.74 M	7.48	5.27	6.58	6.47	11	0.064
Attn→LoRA	14.4 M	29.1 M	32.6 M	6.90	6.47	7.67	6.78	13	0.069
Attn-only FT	42.5 M	159 M	53.2 M	**3.38**	4.67	3.25	**3.91**	**38**	**0.036**
Full FT	101 M	425 M	126 M	5.44	–	7.75	5.74	31	0.055

Table 6: Label-level differences with 95% intervals. Each label (or structure) counts once, with its per-task and joint differences averaged (except the two LoRA − Dec subrows). Intervals are percentile bootstraps (4000 resamples) over cohort–backbone groups and labels within them, without and with a third stage that resamples the held-out cases of each drawn label (Appendix B). Pillar-0 has one cohort, so its interval resamples labels and cases only. The last column counts the cohorts (datasets) with a positive mean difference.

				95% interval, resampling		
Comparison	Backbone	n	Mean	cohorts, labels	+ cases	Cohorts > 0
LoRA − Lin	all	78	+0.034	[+0.019, +0.049]	[+0.017, +0.052]	9/10
LoRA − Lin	TAP-CT	39	+0.032	[+0.009, +0.049]	[+0.007, +0.053]	4/5
LoRA − Lin	SPECTRE	33	+0.037	[+0.013, +0.068]	[+0.009, +0.075]	4/4
LoRA − Lin	Pillar-0	6	+0.033	[+0.021, +0.042]	[+0.021, +0.044]	1/1
LoRA − Lin	VoCo	16	-0.071	[-0.162, -0.011]	[-0.168, -0.005]	0/3
LoRA − Attn	all	78	+0.027	[+0.013, +0.047]	[+0.012, +0.048]	10/10
LoRA − Attn	TAP-CT	39	+0.022	[+0.009, +0.039]	[+0.006, +0.042]	5/5
LoRA − Attn	SPECTRE	33	+0.033	[+0.007, +0.074]	[+0.003, +0.078]	4/4
LoRA − Attn	Pillar-0	6	+0.034	[+0.021, +0.051]	[+0.021, +0.052]	1/1
LoRA − Attn	VoCo	16	-0.031	[-0.150, +0.033]	[-0.145, +0.043]	1/3
Attn-only FT − LoRA	all	78	+0.011	[-0.005, +0.026]	[-0.007, +0.029]	7/10
Attn-only FT − LoRA	TAP-CT	39	+0.019	[+0.002, +0.053]	[-0.001, +0.054]	4/5
Attn-only FT − LoRA	SPECTRE	33	+0.002	[-0.037, +0.023]	[-0.037, +0.026]	2/4
Attn-only FT − LoRA	Pillar-0	6	+0.001	[0.000, +0.003]	[-0.002, +0.004]	1/1
Attn-only FT − LoRA	VoCo	16	+0.036	[-0.020, +0.097]	[-0.024, +0.102]	2/3
Full FT − Lin	all	78	-0.038	[-0.101, +0.017]	[-0.103, +0.019]	5/10
Full FT − Lin	TAP-CT	39	+0.022	[-0.057, +0.064]	[-0.061, +0.068]	4/5
Full FT − Lin	SPECTRE	33	-0.117	[-0.195, -0.063]	[-0.195, -0.059]	0/4
Full FT − Lin	Pillar-0	6	+0.004	[-0.011, +0.019]	[-0.013, +0.021]	1/1
Full FT − Lin	VoCo	16	-0.034	[-0.133, +0.018]	[-0.136, +0.027]	1/3
Joint − per-task	all	78	+0.004	[-0.017, +0.027]	[-0.021, +0.030]	6/10
Joint − per-task	TAP-CT	39	+0.012	[-0.005, +0.032]	[-0.014, +0.038]	3/5
Joint − per-task	SPECTRE	33	-0.003	[-0.035, +0.049]	[-0.044, +0.051]	3/4
Joint − per-task	Pillar-0	6	-0.008	[-0.012, -0.003]	[-0.013, -0.001]	0/1
Joint − per-task	VoCo	16	-0.001	[-0.017, +0.012]	[-0.022, +0.018]	2/3
Dec − Lin	all	74	+0.073	[+0.049, +0.093]	[+0.049, +0.092]	10/10
Dec − Lin	TAP-CT	37	+0.085	[+0.055, +0.109]	[+0.055, +0.109]	5/5
Dec − Lin	SPECTRE	37	+0.061	[+0.028, +0.089]	[+0.027, +0.088]	5/5
Dec − Lin	VoCo	37	+0.680	[+0.430, +0.747]	[+0.433, +0.746]	5/5
LoRA − Dec	all	74	+0.009	[-0.014, +0.046]	[-0.015, +0.046]	8/10
LoRA − Dec	TAP-CT	37	-0.005	[-0.021, +0.023]	[-0.021, +0.024]	4/5
LoRA − Dec	SPECTRE	37	+0.023	[-0.017, +0.079]	[-0.018, +0.076]	4/5
LoRA − Dec	VoCo	37	-0.018	[-0.045, +0.017]	[-0.047, +0.016]	2/5
per-structure models	all	74	-0.007	[-0.021, +0.009]	[-0.022, +0.009]	7/10
per-structure models	TAP-CT	37	-0.021	[-0.035, +0.002]	[-0.036, +0.002]	2/5
per-structure models	SPECTRE	37	+0.006	[-0.001, +0.028]	[-0.003, +0.031]	5/5
per-structure models	VoCo	37	-0.035	[-0.062, -0.008]	[-0.062, -0.007]	0/5
joint models	all	74	+0.026	[-0.020, +0.092]	[-0.021, +0.093]	8/10
joint models	TAP-CT	37	+0.011	[-0.019, +0.067]	[-0.021, +0.066]	4/5
joint models	SPECTRE	37	+0.040	[-0.038, +0.148]	[-0.039, +0.147]	4/5
joint models	VoCo	37	0.000	[-0.034, +0.047]	[-0.034, +0.047]	2/5
Full FT − Dec	all	74	-0.002	[-0.017, +0.009]	[-0.018, +0.009]	6/10
Full FT − Dec	TAP-CT	37	0.000	[-0.025, +0.015]	[-0.024, +0.016]	4/5
Full FT − Dec	SPECTRE	37	-0.003	[-0.033, +0.008]	[-0.035, +0.009]	2/5
Full FT − Dec	VoCo	37	+0.015	[+0.008, +0.023]	[+0.007, +0.023]	4/5
Joint − per-task	all	74	-0.052	[-0.082, -0.021]	[-0.082, -0.022]	0/10
Joint − per-task	TAP-CT	37	-0.046	[-0.088, -0.009]	[-0.090, -0.008]	0/5
Joint − per-task	SPECTRE	37	-0.058	[-0.092, -0.010]	[-0.093, -0.009]	0/5
Joint − per-task	VoCo	37	-0.041	[-0.081, +0.002]	[-0.081, +0.004]	2/5

Table 7: Per-label paired comparisons of held-out AUROC against the linear probe and against full fine-tuning, over the per-task and joint modes (DeLong test; Benjamini–Hochberg correction within each baseline over the comparisons of TAP-CT, SPECTRE and Pillar-0). "Better" is the share of label–mode pairs in which the method's AUROC exceeds the baseline's; "Sig." the share in which, in addition, the corrected $p < 0.05$. SPECTRE Full FT did not train.

Backbone	Baseline	Method	Labels	Mean ΔAUROC	Better (%)	Sig. better (%)	Sig. worse (%)
TAP-CT	vs Lin	Attn	78	+0.011	59	4	0
		LoRA	78	+0.032	78	17	0
		$\text{LoRA}_{\text{late}}$	78	+0.023	69	14	1
		LoRA+ABMIL	78	+0.024	68	12	1
		DoRA	78	+0.028	68	17	0
		rsLoRA	78	+0.003	55	5	5
		Lin→LoRA	78	+0.001	51	5	1
		Attn→LoRA	78	+0.008	56	9	0
		Attn-only FT	78	+0.051	83	24	0
		Full FT	78	+0.022	72	13	8
TAP-CT	vs Full FT	Lin	78	-0.022	28	12	17
		Attn	78	-0.011	40	10	18
		LoRA	78	+0.010	55	18	6
		$\text{LoRA}_{\text{late}}$	78	+0.001	44	18	6
		LoRA+ABMIL	78	+0.002	45	17	12
		DoRA	78	+0.006	51	17	9
		rsLoRA	78	-0.019	47	13	22
		Lin→LoRA	78	-0.021	31	9	18
		Attn→LoRA	78	-0.014	38	13	15
		Attn-only FT	78	+0.029	63	23	5
SPECTRE	vs Lin	Attn	66	+0.004	50	3	2
		LoRA	66	+0.037	74	9	0
		$\text{LoRA}_{\text{late}}$	66	+0.034	65	5	0
		LoRA+ABMIL	66	+0.052	77	12	2
		DoRA	66	+0.037	77	8	0
		rsLoRA	66	+0.028	70	9	0
		Lin→LoRA	66	+0.035	71	6	0
		Attn→LoRA	66	+0.016	59	6	2
		Attn-only FT	66	+0.039	70	3	3
		Full FT	66	-0.117	21	0	32
SPECTRE	vs Full FT	Lin	66	+0.117	79	33	0
		Attn	66	+0.121	86	35	0
		LoRA	66	+0.153	94	44	2
		$\text{LoRA}_{\text{late}}$	66	+0.151	95	44	0
		LoRA+ABMIL	66	+0.168	94	52	0
		DoRA	66	+0.154	92	44	2
		rsLoRA	66	+0.145	89	39	0
		Lin→LoRA	66	+0.152	94	41	0
		Attn→LoRA	66	+0.133	88	44	0
		Attn-only FT	66	+0.156	89	48	0
Pillar-0	vs Lin	Attn	12	-0.001	42	33	8
		LoRA	12	+0.033	92	83	0
		$\text{LoRA}_{\text{late}}$	12	+0.016	75	67	0
		LoRA+ABMIL	12	+0.026	83	75	8
		DoRA	12	+0.034	92	83	0
		rsLoRA	12	+0.032	100	83	0
		Lin→LoRA	12	+0.022	100	83	0
		Attn→LoRA	12	+0.013	92	58	0
		Attn-only FT	12	+0.034	92	83	0
		Full FT	12	+0.004	67	50	17
Pillar-0	vs Full FT	Lin	12	-0.004	33	17	50
		Attn	12	-0.005	33	17	58
		LoRA	12	+0.029	100	67	0
		$\text{LoRA}_{\text{late}}$	12	+0.012	58	33	25
		LoRA+ABMIL	12	+0.022	58	58	8
		DoRA	12	+0.030	83	67	0
		rsLoRA	12	+0.028	92	67	0
		Lin→LoRA	12	+0.018	67	42	25
		Attn→LoRA	12	+0.009	50	33	25
		Attn-only FT	12	+0.030	100	83	0

Table 8: Per-structure paired comparisons of held-out Dice over the per-structure and joint modes against the linear probe and the frozen decoder (paired bootstrap of the mean per-case difference, 4000 resamples; Benjamini–Hochberg correction within each baseline). Pairing keeps the cases on which a structure enters the Dice of both methods (present in the reference or in either prediction), so the mean differences here can differ from the differences of structure means in Table 19.

Backbone	Baseline	Method	Pairs	Mean ΔDice	Better (%)	Sig. better (%)	Sig. worse (%)
TAP-CT	vs Lin	Dec	74	+0.086	96	93	4
		LoRA	74	+0.088	96	93	3
		$\text{LoRA}_{\text{late}}$	59	+0.077	95	92	5
		DoRA	59	+0.078	95	90	5
		Full FT	74	+0.085	93	93	7
		Dec→LoRA	59	+0.066	92	88	8
TAP-CT	vs Dec	Lin	74	-0.086	4	4	93
		LoRA	74	+0.002	54	31	30
		$\text{LoRA}_{\text{late}}$	59	-0.006	53	29	32
		DoRA	59	-0.006	53	25	25
		Full FT	74	-0.001	76	53	8
		Dec→LoRA	59	-0.017	29	7	44
SPECTRE	vs Lin	Dec	74	+0.061	92	86	5
		LoRA	74	+0.095	93	93	4
		$\text{LoRA}_{\text{late}}$	59	+0.079	92	90	5
		DoRA	59	+0.088	93	90	7
		Full FT	74	+0.059	89	81	8
		Dec→LoRA	59	+0.051	90	85	7
SPECTRE	vs Dec	Lin	74	-0.061	8	5	86
		LoRA	74	+0.034	80	57	8
		$\text{LoRA}_{\text{late}}$	59	+0.032	69	46	8
		DoRA	59	+0.041	76	54	12
		Full FT	74	-0.002	54	38	20
		Dec→LoRA	59	+0.004	69	20	12

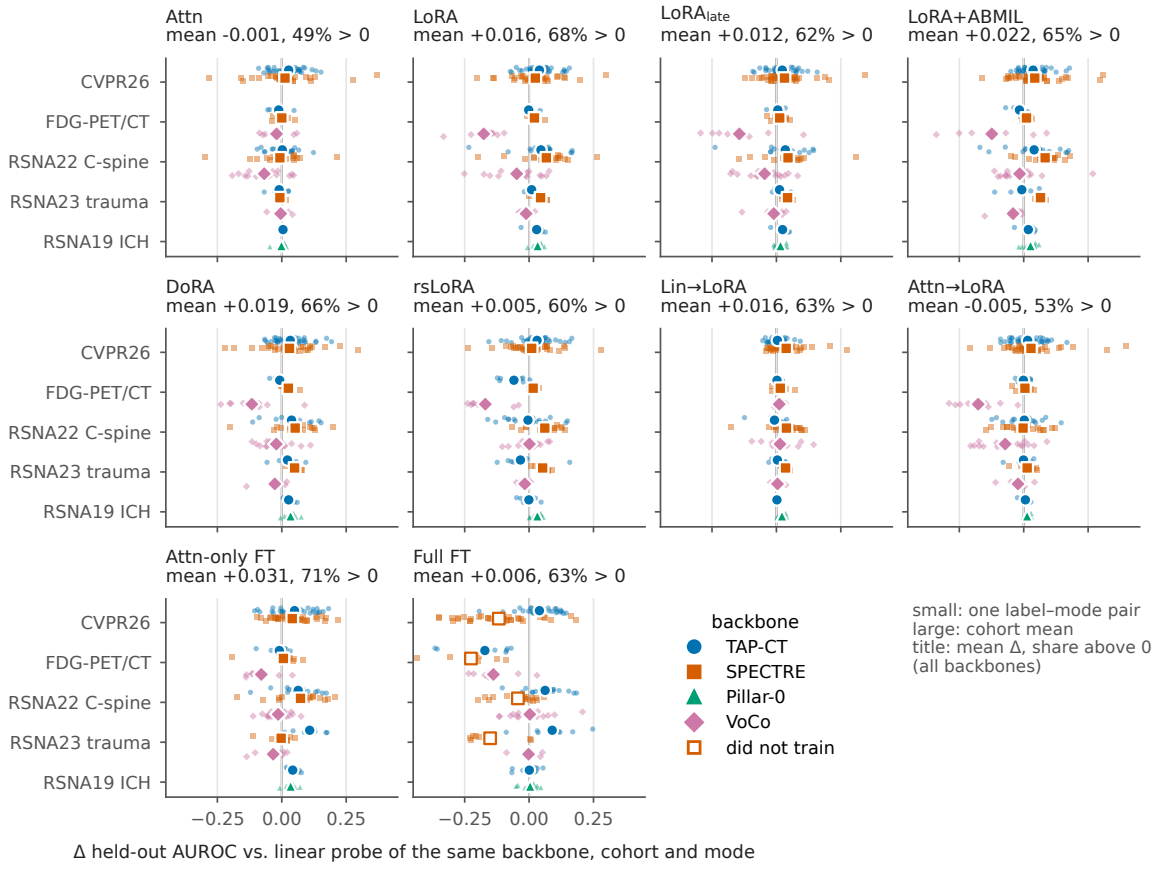


Figure 8: Per-label change in held-out AUROC of each classification method relative to the linear probe of the same backbone, cohort and mode, one panel per method; rows are cohorts, color is the backbone, small markers are single label–mode comparisons and large markers the mean per cohort and backbone. Titles give the mean change and the share above zero over the pairs of all four backbones that trained. SPECTRE Full FT did not train; its cohort means are drawn open and left out of the title.

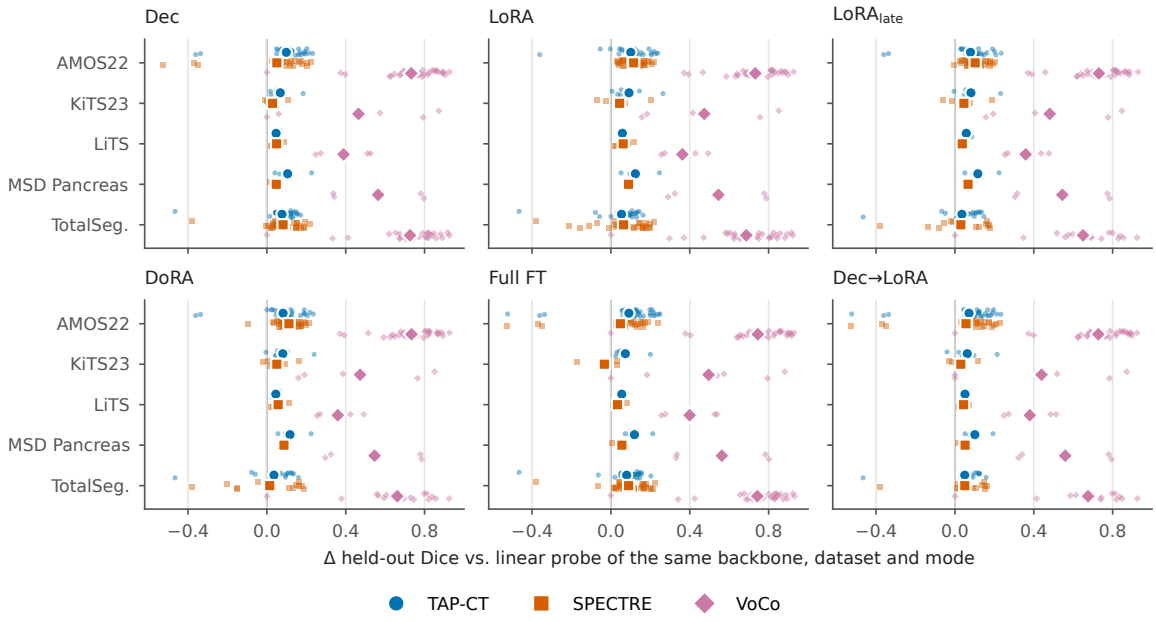


Figure 9: Per-structure change in held-out Dice of each segmentation method relative to the linear probe of the same backbone, dataset and mode, one panel per method; rows are datasets, color is the backbone, small markers are single structure–mode comparisons and large markers the mean per dataset and backbone. VoCo's linear probe reads one token per 48 mm and segments almost nothing, so its changes are far larger than those of the two ViTs.

Table 9: Encoder adaptation on top of the frozen decoder per dataset, backbone and mode: difference of structure-mean held-out Dice. Joint rows of AMOS22 and TotalSegmentator use the 24 000-step schedule; the 8000-step rows show the same models under the shared budget.

		LoRA − Dec		Full FT − Dec	
Dataset	Mode	TAP-CT	SPECTRE	TAP-CT	SPECTRE
AMOS22 (15)	per-structure	−0.026	+0.004	+0.012	−0.016
	joint	+0.031	+0.126	−0.027	+0.011
	joint, 8000 steps	−0.125	−0.071	+0.007	+0.010
KiTS23 (3)	per-structure	−0.021	+0.014	+0.005	−0.071
	joint	+0.068	+0.016	+0.006	−0.052
LiTS (2)	per-structure	+0.007	+0.015	+0.001	−0.002
	joint	+0.016	+0.011	+0.016	−0.030
MSD Pancreas (2)	per-structure	+0.013	+0.047	+0.021	+0.000
	joint	+0.025	+0.037	+0.008	+0.016
TotalSegmentator (15)	per-structure	−0.024	+0.001	−0.011	−0.009
	joint	−0.022	−0.036	+0.017	+0.025
	joint, 8000 steps	−0.199	−0.254	+0.022	+0.037

Table 10: The training budget of the 15-structure joint models (AMOS22, TotalSegmentator): held-out Dice, mean over structures, for one model per structure (8000 steps each), one joint model under the shared 8000 steps, and one joint model with three times the steps (Dec→LoRA: 12 000 + 12 000).

Dataset	Backbone	Training	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→LoRA
AMOS22	TAP-CT	per-structure, 8000 steps	0.658	0.793	0.767	0.764	0.761	0.805	0.791
AMOS22	TAP-CT	joint, 8000 steps	0.632	0.595	0.470	0.412	0.455	0.602	0.591
AMOS22	TAP-CT	joint, 24 000 steps	0.646	0.708	0.739	0.698	0.708	0.681	0.656
AMOS22	SPECTRE	per-structure, 8000 steps	0.683	0.808	0.812	0.806	0.814	0.792	0.814
AMOS22	SPECTRE	joint, 8000 steps	0.670	0.640	0.568	0.527	0.567	0.649	0.627
AMOS22	SPECTRE	joint, 24 000 steps	0.682	0.658	0.784	0.764	0.775	0.669	0.664
TotalSegmentator	TAP-CT	per-structure, 8000 steps	0.747	0.843	0.819	–	–	0.832	–
TotalSegmentator	TAP-CT	joint, 8000 steps	0.731	0.781	0.583	0.594	0.591	0.803	0.776
TotalSegmentator	TAP-CT	joint, 24 000 steps	0.740	0.798	0.776	0.776	0.776	0.815	0.790
TotalSegmentator	SPECTRE	per-structure, 8000 steps	0.731	0.848	0.848	–	–	0.838	–
TotalSegmentator	SPECTRE	joint, 8000 steps	0.712	0.679	0.425	0.387	0.408	0.716	0.637
TotalSegmentator	SPECTRE	joint, 24 000 steps	0.722	0.770	0.733	0.752	0.736	0.795	0.772

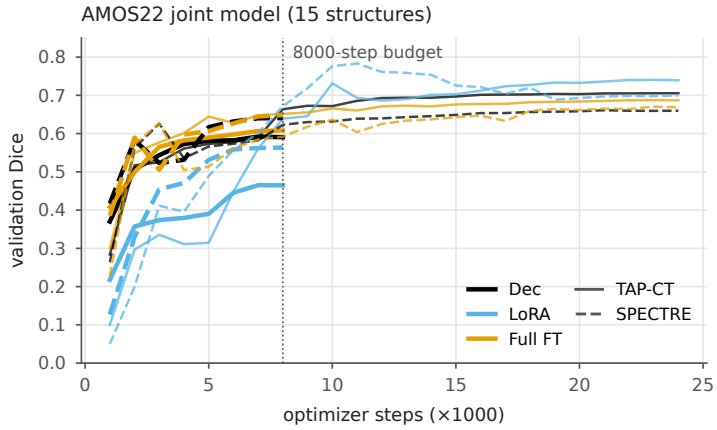


Figure 10: Validation Dice during training of the 15-structure AMOS22 joint model with the frozen decoder, LoRA and full fine-tuning, on TAP-CT (solid) and SPECTRE (dashed); thick lines stop at the shared 8000 steps, thin lines run 24 000 steps.

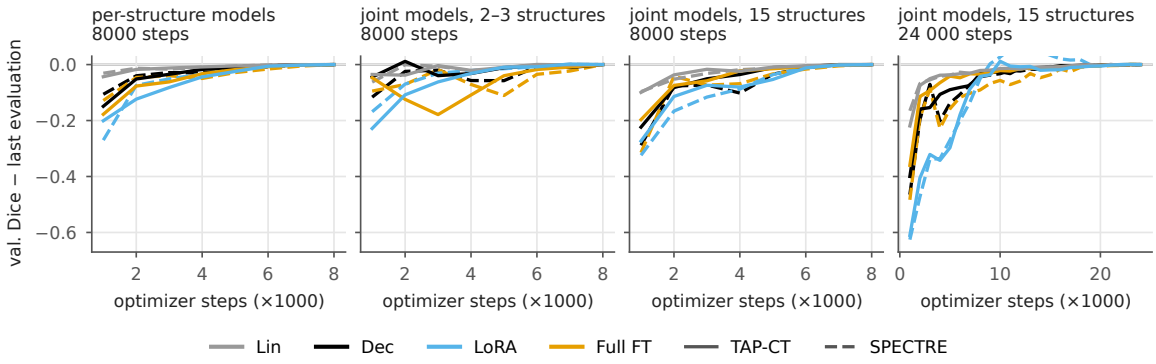


Figure 11: Validation Dice at each evaluation minus that at the run's last evaluation, median over the runs of each schedule group, for Lin, Dec, LoRA and Full FT on TAP-CT (solid) and SPECTRE (dashed). Every curve ends at zero by construction; a curve still rising at the right edge had not converged.

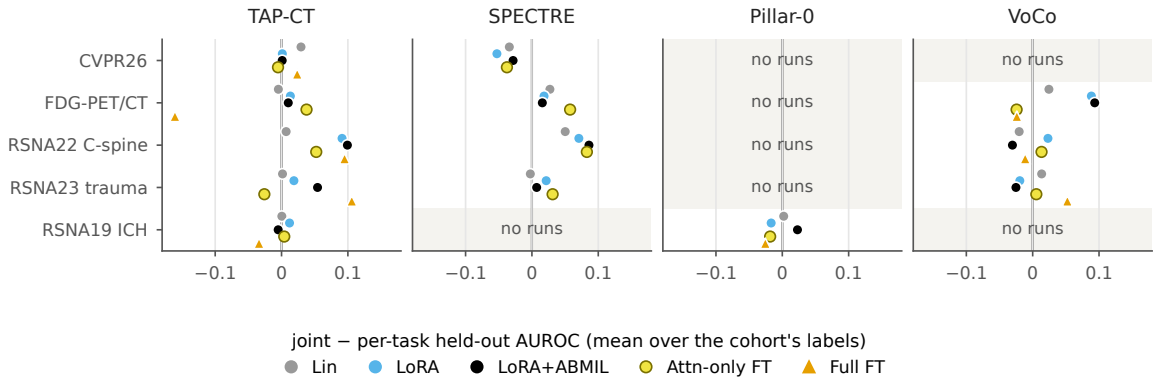


Figure 12: Joint minus per-task held-out AUROC, mean over the labels of each cohort, for Lin, LoRA, LoRA+ABMIL, Attn-only FT and Full FT, per backbone (SPECTRE Full FT, which did not train, is not shown). Right of zero favors one joint model per cohort.

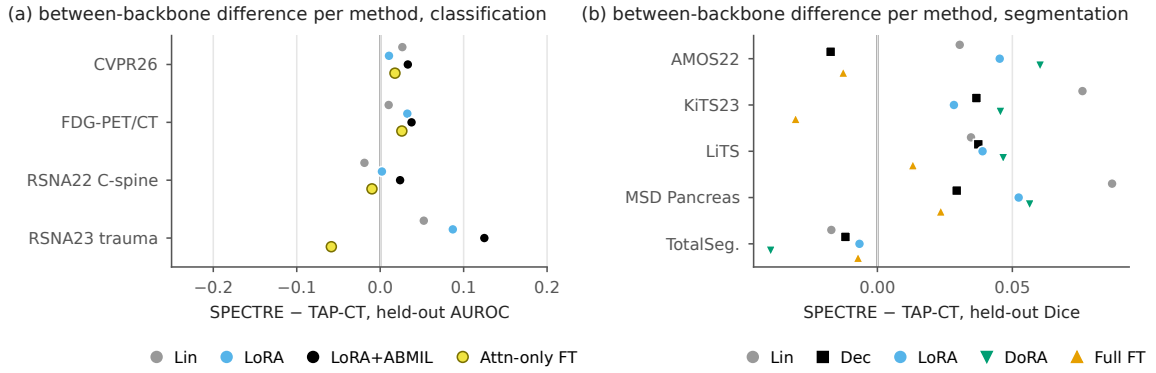


Figure 13: Between-backbone difference per method: SPECTRE minus TAP-CT held-out AUROC per cohort (a) and held-out Dice per dataset (b), on the same labels or structures and modes. SPECTRE Full FT, which did not train in classification, is not shown in (a).

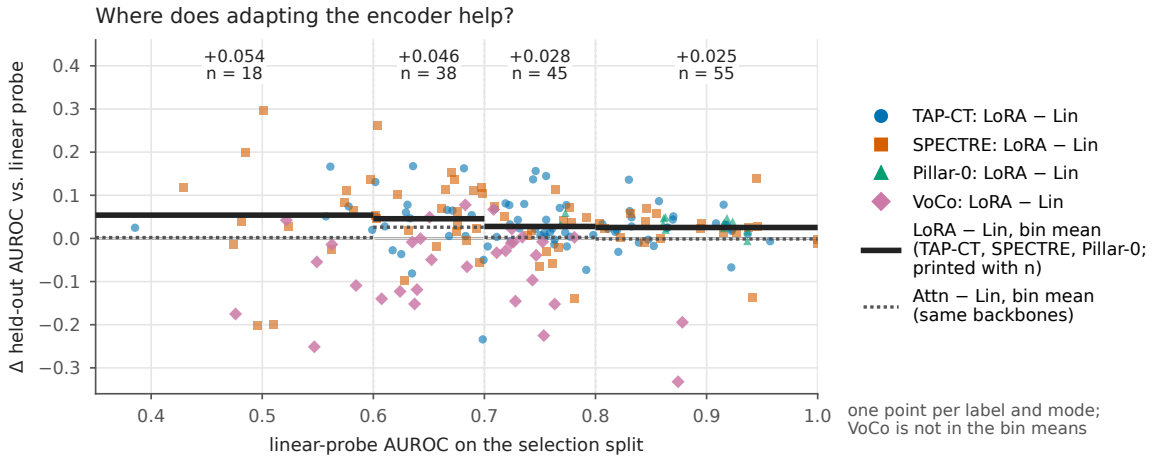


Figure 14: Where does adapting the encoder help? LoRA minus linear-probe held-out AUROC per label–mode pair against the linear probe's selection AUROC (markers: backbone), with bin means (bars, printed) and, as a negative control, the bin means of the frozen attentive probe minus the linear probe (dotted).

Table 11: Median over label–mode pairs of the standard deviation of the held-out probabilities per backbone and method (the did-not-train rule flags values below 0.010; applied to TAP-CT, SPECTRE and Pillar-0 only).

Backbone	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ABMIL	DoRA	rsLoRA	Lin→LoRA	Attn→LoRA	Attn-only FT	Full FT
TAP-CT	0.227	0.239	0.239	0.235	0.233	0.237	0.167	0.222	0.225	0.247	0.170
SPECTRE	0.187	0.114	0.161	0.185	0.187	0.185	0.150	0.178	0.129	0.148	**0.004**†
Pillar-0	0.294	0.321	0.359	0.350	0.352	0.354	0.359	0.337	0.338	0.355	0.347
VoCo	0.152	0.023	0.035	0.032	0.008	0.030	0.036	0.150	0.020	0.041	0.036

Appendix D. Seed repeats, full fine-tuning settings and compute-matched models

Table 12: Held-out AUROC of the joint classification models across seeds 42, 43 and 44: mean over the cohort's labels, given as mean ± standard deviation over the seeds (their number in brackets), and the number of seeds in which the mean label difference between two methods is positive. SPECTRE Full FT, which did not train, has seed 42 only.

Cohort	Lin	LoRA	Attn-only FT	Full FT	LoRA > Lin	Attn-only FT > LoRA
TAP-CT						
CVPR26	0.705 ± .004 (3)	0.719 ± .016 (3)	0.740 ± .009 (3)	0.711 ± .029 (3)	2/3	3/3
FDG-PET/CT	0.860 ± .003 (3)	0.864 ± .008 (3)	0.875 ± .012 (3)	0.661 ± .130 (3)	2/3	3/3
RSNA23 trauma	0.738 ± .008 (3)	0.760 ± .002 (3)	0.842 ± .030 (3)	0.793 ± .157 (3)	3/3	3/3
RSNA22 C-spine	0.623 ± .006 (3)	0.659 ± .050 (3)	0.691 ± .019 (3)	0.695 ± .035 (3)	2/3	2/3
RSNA19 ICH	0.871 ± .002 (3)	0.896 ± .009 (3)	0.904 ± .019 (3)	0.836 ± .086 (3)	3/3	2/3
SPECTRE						
CVPR26	0.711 ± .008 (3)	0.741 ± .024 (3)	0.759 ± .023 (3)	0.575 (1)	3/3	3/3
FDG-PET/CT	0.887 ± .003 (3)	0.901 ± .000 (3)	0.910 ± .006 (3)	0.673 (1)	3/3	3/3
RSNA23 trauma	0.792 ± .004 (3)	0.846 ± .001 (3)	0.834 ± .028 (3)	0.654 (1)	3/3	1/3
RSNA22 C-spine	0.592 ± .046 (3)	0.696 ± .009 (3)	0.683 ± .052 (3)	0.554 (1)	3/3	2/3
Pillar-0						
RSNA19 ICH	0.888 ± .000 (3)	0.916 ± .004 (3)	0.921 ± .007 (3)	0.894 ± .015 (3)	3/3	3/3

Table 13: Segmentation across seeds: held-out Dice of the joint models (AMOS22 and TotalSegmentator: 24 000 steps), mean over structures, as in Table 12.

Dataset	Dec	LoRA	Full FT	LoRA > Dec	Full FT > Dec
TAP-CT					
AMOS22	0.675 ± .029 (3)	0.709 ± .059 (3)	0.680 ± .001 (3)	2/3	2/3
KiTS23	0.516 ± .003 (3)	0.576 ± .005 (3)	0.529 ± .014 (3)	3/3	3/3
LiTS	0.715 ± .016 (3)	0.718 ± .004 (3)	0.717 ± .007 (3)	2/3	2/3
MSD Pancreas	0.480 ± .002 (3)	0.496 ± .009 (3)	0.468 ± .027 (3)	3/3	2/3
TotalSegmentator	0.783 ± .026 (3)	0.773 ± .008 (3)	0.815 ± .000 (3)	1/3	3/3
SPECTRE					
AMOS22	0.658 ± .000 (3)	0.775 ± .011 (3)	0.669 ± .003 (3)	3/3	3/3
KiTS23	0.579 ± .010 (3)	0.616 ± .024 (3)	0.529 ± .015 (3)	3/3	0/3
LiTS	0.741 ± .001 (3)	0.740 ± .012 (3)	0.707 ± .019 (3)	1/3	0/3
MSD Pancreas	0.518 ± .004 (3)	0.551 ± .008 (3)	0.517 ± .011 (3)	3/3	1/3
TotalSegmentator	0.741 ± .025 (3)	0.752 ± .020 (3)	0.762 ± .028 (3)	2/3	3/3

Table 14: Held-out AUROC of the joint classification models under the default and three alternative full fine-tuning settings, mean over the cohort's labels, next to Lin and LoRA. **Bold**: the setting with the best mean selection-split AUROC on the backbone. Grey: runs whose held-out predictions barely vary (median over labels of the standard deviation of the predicted probabilities below 0.010).

			Full FT			
Cohort	Lin	LoRA	default	lower LR	layer decay	accum. + clip
TAP-CT						
CVPR26	0.707	0.733	0.744	0.730	**0.750**	0.685
FDG-PET/CT	0.858	0.866	0.608	0.861	**0.867**	0.867
RSNA23 trauma	0.739	0.757	0.881	0.860	**0.869**	0.758
RSNA22 C-spine	0.626	0.714	0.732	0.654	**0.618**	0.687
RSNA19 ICH	0.871	0.906	0.854	0.889	**0.917**	0.909
SPECTRE						
CVPR26	0.702	0.716	0.575	0.713	**0.758**	0.511
FDG-PET/CT	0.884	0.901	0.673	0.884	**0.865**	0.883
RSNA23 trauma	0.789	0.846	0.654	0.793	**0.863**	0.578
RSNA22 C-spine	0.629	0.706	0.554	0.568	**0.700**	0.553
Pillar-0						
RSNA19 ICH	0.888	0.912	0.878	**0.924**	0.920	0.918

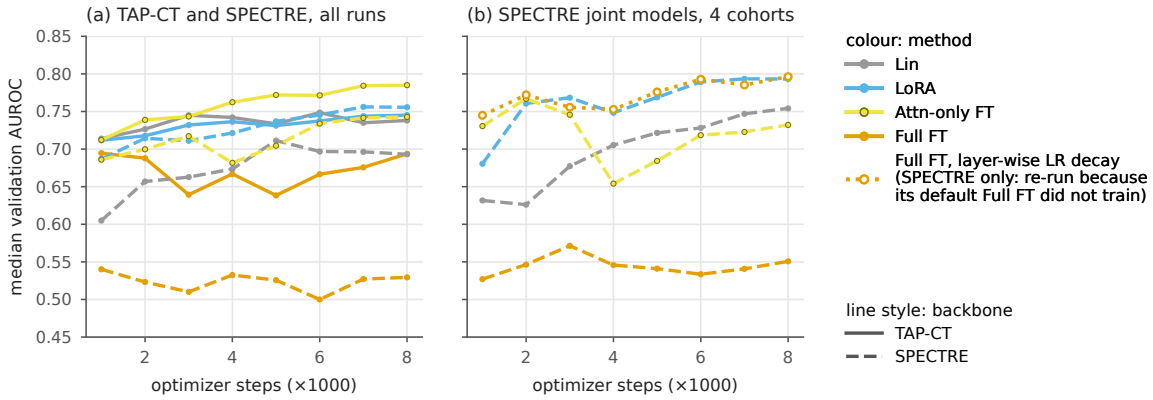


Figure 15: Full fine-tuning on SPECTRE. (a) Median validation AUROC at each evaluation with the default settings, for TAP-CT and SPECTRE, over all runs of each method. (b) SPECTRE joint models on four cohorts, including Full FT with layer-wise learning-rate decay, the setting selected in Table 14; the alternative settings were run only on SPECTRE, the one backbone on which Full FT failed.

Table 15: Controls for two confounds of the main comparison: rsLoRA at a quarter of the learning rate, which matches LoRA's update scale at initialization, and fine-tuning the MLP instead of the attention weights. Held-out AUROC of the joint models, mean over the cohort's labels.

Cohort	LoRA	rsLoRA	rsLoRA (LR/4)	Attn-only FT	MLP-only FT
TAP-CT					
CVPR26	0.733	0.722	0.741	0.739	0.745
FDG-PET/CT	0.866	0.785	0.855	0.870	0.866
RSNA23 trauma	0.757	0.684	0.751	0.833	0.894
RSNA22 C-spine	0.714	0.601	0.611	0.712	0.627
RSNA19 ICH	0.906	0.869	0.901	0.915	0.907
SPECTRE					
CVPR26	0.716	0.690	0.739	0.740	0.681
FDG-PET/CT	0.901	0.900	0.908	0.906	0.814
RSNA23 trauma	0.846	0.850	0.834	0.803	0.838
RSNA22 C-spine	0.706	0.691	0.672	0.718	0.626
Pillar-0					
RSNA19 ICH	0.912	0.918	0.920	0.912	0.912

Table 16: Joint against compute-matched per-structure segmentation: held-out Dice, mean over the 15 structures, of per-structure models trained 1600 steps (matched) or 8000 steps, and of the 24 000-step joint model on the same structures; the difference joint − matched and the number of structures where it is positive.

			per-structure			joint − matched	
Backbone	Method	Struct. finished	matched	8000 steps	joint	mean	> 0
AMOS22 (15 structures)							
TAP-CT	Dec	15	0.763	0.793	0.708	−0.055	11/15
TAP-CT	LoRA	15	0.617	0.767	0.739	+0.122	14/15
TAP-CT	Full FT	15	0.766	0.805	0.681	−0.085	12/15
SPECTRE	Dec	15	0.760	0.808	0.658	−0.101	6/15
SPECTRE	LoRA	15	0.648	0.812	0.784	+0.136	14/15
SPECTRE	Full FT	15	0.793	0.792	0.669	−0.123	5/15
TotalSegmentator (15 structures)							
TAP-CT	Dec	15	0.783	0.843	0.798	+0.015	13/15
TAP-CT	LoRA	15	0.657	0.819	0.776	+0.119	14/15
TAP-CT	Full FT	15	0.803	0.832	0.815	+0.012	14/15
SPECTRE	Dec	15	0.806	0.848	0.770	−0.037	6/15
SPECTRE	LoRA	15	0.691	0.848	0.733	+0.042	11/15
SPECTRE	Full FT	15	0.819	0.838	0.795	−0.024	10/15

Table 17: From-scratch control: TAP-CT architecture with a randomly initialised encoder, trained as the Full FT joint model, next to Full FT and Dec, per dataset (held-out Dice, mean over structures).

Dataset (structures)	From scratch	Dec	Full FT	Scratch − Dec	Scratch − Full FT
AMOS22 (15)	0.496	0.708	0.681	-0.212	-0.185
KiTS23 (3)	0.400	0.514	0.519	-0.114	-0.120
LiTS (2)	0.612	0.699	0.715	-0.087	-0.103
MSD Pancreas (2)	0.308	0.480	0.487	-0.172	-0.180
TotalSegmentator (15)	0.579	0.798	0.815	-0.219	-0.236

Appendix E. Per-cohort, per-label and per-structure results

Table 18: Classification: held-out AUROC, mean over the labels of each cohort (number of labels in brackets), per backbone, training mode and method. Within each row, **bold** is the best and underlined the second-best method. A shaded cell marks the better of per-task and joint training for that method, backbone and cohort. SPECTRE Full FT did not train with the default setting (Section 4.1). VoCo was run on three cohorts.

Cohort (labels)	Backbone	Mode	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ ABMIL	DoRA	rsLoRA	Lin→ LoRA	Attn→ LoRA	Attn-only FT	Full FT
CVPR26 (17)	TAP-CT	per-task	0.677	0.718	0.732	0.704	0.728	0.717	0.724	0.697	0.707	**0.744**	0.720
		joint	0.707	0.720	0.733	0.726	0.729	0.733	0.722	0.695	0.708	0.739	**0.744**
CVPR26 (17)	SPECTRE	per-task	0.736	0.725	0.769	0.769	0.776	0.766	0.768	0.769	0.765	**0.778**	0.627
		joint	0.702	0.738	0.716	0.731	**0.747**	0.731	0.690	0.744	0.728	0.740	0.575
FDG (4)	TAP-CT	per-task	0.863	0.836	0.852	**0.868**	0.839	0.847	0.817	0.866	0.856	0.832	0.769
		joint	0.858	0.860	0.866	0.864	0.849	0.858	0.785	0.860	0.862	**0.870**	0.608
FDG (4)	SPECTRE	per-task	0.857	0.862	0.882	0.879	0.874	**0.892**	0.872	0.887	0.867	0.848	0.617
		joint	0.884	0.880	0.901	0.888	0.889	0.900	0.900	0.884	0.884	**0.906**	0.673
FDG (4)	VoCo	per-task	0.720	0.685	0.512	0.671	0.562	0.669	0.598	**0.736**	0.591	0.665	0.606
		joint	0.745	0.739	0.600	0.506	0.655	0.564	0.527	**0.750**	0.520	0.641	0.582
RSNA22 (8)	TAP-CT	per-task	0.619	0.625	0.623	0.608	0.613	0.641	0.635	0.606	0.629	**0.660**	0.637
		joint	0.626	0.624	0.714	0.706	0.713	0.680	0.601	0.624	0.620	0.712	**0.732**
RSNA22 (8)	SPECTRE	per-task	0.579	0.591	0.635	**0.654**	0.644	0.619	0.638	0.615	0.591	0.635	0.566
		joint	0.629	0.601	0.706	0.644	**0.730**	0.692	0.691	0.670	0.611	0.718	0.554
RSNA22 (8)	VoCo	per-task	0.606	0.508	0.537	0.551	0.596	0.561	**0.617**	0.602	0.521	0.576	0.605
		joint	0.586	0.548	0.560	0.550	0.565	0.589	0.578	**0.619**	0.527	0.589	0.593
RSNA23 (4)	TAP-CT	per-task	0.737	0.718	0.739	0.756	0.703	0.752	0.725	0.738	0.716	**0.859**	0.775
		joint	0.739	0.738	0.757	0.744	0.758	0.768	0.684	0.746	0.758	0.833	**0.881**
RSNA23 (4)	SPECTRE	per-task	0.791	0.778	0.824	0.832	**0.852**	0.842	0.836	0.825	0.798	0.772	0.624
		joint	0.789	0.788	0.846	0.835	**0.859**	0.838	0.850	0.827	0.810	0.803	0.654
RSNA23 (4)	VoCo	per-task	0.617	0.625	0.622	0.601	0.595	0.611	0.608	**0.631**	0.569	0.588	0.596
		joint	0.631	0.616	0.603	0.627	0.570	0.583	0.607	0.626	0.635	0.593	**0.648**
ICH (6)	TAP-CT	per-task	0.870	0.873	0.894	0.894	0.891	0.897	0.870	0.877	0.878	**0.911**	0.888
		joint	0.871	0.877	0.906	0.895	0.886	0.897	0.869	0.868	0.876	**0.915**	0.854
ICH (6)	Pillar-0	per-task	0.886	0.884	0.929	0.914	0.902	**0.931**	0.920	0.912	0.903	**0.931**	0.904
		joint	0.888	0.888	0.912	0.892	**0.925**	0.912	0.918	0.905	0.898	0.912	0.878

Table 19: Segmentation: held-out Dice, mean over the structures of each dataset (number of structures in brackets), per backbone, training mode and method; – marks methods not run per structure on TotalSegmentator. Within each row, **bold** is the best and underlined the second-best method. A shaded cell marks the better of per-structure and joint training for that method, backbone and dataset. The AMOS22 and TotalSegmentator joint models train 24 000 steps. VoCo's Lin reads stage-4 tokens at a stride of 32 voxels.

Dataset (structures)	Backbone	Mode	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→LoRA
AMOS22 (15)	TAP-CT	per-task	0.658	0.793	0.767	0.764	0.761	**0.805**	0.791
		joint	0.646	0.708	**0.739**	0.698	0.708	0.681	0.656
AMOS22 (15)	SPECTRE	per-task	0.683	0.808	0.812	0.806	**0.814**	0.792	**0.814**
		joint	0.682	0.658	**0.784**	0.764	0.775	0.669	0.664
AMOS22 (15)	VoCo	per-task	0.041	0.805	0.782	0.783	0.783	**0.818**	0.799
		joint	0.038	0.739	0.762	0.758	**0.764**	0.752	0.736
KiTS23 (3)	TAP-CT	per-task	0.531	0.584	0.563	0.567	0.563	**0.588**	0.587
		joint	0.430	0.514	**0.582**	0.556	0.560	0.519	0.500
KiTS23 (3)	SPECTRE	per-task	0.583	0.599	0.613	0.617	**0.622**	0.528	0.603
		joint	0.530	0.572	0.588	0.588	**0.593**	0.519	0.570
KiTS23 (3)	VoCo	per-task	0.017	0.519	0.501	0.500	0.493	**0.564**	0.481
		joint	0.037	0.465	0.501	**0.516**	0.507	0.481	0.452
LiTS (2)	TAP-CT	per-task	0.659	0.709	**0.716**	0.709	0.714	0.710	0.714
		joint	0.656	0.699	0.715	**0.721**	0.692	0.715	0.703
LiTS (2)	SPECTRE	per-task	0.688	0.741	**0.756**	0.723	**0.756**	0.739	0.729
		joint	0.696	0.742	**0.753**	0.736	0.743	0.713	0.743
LiTS (2)	VoCo	per-task	0.342	0.733	0.692	0.694	0.688	**0.748**	0.720
		joint	0.345	0.732	0.720	0.710	0.718	**0.737**	0.726
MSD Pancreas (2)	TAP-CT	per-task	0.427	0.491	0.504	0.501	0.502	**0.512**	0.495
		joint	0.334	0.480	**0.505**	0.492	0.493	0.487	0.468
MSD Pancreas (2)	SPECTRE	per-task	0.475	0.516	**0.563**	0.538	0.557	0.516	0.518
		joint	0.459	0.514	**0.550**	0.529	**0.550**	0.530	0.520
MSD Pancreas (2)	VoCo	per-task	0.000	**0.569**	0.539	0.545	0.541	0.566	0.558
		joint	0.000	0.559	0.552	0.545	0.552	0.560	**0.561**
TotalSegmentator (15)	TAP-CT	per-task	0.747	**0.843**	0.819	–	–	0.832	–
		joint	0.740	0.798	0.776	0.776	0.776	**0.815**	0.790
TotalSegmentator (15)	SPECTRE	per-task	0.731	**0.848**	**0.848**	–	–	0.838	–
		joint	0.722	0.770	0.733	0.752	0.736	**0.795**	0.772
TotalSegmentator (15)	VoCo	per-task	0.055	0.822	0.771	–	–	**0.838**	–
		joint	0.047	0.732	0.705	0.697	0.709	**0.750**	0.723

Table 20: Held-out AUROC as in Table 18, with the longer CVPR26 joint schedule (26 000 steps) as an extra row. **Bold**: best, underlined: second best of the row; shaded: the better of per-task and joint training (8000 steps).

Cohort	Backbone	Mode (labels)	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ABMIL	DoRA	rsLoRA	Lin→LoRA	Attn→LoRA	Attn-only FT	Full FT
CVPR26	TAP-CT	per-task (17)	0.677	0.718	0.732	0.704	0.728	0.717	0.724	0.697	0.707	**0.744**	0.720
CVPR26	TAP-CT	joint (17)	0.707	0.720	0.733	0.726	0.729	0.733	0.722	0.695	0.708	0.739	**0.744**
CVPR26	TAP-CT	joint (26 000 steps) (17)	0.693	0.722	0.692	0.728	0.727	0.720	0.701	0.698	0.726	**0.756**	0.650
CVPR26	SPECTRE	per-task (17)	0.736	0.725	0.769	0.769	0.776	0.766	0.768	0.769	0.765	**0.778**	0.627
CVPR26	SPECTRE	joint (17)	0.702	0.738	0.716	0.731	**0.747**	0.731	0.690	0.744	0.728	0.740	0.575
CVPR26	SPECTRE	joint (26 000 steps) (17)	0.744	0.713	0.755	0.762	0.760	0.752	**0.763**	0.746	0.758	0.756	0.574
FDG-PET/CT	TAP-CT	per-task (4)	0.863	0.836	0.852	**0.868**	0.839	0.847	0.817	0.866	0.856	0.832	0.769
FDG-PET/CT	TAP-CT	joint (4)	0.858	0.860	0.866	0.864	0.849	0.858	0.785	0.860	0.862	**0.870**	0.608
FDG-PET/CT	SPECTRE	per-task (4)	0.857	0.862	0.882	0.879	0.874	**0.892**	0.872	0.887	0.867	0.848	0.617
FDG-PET/CT	SPECTRE	joint (4)	0.884	0.880	0.901	0.888	0.889	0.900	0.900	0.884	0.884	**0.906**	0.673
FDG-PET/CT	VoCo	per-task (4)	0.720	0.685	0.512	0.671	0.562	0.669	0.598	**0.736**	0.591	0.665	0.606
FDG-PET/CT	VoCo	joint (4)	0.745	0.739	0.600	0.506	0.655	0.564	0.527	**0.750**	0.520	0.641	0.582
RSNA 2022 C-spine	TAP-CT	per-task (8)	0.619	0.625	0.623	0.608	0.613	0.641	0.635	0.606	0.629	**0.660**	0.637
RSNA 2022 C-spine	TAP-CT	joint (8)	0.626	0.624	0.714	0.706	0.713	0.680	0.601	0.624	0.620	0.712	**0.732**
RSNA 2022 C-spine	SPECTRE	per-task (8)	0.579	0.591	0.635	**0.654**	0.644	0.619	0.638	0.615	0.591	0.635	0.566
RSNA 2022 C-spine	SPECTRE	joint (8)	0.629	0.601	0.706	0.644	**0.730**	0.692	0.691	0.670	0.611	0.718	0.554
RSNA 2022 C-spine	VoCo	per-task (8)	0.606	0.508	0.537	0.551	0.596	0.561	**0.617**	0.602	0.521	0.576	0.605
RSNA 2022 C-spine	VoCo	joint (8)	0.586	0.548	0.560	0.550	0.565	0.589	0.578	**0.619**	0.527	0.589	0.593
RSNA 2023 trauma	TAP-CT	per-task (4)	0.737	0.718	0.739	0.756	0.703	0.752	0.725	0.738	0.716	**0.859**	0.775
RSNA 2023 trauma	TAP-CT	joint (4)	0.739	0.738	0.757	0.744	0.758	0.768	0.684	0.746	0.758	0.833	**0.881**
RSNA 2023 trauma	SPECTRE	per-task (4)	0.791	0.778	0.824	0.832	**0.852**	0.842	0.836	0.825	0.798	0.772	0.624
RSNA 2023 trauma	SPECTRE	joint (4)	0.789	0.788	0.846	0.835	**0.859**	0.838	0.850	0.827	0.810	0.803	0.654
RSNA 2023 trauma	VoCo	per-task (4)	0.617	0.625	0.622	0.601	0.595	0.611	0.608	**0.631**	0.569	0.588	0.596
RSNA 2023 trauma	VoCo	joint (4)	0.631	0.616	0.603	0.627	0.570	0.583	0.607	0.626	0.635	0.593	**0.648**
RSNA 2019 ICH	TAP-CT	per-task (6)	0.870	0.873	0.894	0.894	0.891	0.897	0.870	0.877	0.878	**0.911**	0.888
RSNA 2019 ICH	TAP-CT	joint (6)	0.871	0.877	0.906	0.895	0.886	0.897	0.869	0.868	0.876	**0.915**	0.854
RSNA 2019 ICH	Pillar-0	per-task (6)	0.886	0.884	0.929	0.914	0.902	**0.931**	0.920	0.912	0.903	**0.931**	0.904
RSNA 2019 ICH	Pillar-0	joint (6)	0.888	0.888	0.912	0.892	**0.925**	0.912	0.918	0.905	0.898	0.912	0.878

CVPR26, TAP-CT

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ABMIL	DoRA	rsLoRA	Lin→LoRA	Attn→LoRA	Attn-only FT	Full FT
TAP-CT, per-task											
adrenal_hyperplasia	0.80 [0.62,0.95]	0.84 [0.68,0.97]	0.80 [0.62,0.96]	0.79 [0.62,0.95]	0.81 [0.64,0.95]	0.77 [0.59,0.92]	0.79 [0.60,0.94]	0.80 [0.62,0.95]	0.82 [0.67,0.95]	0.84 [0.68,0.97]	0.87 [0.72,0.98]
ascites	0.78 [0.60,0.92]	0.84 [0.67,0.97]	0.77 [0.59,0.92]	0.74 [0.55,0.90]	0.78 [0.61,0.92]	0.71 [0.50,0.87]	0.77 [0.60,0.93]	0.76 [0.58,0.92]	0.75 [0.56,0.92]	0.68 [0.47,0.85]	0.82 [0.66,0.95]
atherosclerosis	0.75 [0.57,0.90]	0.87 [0.73,0.97]	0.89 [0.76,0.97]	0.83 [0.68,0.96]	0.83 [0.68,0.95]	0.84 [0.68,0.96]	0.86 [0.71,0.97]	0.75 [0.56,0.90]	0.86 [0.71,0.97]	0.81 [0.64,0.93]	0.88 [0.74,0.97]
cholecystitis	0.73 [0.56,0.88]	0.78 [0.62,0.92]	0.74 [0.55,0.90]	0.69 [0.49,0.85]	0.73 [0.54,0.89]	0.75 [0.60,0.90]	0.74 [0.56,0.89]	0.75 [0.56,0.91]	0.72 [0.56,0.88]	0.77 [0.60,0.91]	0.74 [0.57,0.89]
colorectal_cancer	0.56 [0.33,0.77]	0.65 [0.44,0.85]	0.65 [0.43,0.88]	0.65 [0.43,0.85]	0.67 [0.43,0.86]	0.63 [0.40,0.84]	0.62 [0.38,0.84]	0.60 [0.37,0.84]	0.64 [0.42,0.85]	0.70 [0.48,0.88]	0.62 [0.41,0.82]
fatty_liver	0.87 [0.75,0.97]	0.92 [0.82,0.99]	0.95 [0.88,1.00]	0.95 [0.86,1.00]	0.95 [0.88,1.00]	0.94 [0.85,0.99]	0.99 [0.97,1.00]	0.90 [0.78,0.99]	0.92 [0.82,0.98]	0.99 [0.97,1.00]	0.98 [0.94,1.00]
gallstone	0.60 [0.46,0.74]	0.65 [0.49,0.78]	0.66 [0.52,0.80]	0.64 [0.50,0.78]	0.68 [0.54,0.81]	0.68 [0.53,0.82]	0.60 [0.46,0.74]	0.61 [0.46,0.75]	0.69 [0.55,0.82]	0.65 [0.51,0.78]	0.69 [0.56,0.82]
hydronephrosis	0.86 [0.71,0.97]	0.82 [0.64,0.95]	0.90 [0.78,0.98]	0.86 [0.70,0.97]	0.86 [0.71,0.96]	0.85 [0.70,0.96]	0.78 [0.61,0.92]	0.87 [0.71,0.97]	0.90 [0.78,0.98]	0.86 [0.71,0.97]	0.67 [0.47,0.85]
kidney_stone	0.62 [0.49,0.75]	0.69 [0.57,0.80]	0.63 [0.50,0.75]	0.65 [0.52,0.77]	0.64 [0.50,0.76]	0.62 [0.51,0.74]	0.70 [0.58,0.81]	0.70 [0.57,0.81]	0.71 [0.59,0.83]	0.70 [0.58,0.80]	0.67 [0.56,0.79]
liver_calcifications	0.44 [0.25,0.63]	0.37 [0.20,0.56]	0.49 [0.30,0.69]	0.44 [0.26,0.63]	0.47 [0.28,0.67]	0.51 [0.32,0.69]	0.51 [0.31,0.71]	0.43 [0.26,0.63]	0.39 [0.22,0.59]	0.58 [0.38,0.75]	0.57 [0.40,0.75]
liver_cyst	0.64 [0.54,0.74]	0.67 [0.57,0.76]	0.67 [0.56,0.77]	0.64 [0.54,0.74]	0.65 [0.54,0.75]	0.62 [0.53,0.73]	0.68 [0.57,0.77]	0.66 [0.57,0.76]	0.64 [0.55,0.73]	0.69 [0.60,0.78]	0.66 [0.55,0.75]
liver_lesion	0.57 [0.42,0.72]	0.64 [0.48,0.77]	0.71 [0.57,0.83]	0.69 [0.56,0.81]	0.61 [0.47,0.75]	0.66 [0.52,0.79]	0.65 [0.51,0.78]	0.59 [0.45,0.73]	0.64 [0.51,0.77]	0.66 [0.52,0.80]	0.70 [0.57,0.82]
lymphadenopathy	0.60 [0.41,0.78]	0.73 [0.57,0.88]	0.64 [0.48,0.81]	0.67 [0.48,0.85]	0.76 [0.61,0.91]	0.73 [0.55,0.88]	0.66 [0.46,0.82]	0.69 [0.51,0.85]	0.70 [0.52,0.85]	0.76 [0.60,0.90]	0.54 [0.36,0.72]
renal_cyst	0.66 [0.57,0.75]	0.64 [0.55,0.73]	0.67 [0.58,0.76]	0.69 [0.60,0.77]	0.65 [0.56,0.74]	0.64 [0.55,0.73]	0.69 [0.60,0.77]	0.64 [0.55,0.73]	0.64 [0.54,0.73]	0.66 [0.57,0.75]	0.67 [0.57,0.75]
splenomegaly	0.68 [0.52,0.85]	0.75 [0.59,0.88]	0.71 [0.54,0.85]	0.72 [0.57,0.86]	0.71 [0.55,0.88]	0.71 [0.54,0.87]	0.73 [0.57,0.87]	0.70 [0.53,0.87]	0.71 [0.54,0.86]	0.72 [0.55,0.87]	0.71 [0.55,0.86]
covid	0.81 [0.65,0.96]	0.76 [0.58,0.91]	0.97 [0.92,1.00]	0.75 [0.57,0.91]	0.99 [0.96,1.00]	0.99 [0.95,1.00]	0.98 [0.93,1.00]	0.84 [0.68,0.96]	0.77 [0.60,0.93]	1.00 [0.98,1.00]	0.92 [0.78,1.00]
lung_nodule_malignancy	0.53 [0.42,0.63]	0.57 [0.45,0.69]	0.59 [0.47,0.71]	0.56 [0.46,0.67]	0.57 [0.45,0.68]	0.55 [0.43,0.66]	0.57 [0.46,0.69]	0.56 [0.45,0.67]	0.50 [0.39,0.62]	0.59 [0.48,0.70]	0.55 [0.43,0.66]

CVPR26, TAP-CT

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ABMIL	DoRA	rsLoRA	Lin→LoRA	Attn→LoRA	Attn-only FT	Full FT
TAP-CT, joint											
adrenal_hyperplasia	0.84 [0.69,0.96]	0.83 [0.67,0.97]	0.85 [0.69,0.98]	0.84 [0.68,0.96]	0.87 [0.74,0.97]	0.82 [0.63,0.95]	0.82 [0.64,0.94]	0.87 [0.74,0.97]	0.83 [0.68,0.95]	0.80 [0.63,0.95]	0.87 [0.71,0.98]
ascites	0.77 [0.58,0.93]	0.80 [0.62,0.94]	0.84 [0.69,0.97]	0.80 [0.61,0.95]	0.75 [0.56,0.91]	0.81 [0.65,0.96]	0.81 [0.65,0.95]	0.77 [0.58,0.91]	0.68 [0.47,0.85]	0.89 [0.76,0.98]	0.81 [0.64,0.94]
atherosclerosis	0.83 [0.70,0.95]	0.81 [0.64,0.93]	0.87 [0.73,0.97]	0.83 [0.69,0.95]	0.81 [0.64,0.94]	0.81 [0.64,0.94]	0.85 [0.70,0.96]	0.82 [0.65,0.94]	0.78 [0.60,0.91]	0.83 [0.66,0.97]	0.84 [0.67,0.96]
cholecystitis	0.76 [0.57,0.91]	0.70 [0.52,0.87]	0.71 [0.54,0.87]	0.77 [0.58,0.92]	0.76 [0.58,0.89]	0.73 [0.56,0.88]	0.73 [0.53,0.88]	0.71 [0.53,0.88]	0.71 [0.55,0.87]	0.72 [0.56,0.87]	0.72 [0.55,0.88]
colorectal_cancer	0.62 [0.39,0.83]	0.64 [0.40,0.86]	0.70 [0.46,0.89]	0.67 [0.43,0.87]	0.67 [0.44,0.88]	0.66 [0.41,0.88]	0.67 [0.43,0.86]	0.61 [0.38,0.81]	0.68 [0.44,0.88]	0.67 [0.44,0.86]	0.64 [0.39,0.83]
fatty_liver	0.84 [0.69,0.96]	0.80 [0.62,0.94]	0.78 [0.60,0.91]	0.76 [0.57,0.90]	0.82 [0.67,0.93]	0.79 [0.62,0.94]	0.78 [0.61,0.92]	0.82 [0.67,0.94]	0.75 [0.58,0.91]	0.81 [0.65,0.93]	0.88 [0.75,0.97]
gallstone	0.67 [0.53,0.79]	0.66 [0.52,0.80]	0.65 [0.50,0.78]	0.63 [0.48,0.77]	0.66 [0.51,0.79]	0.67 [0.53,0.81]	0.66 [0.52,0.79]	0.67 [0.52,0.80]	0.66 [0.51,0.79]	0.68 [0.54,0.80]	0.67 [0.53,0.80]
hydronephrosis	0.86 [0.71,0.98]	0.83 [0.69,0.95]	0.79 [0.63,0.93]	0.77 [0.59,0.92]	0.81 [0.65,0.94]	0.82 [0.65,0.96]	0.70 [0.49,0.88]	0.83 [0.67,0.96]	0.79 [0.61,0.93]	0.78 [0.58,0.94]	0.76 [0.58,0.91]
kidney_stone	0.74 [0.62,0.84]	0.69 [0.56,0.80]	0.70 [0.59,0.82]	0.71 [0.58,0.82]	0.71 [0.58,0.81]	0.67 [0.56,0.79]	0.63 [0.50,0.74]	0.69 [0.58,0.81]	0.68 [0.56,0.81]	0.63 [0.50,0.74]	0.71 [0.60,0.82]
liver_calcifications	0.43 [0.25,0.62]	0.51 [0.32,0.71]	0.46 [0.26,0.65]	0.51 [0.32,0.72]	0.50 [0.33,0.71]	0.52 [0.34,0.72]	0.47 [0.29,0.68]	0.40 [0.22,0.58]	0.51 [0.33,0.71]	0.54 [0.34,0.76]	0.58 [0.38,0.77]
liver_cyst	0.63 [0.53,0.73]	0.67 [0.57,0.76]	0.65 [0.55,0.75]	0.66 [0.56,0.75]	0.66 [0.56,0.75]	0.66 [0.56,0.76]	0.65 [0.55,0.74]	0.63 [0.52,0.72]	0.63 [0.53,0.72]	0.67 [0.57,0.77]	0.67 [0.56,0.76]
liver_lesion	0.63 [0.48,0.77]	0.71 [0.58,0.83]	0.70 [0.56,0.81]	0.69 [0.55,0.82]	0.70 [0.57,0.82]	0.68 [0.54,0.81]	0.70 [0.56,0.82]	0.66 [0.52,0.79]	0.70 [0.56,0.82]	0.69 [0.55,0.80]	0.66 [0.52,0.79]
lymphadenopathy	0.73 [0.55,0.88]	0.77 [0.60,0.91]	0.78 [0.61,0.92]	0.74 [0.56,0.89]	0.71 [0.55,0.86]	0.75 [0.56,0.89]	0.78 [0.61,0.92]	0.73 [0.58,0.88]	0.68 [0.51,0.85]	0.74 [0.55,0.89]	0.78 [0.59,0.93]
renal_cyst	0.60 [0.50,0.69]	0.64 [0.54,0.72]	0.66 [0.57,0.75]	0.66 [0.57,0.75]	0.68 [0.60,0.78]	0.68 [0.60,0.76]	0.66 [0.56,0.74]	0.61 [0.52,0.70]	0.65 [0.55,0.74]	0.70 [0.61,0.79]	0.67 [0.58,0.76]
splenomegaly	0.72 [0.56,0.86]	0.78 [0.63,0.91]	0.86 [0.73,0.95]	0.82 [0.66,0.95]	0.80 [0.65,0.92]	0.91 [0.81,0.98]	0.87 [0.75,0.97]	0.68 [0.52,0.84]	0.78 [0.64,0.91]	0.86 [0.74,0.96]	0.84 [0.70,0.95]
covid	0.83 [0.66,0.97]	0.74 [0.56,0.89]	0.83 [0.67,0.97]	0.82 [0.64,0.96]	0.84 [0.69,0.97]	0.82 [0.65,0.96]	0.86 [0.69,0.98]	0.82 [0.65,0.95]	0.84 [0.69,0.97]	0.89 [0.75,1.00]	0.93 [0.82,1.00]
lung_nodule_malignancy	0.51 [0.39,0.63]	0.67 [0.55,0.76]	0.65 [0.53,0.75]	0.67 [0.57,0.77]	0.64 [0.53,0.74]	0.65 [0.53,0.75]	0.65 [0.55,0.75]	0.48 [0.37,0.60]	0.68 [0.58,0.78]	0.66 [0.54,0.76]	0.62 [0.51,0.73]

CVPR26, TAP-CT

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ABMIL	DoRA	rsLoRA	Lin→LoRA	Attn→LoRA	Attn-only FT	Full FT
TAP-CT, joint (26 000 steps)											
adrenal_hyperplasia	0.80 [0.61,0.94]	0.76 [0.59,0.92]	0.62 [0.42,0.80]	0.80 [0.62,0.94]	0.76 [0.59,0.92]	0.81 [0.64,0.94]	0.85 [0.70,0.97]	0.81 [0.64,0.95]	0.80 [0.62,0.95]	0.88 [0.75,0.99]	0.88 [0.74,0.98]
ascites	0.79 [0.61,0.93]	0.81 [0.63,0.94]	0.82 [0.65,0.96]	0.85 [0.68,0.97]	0.77 [0.58,0.92]	0.86 [0.69,1.00]	0.82 [0.66,0.96]	0.78 [0.59,0.93]	0.72 [0.52,0.87]	0.76 [0.59,0.91]	0.62 [0.41,0.80]
atherosclerosis	0.80 [0.61,0.93]	0.82 [0.65,0.95]	0.84 [0.69,0.96]	0.82 [0.67,0.95]	0.74 [0.54,0.90]	0.77 [0.59,0.92]	0.80 [0.63,0.93]	0.79 [0.62,0.92]	0.78 [0.60,0.92]	0.85 [0.68,0.97]	0.62 [0.42,0.81]
cholecystitis	0.81 [0.64,0.94]	0.70 [0.53,0.85]	0.69 [0.50,0.83]	0.71 [0.53,0.87]	0.73 [0.57,0.88]	0.71 [0.51,0.87]	0.68 [0.50,0.84]	0.87 [0.75,0.97]	0.84 [0.70,0.94]	0.73 [0.56,0.88]	0.66 [0.48,0.83]
colorectal_cancer	0.60 [0.38,0.85]	0.71 [0.49,0.91]	0.59 [0.37,0.82]	0.70 [0.48,0.91]	0.61 [0.38,0.83]	0.63 [0.40,0.85]	0.71 [0.50,0.89]	0.58 [0.36,0.82]	0.72 [0.47,0.90]	0.68 [0.46,0.88]	0.58 [0.35,0.80]
fatty_liver	0.85 [0.71,0.97]	0.85 [0.72,0.96]	0.79 [0.62,0.93]	0.85 [0.71,0.95]	0.88 [0.75,0.98]	0.80 [0.64,0.93]	0.72 [0.54,0.88]	0.86 [0.73,0.96]	0.85 [0.70,0.96]	0.90 [0.78,0.97]	0.73 [0.56,0.88]
gallstone	0.62 [0.47,0.76]	0.62 [0.47,0.75]	0.66 [0.51,0.80]	0.62 [0.47,0.76]	0.65 [0.51,0.80]	0.68 [0.53,0.82]	0.68 [0.55,0.81]	0.65 [0.51,0.79]	0.63 [0.48,0.77]	0.66 [0.52,0.80]	0.65 [0.52,0.79]
hydronephrosis	0.83 [0.67,0.96]	0.84 [0.69,0.95]	0.76 [0.59,0.92]	0.74 [0.55,0.90]	0.79 [0.62,0.93]	0.76 [0.57,0.91]	0.69 [0.50,0.88]	0.84 [0.68,0.95]	0.80 [0.64,0.93]	0.77 [0.57,0.93]	0.70 [0.49,0.87]
kidney_stone	0.68 [0.55,0.79]	0.72 [0.61,0.83]	0.65 [0.53,0.77]	0.67 [0.56,0.79]	0.67 [0.55,0.79]	0.65 [0.53,0.77]	0.65 [0.53,0.77]	0.69 [0.56,0.80]	0.73 [0.61,0.83]	0.67 [0.54,0.77]	0.59 [0.46,0.72]
liver_calcifications	0.49 [0.30,0.69]	0.43 [0.25,0.62]	0.44 [0.25,0.63]	0.54 [0.34,0.73]	0.44 [0.24,0.62]	0.52 [0.34,0.71]	0.48 [0.30,0.67]	0.39 [0.20,0.58]	0.42 [0.24,0.63]	0.52 [0.33,0.72]	0.40 [0.22,0.58]
liver_cyst	0.65 [0.54,0.75]	0.69 [0.59,0.78]	0.63 [0.53,0.73]	0.71 [0.61,0.80]	0.72 [0.63,0.81]	0.72 [0.63,0.81]	0.63 [0.51,0.72]	0.63 [0.52,0.72]	0.68 [0.58,0.77]	0.67 [0.57,0.76]	0.59 [0.48,0.69]
liver_lesion	0.54 [0.39,0.68]	0.71 [0.58,0.83]	0.70 [0.56,0.82]	0.66 [0.52,0.79]	0.70 [0.57,0.82]	0.70 [0.57,0.82]	0.66 [0.52,0.79]	0.60 [0.46,0.75]	0.71 [0.58,0.82]	0.70 [0.57,0.83]	0.67 [0.54,0.80]
lymphadenopathy	0.69 [0.51,0.86]	0.73 [0.55,0.88]	0.63 [0.45,0.79]	0.74 [0.57,0.90]	0.78 [0.60,0.93]	0.78 [0.60,0.91]	0.68 [0.48,0.84]	0.69 [0.52,0.85]	0.74 [0.56,0.89]	0.77 [0.59,0.92]	0.67 [0.47,0.83]
renal_cyst	0.61 [0.52,0.70]	0.66 [0.57,0.75]	0.63 [0.54,0.73]	0.69 [0.59,0.77]	0.72 [0.64,0.81]	0.66 [0.56,0.74]	0.68 [0.59,0.77]	0.62 [0.53,0.72]	0.65 [0.56,0.75]	0.68 [0.59,0.77]	0.58 [0.48,0.67]
splenomegaly	0.73 [0.58,0.87]	0.83 [0.69,0.93]	0.87 [0.77,0.96]	0.81 [0.65,0.93]	0.81 [0.67,0.92]	0.88 [0.76,0.97]	0.75 [0.60,0.89]	0.73 [0.56,0.88]	0.79 [0.65,0.91]	0.82 [0.67,0.93]	0.76 [0.61,0.90]
covid	0.78 [0.61,0.94]	0.76 [0.59,0.91]	0.81 [0.64,0.95]	0.80 [0.61,0.94]	0.94 [0.80,1.00]	0.78 [0.59,0.95]	0.85 [0.69,0.99]	0.81 [0.66,0.94]	0.82 [0.65,0.96]	0.99 [0.96,1.00]	0.79 [0.62,0.94]
lung_nodule_malignancy	0.52 [0.40,0.64]	0.64 [0.53,0.74]	0.63 [0.52,0.74]	0.68 [0.58,0.78]	0.65 [0.54,0.75]	0.53 [0.42,0.64]	0.58 [0.47,0.69]	0.53 [0.41,0.64]	0.65 [0.54,0.76]	0.78 [0.70,0.87]	0.56 [0.45,0.68]

CVPR26, SPECTRE

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ABMIL	DoRA	rsLoRA	Lin→LoRA	Attn→LoRA	Attn-only FT	Full FT
SPECTRE, per-task											
adrenal_hyperplasia	0.81 [0.64,0.93]	0.52 [0.33,0.75]	0.83 [0.66,0.94]	0.80 [0.62,0.95]	0.86 [0.71,0.96]	0.59 [0.37,0.80]	0.76 [0.58,0.91]	0.80 [0.64,0.94]	0.76 [0.57,0.92]	0.78 [0.60,0.93]	0.76 [0.57,0.93]
ascites	0.89 [0.75,0.99]	0.90 [0.78,0.98]	0.92 [0.80,1.00]	0.88 [0.73,0.98]	0.88 [0.73,0.98]	0.90 [0.76,1.00]	0.88 [0.75,0.98]	0.92 [0.80,1.00]	0.89 [0.75,0.98]	0.80 [0.63,0.93]	0.55 [0.39,0.70]
atherosclerosis	0.76 [0.58,0.92]	0.78 [0.61,0.93]	0.80 [0.63,0.94]	0.83 [0.68,0.95]	0.84 [0.68,0.96]	0.84 [0.69,0.96]	0.79 [0.62,0.94]	0.74 [0.55,0.90]	0.70 [0.52,0.87]	0.78 [0.61,0.93]	0.82 [0.67,0.95]
cholecystitis	0.73 [0.57,0.88]	0.72 [0.55,0.87]	0.80 [0.67,0.92]	0.80 [0.65,0.93]	0.88 [0.77,0.97]	0.84 [0.71,0.94]	0.82 [0.67,0.93]	0.83 [0.70,0.93]	0.79 [0.64,0.93]	0.80 [0.65,0.92]	0.65 [0.48,0.82]
colorectal_cancer	0.60 [0.37,0.83]	0.70 [0.48,0.90]	0.71 [0.50,0.90]	0.65 [0.43,0.86]	0.56 [0.33,0.79]	0.72 [0.51,0.90]	0.67 [0.48,0.82]	0.74 [0.53,0.91]	0.75 [0.54,0.90]	0.74 [0.52,0.90]	0.50 [0.50,0.50]
fatty_liver	0.81 [0.65,0.93]	0.79 [0.63,0.93]	0.95 [0.86,1.00]	0.97 [0.89,1.00]	0.98 [0.93,1.00]	0.92 [0.80,0.99]	0.95 [0.83,1.00]	0.90 [0.79,0.98]	0.94 [0.83,1.00]	0.94 [0.86,1.00]	0.77 [0.60,0.90]
gallstone	0.66 [0.52,0.80]	0.70 [0.55,0.81]	0.64 [0.50,0.78]	0.69 [0.56,0.82]	0.79 [0.66,0.91]	0.69 [0.54,0.82]	0.78 [0.64,0.89]	0.64 [0.49,0.78]	0.75 [0.61,0.87]	0.83 [0.71,0.92]	0.57 [0.42,0.71]
hydronephrosis	0.85 [0.69,0.96]	0.85 [0.70,0.95]	0.91 [0.79,0.99]	0.93 [0.82,1.00]	0.94 [0.85,1.00]	0.90 [0.77,0.99]	0.87 [0.72,0.98]	0.93 [0.82,0.99]	0.90 [0.80,0.98]	0.94 [0.85,1.00]	0.56 [0.34,0.76]
kidney_stone	0.66 [0.54,0.78]	0.68 [0.56,0.81]	0.71 [0.60,0.82]	0.74 [0.63,0.84]	0.72 [0.61,0.83]	0.67 [0.54,0.79]	0.69 [0.56,0.81]	0.67 [0.54,0.78]	0.68 [0.55,0.79]	0.76 [0.66,0.86]	0.59 [0.46,0.70]
liver_calcifications	0.58 [0.39,0.77]	0.69 [0.49,0.86]	0.63 [0.42,0.83]	0.52 [0.32,0.71]	0.52 [0.32,0.71]	0.63 [0.44,0.81]	0.56 [0.37,0.75]	0.62 [0.43,0.81]	0.62 [0.43,0.81]	0.57 [0.41,0.72]	0.55 [0.46,0.65]
liver_cyst	0.58 [0.48,0.69]	0.63 [0.53,0.73]	0.65 [0.54,0.74]	0.65 [0.55,0.75]	0.65 [0.56,0.74]	0.66 [0.55,0.75]	0.64 [0.54,0.74]	0.65 [0.56,0.75]	0.66 [0.56,0.75]	0.66 [0.57,0.75]	0.62 [0.53,0.71]
liver_lesion	0.68 [0.55,0.81]	0.64 [0.51,0.77]	0.62 [0.48,0.76]	0.71 [0.57,0.83]	0.70 [0.57,0.83]	0.70 [0.56,0.82]	0.75 [0.62,0.86]	0.69 [0.57,0.81]	0.69 [0.54,0.82]	0.72 [0.60,0.84]	0.60 [0.47,0.74]
lymphadenopathy	0.76 [0.61,0.89]	0.61 [0.43,0.78]	0.78 [0.62,0.91]	0.74 [0.57,0.88]	0.77 [0.61,0.90]	0.83 [0.69,0.94]	0.78 [0.63,0.91]	0.79 [0.64,0.91]	0.73 [0.56,0.88]	0.70 [0.50,0.85]	0.53 [0.50,0.58]
renal_cyst	0.63 [0.54,0.72]	0.65 [0.55,0.74]	0.57 [0.48,0.67]	0.72 [0.63,0.80]	0.71 [0.62,0.79]	0.58 [0.48,0.67]	0.57 [0.47,0.66]	0.60 [0.51,0.69]	0.67 [0.57,0.75]	0.70 [0.61,0.78]	0.48 [0.41,0.55]
splenomegaly	0.76 [0.59,0.89]	0.70 [0.52,0.85]	0.87 [0.75,0.96]	0.77 [0.61,0.90]	0.70 [0.53,0.83]	0.87 [0.75,0.96]	0.83 [0.68,0.94]	0.82 [0.67,0.93]	0.77 [0.61,0.89]	0.76 [0.62,0.89]	0.57 [0.41,0.75]
covid	1.00 [1.00,1.00]	1.00 [0.98,1.00]	0.99 [0.95,1.00]	0.98 [0.93,1.00]	0.93 [0.81,1.00]	0.98 [0.94,1.00]	0.99 [0.95,1.00]	1.00 [0.98,1.00]	1.00 [0.98,1.00]	0.98 [0.94,1.00]	0.72 [0.54,0.87]
lung_nodule_malignancy	0.74 [0.63,0.84]	0.74 [0.63,0.84]	0.68 [0.57,0.78]	0.71 [0.60,0.80]	0.75 [0.66,0.84]	0.71 [0.60,0.81]	0.74 [0.64,0.83]	0.72 [0.61,0.81]	0.73 [0.63,0.82]	0.76 [0.66,0.85]	0.82 [0.73,0.89]

CVPR26, SPECTRE

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ABMIL	DoRA	rsLoRA	Lin→LoRA	Attn→LoRA	Attn-only FT	Full FT
SPECTRE, joint											
adrenal_hyperplasia	0.72 [0.55,0.89]	0.57 [0.37,0.77]	0.52 [0.31,0.73]	0.63 [0.42,0.80]	0.67 [0.47,0.84]	0.54 [0.32,0.73]	0.69 [0.52,0.86]	0.78 [0.60,0.93]	0.65 [0.46,0.83]	0.74 [0.55,0.89]	0.50 [0.50,0.50]
ascites	0.83 [0.67,0.95]	0.87 [0.72,0.98]	0.84 [0.68,0.96]	0.81 [0.63,0.94]	0.84 [0.67,0.95]	0.86 [0.71,0.98]	0.81 [0.66,0.94]	0.78 [0.62,0.92]	0.82 [0.67,0.95]	0.87 [0.72,0.97]	0.60 [0.43,0.75]
atherosclerosis	0.78 [0.60,0.93]	0.76 [0.58,0.91]	0.76 [0.59,0.91]	0.78 [0.60,0.93]	0.79 [0.62,0.93]	0.74 [0.57,0.89]	0.72 [0.54,0.88]	0.81 [0.64,0.94]	0.77 [0.59,0.91]	0.74 [0.56,0.91]	0.53 [0.50,0.59]
cholecystitis	0.68 [0.51,0.83]	0.80 [0.64,0.92]	0.76 [0.58,0.91]	0.68 [0.50,0.85]	0.77 [0.62,0.90]	0.73 [0.56,0.86]	0.71 [0.53,0.86]	0.71 [0.52,0.87]	0.69 [0.53,0.85]	0.78 [0.63,0.91]	0.66 [0.51,0.80]
colorectal_cancer	0.69 [0.49,0.89]	0.71 [0.48,0.89]	0.67 [0.46,0.86]	0.67 [0.43,0.86]	0.71 [0.49,0.89]	0.71 [0.50,0.89]	0.66 [0.43,0.86]	0.72 [0.51,0.92]	0.70 [0.49,0.89]	0.65 [0.42,0.85]	0.35 [0.18,0.53]
fatty_liver	0.83 [0.68,0.95]	0.71 [0.53,0.88]	0.69 [0.51,0.85]	0.77 [0.60,0.91]	0.80 [0.63,0.93]	0.75 [0.57,0.90]	0.59 [0.40,0.77]	0.80 [0.64,0.93]	0.70 [0.52,0.86]	0.80 [0.65,0.93]	0.67 [0.52,0.80]
gallstone	0.48 [0.33,0.63]	0.75 [0.62,0.87]	0.68 [0.53,0.80]	0.68 [0.53,0.81]	0.77 [0.65,0.88]	0.77 [0.65,0.89]	0.75 [0.63,0.87]	0.75 [0.63,0.87]	0.80 [0.69,0.90]	0.69 [0.55,0.82]	0.66 [0.53,0.78]
hydronephrosis	0.90 [0.78,0.98]	0.81 [0.66,0.94]	0.76 [0.58,0.92]	0.78 [0.60,0.93]	0.74 [0.54,0.89]	0.78 [0.61,0.93]	0.75 [0.59,0.90]	0.86 [0.72,0.97]	0.69 [0.51,0.85]	0.82 [0.66,0.95]	0.74 [0.59,0.88]
kidney_stone	0.63 [0.51,0.74]	0.67 [0.54,0.78]	0.66 [0.55,0.78]	0.70 [0.59,0.81]	0.70 [0.58,0.82]	0.66 [0.56,0.78]	0.66 [0.54,0.77]	0.61 [0.48,0.73]	0.70 [0.58,0.81]	0.68 [0.56,0.79]	0.53 [0.45,0.60]
liver_calcifications	0.34 [0.16,0.53]	0.70 [0.52,0.86]	0.63 [0.44,0.81]	0.68 [0.49,0.85]	0.65 [0.47,0.82]	0.56 [0.37,0.76]	0.47 [0.28,0.66]	0.54 [0.36,0.76]	0.73 [0.55,0.88]	0.49 [0.29,0.68]	0.50 [0.43,0.58]
liver_cyst	0.62 [0.52,0.71]	0.65 [0.55,0.75]	0.66 [0.54,0.75]	0.65 [0.55,0.75]	0.69 [0.59,0.77]	0.71 [0.62,0.80]	0.62 [0.51,0.72]	0.65 [0.55,0.76]	0.63 [0.53,0.73]	0.69 [0.59,0.78]	0.55 [0.47,0.63]
liver_lesion	0.61 [0.46,0.76]	0.63 [0.48,0.76]	0.64 [0.50,0.77]	0.61 [0.46,0.75]	0.65 [0.51,0.79]	0.63 [0.49,0.76]	0.68 [0.55,0.80]	0.66 [0.52,0.79]	0.59 [0.44,0.73]	0.65 [0.50,0.78]	0.52 [0.42,0.61]
lymphadenopathy	0.80 [0.64,0.93]	0.78 [0.63,0.91]	0.81 [0.65,0.94]	0.75 [0.58,0.90]	0.79 [0.63,0.92]	0.78 [0.62,0.92]	0.73 [0.54,0.88]	0.80 [0.65,0.93]	0.74 [0.57,0.89]	0.78 [0.61,0.91]	0.50 [0.50,0.50]
renal_cyst	0.64 [0.54,0.73]	0.63 [0.53,0.72]	0.63 [0.54,0.72]	0.70 [0.61,0.78]	0.64 [0.55,0.73]	0.68 [0.60,0.77]	0.62 [0.52,0.71]	0.67 [0.58,0.77]	0.60 [0.51,0.69]	0.67 [0.59,0.77]	0.52 [0.44,0.60]
splenomegaly	0.66 [0.49,0.81]	0.78 [0.63,0.91]	0.79 [0.64,0.92]	0.85 [0.71,0.95]	0.82 [0.68,0.93]	0.82 [0.68,0.93]	0.61 [0.44,0.78]	0.76 [0.60,0.89]	0.85 [0.71,0.95]	0.83 [0.68,0.94]	0.72 [0.58,0.84]
covid	0.99 [0.96,1.00]	1.00 [1.00,1.00]	0.99 [0.95,1.00]	1.00 [0.98,1.00]	1.00 [0.98,1.00]	1.00 [0.98,1.00]	1.00 [0.98,1.00]	0.99 [0.96,1.00]	1.00 [1.00,1.00]	0.98 [0.95,1.00]	0.71 [0.53,0.89]
lung_nodule_malignancy	0.74 [0.64,0.83]	0.73 [0.62,0.82]	0.71 [0.60,0.80]	0.71 [0.61,0.81]	0.70 [0.60,0.81]	0.69 [0.58,0.80]	0.65 [0.54,0.76]	0.74 [0.63,0.83]	0.72 [0.61,0.83]	0.72 [0.61,0.81]	0.54 [0.44,0.63]

CVPR26, SPECTRE

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ABMIL	DoRA	rsLoRA	Lin→LoRA	Attn→LoRA	Attn-only FT	Full FT
SPECTRE, joint (26 000 steps)											
adrenal_hyperplasia	0.73 [0.53,0.90]	0.44 [0.24,0.66]	0.70 [0.52,0.88]	0.72 [0.53,0.89]	0.71 [0.52,0.87]	0.64 [0.44,0.84]	0.65 [0.45,0.83]	0.70 [0.50,0.86]	0.67 [0.49,0.85]	0.74 [0.55,0.90]	0.29 [0.15,0.46]
ascites	0.88 [0.76,0.99]	0.89 [0.77,0.98]	0.85 [0.70,0.97]	0.88 [0.75,0.98]	0.89 [0.76,0.98]	0.87 [0.73,0.98]	0.87 [0.74,0.97]	0.85 [0.69,0.96]	0.86 [0.71,0.96]	0.87 [0.72,0.97]	0.49 [0.31,0.69]
atherosclerosis	0.83 [0.67,0.96]	0.74 [0.55,0.89]	0.82 [0.67,0.95]	0.79 [0.63,0.92]	0.72 [0.54,0.88]	0.77 [0.59,0.92]	0.81 [0.65,0.94]	0.79 [0.63,0.93]	0.85 [0.69,0.96]	0.81 [0.64,0.94]	0.67 [0.47,0.84]
cholecystitis	0.76 [0.59,0.90]	0.70 [0.52,0.85]	0.77 [0.61,0.90]	0.78 [0.65,0.91]	0.76 [0.60,0.89]	0.71 [0.54,0.85]	0.76 [0.62,0.90]	0.62 [0.43,0.80]	0.72 [0.56,0.87]	0.66 [0.49,0.82]	0.47 [0.33,0.60]
colorectal_cancer	0.71 [0.51,0.89]	0.58 [0.35,0.83]	0.64 [0.42,0.83]	0.74 [0.54,0.91]	0.73 [0.50,0.91]	0.71 [0.51,0.90]	0.77 [0.57,0.93]	0.81 [0.61,0.96]	0.77 [0.57,0.93]	0.67 [0.44,0.88]	0.70 [0.49,0.88]
fatty_liver	0.82 [0.69,0.96]	0.64 [0.46,0.81]	0.90 [0.76,0.99]	0.92 [0.82,1.00]	0.92 [0.82,0.99]	0.91 [0.79,0.99]	0.88 [0.73,0.99]	0.84 [0.70,0.96]	0.85 [0.69,0.97]	0.84 [0.68,0.96]	0.79 [0.64,0.92]
gallstone	0.70 [0.56,0.83]	0.78 [0.65,0.90]	0.66 [0.51,0.79]	0.69 [0.55,0.82]	0.77 [0.64,0.88]	0.67 [0.52,0.80]	0.66 [0.53,0.80]	0.69 [0.54,0.81]	0.69 [0.55,0.81]	0.71 [0.56,0.83]	0.60 [0.45,0.72]
hydronephrosis	0.81 [0.65,0.95]	0.77 [0.60,0.92]	0.84 [0.69,0.95]	0.88 [0.74,0.97]	0.83 [0.68,0.96]	0.78 [0.60,0.92]	0.85 [0.70,0.96]	0.83 [0.69,0.95]	0.89 [0.76,0.98]	0.82 [0.65,0.95]	0.56 [0.35,0.75]
kidney_stone	0.61 [0.47,0.73]	0.68 [0.55,0.80]	0.71 [0.58,0.81]	0.69 [0.57,0.80]	0.70 [0.57,0.80]	0.73 [0.62,0.84]	0.74 [0.63,0.84]	0.72 [0.60,0.83]	0.73 [0.61,0.83]	0.69 [0.57,0.80]	0.54 [0.45,0.63]
liver_calcifications	0.51 [0.32,0.70]	0.74 [0.57,0.90]	0.54 [0.36,0.73]	0.51 [0.33,0.69]	0.52 [0.32,0.70]	0.60 [0.42,0.78]	0.56 [0.37,0.75]	0.54 [0.34,0.74]	0.63 [0.46,0.80]	0.59 [0.39,0.79]	0.52 [0.34,0.70]
liver_cyst	0.64 [0.55,0.74]	0.64 [0.54,0.74]	0.72 [0.62,0.81]	0.71 [0.61,0.79]	0.69 [0.59,0.78]	0.70 [0.61,0.79]	0.70 [0.60,0.79]	0.67 [0.57,0.76]	0.68 [0.58,0.77]	0.68 [0.58,0.77]	0.43 [0.33,0.52]
liver_lesion	0.62 [0.48,0.75]	0.62 [0.48,0.76]	0.68 [0.53,0.80]	0.67 [0.54,0.80]	0.67 [0.53,0.80]	0.66 [0.52,0.80]	0.66 [0.53,0.79]	0.66 [0.51,0.79]	0.62 [0.47,0.76]	0.68 [0.54,0.80]	0.62 [0.48,0.75]
lymphadenopathy	0.76 [0.58,0.90]	0.73 [0.57,0.88]	0.82 [0.68,0.94]	0.75 [0.59,0.90]	0.75 [0.57,0.89]	0.80 [0.63,0.94]	0.79 [0.63,0.92]	0.76 [0.58,0.90]	0.78 [0.61,0.91]	0.80 [0.63,0.94]	0.47 [0.28,0.66]
renal_cyst	0.61 [0.52,0.71]	0.64 [0.54,0.73]	0.65 [0.56,0.75]	0.69 [0.60,0.78]	0.67 [0.58,0.76]	0.70 [0.62,0.79]	0.68 [0.59,0.78]	0.60 [0.50,0.69]	0.66 [0.56,0.75]	0.68 [0.59,0.77]	0.52 [0.44,0.59]
splenomegaly	0.84 [0.71,0.95]	0.78 [0.62,0.90]	0.87 [0.75,0.96]	0.82 [0.68,0.94]	0.82 [0.68,0.93]	0.83 [0.70,0.94]	0.88 [0.77,0.97]	0.83 [0.68,0.94]	0.80 [0.65,0.93]	0.87 [0.74,0.96]	0.50 [0.33,0.68]
covid	1.00 [0.98,1.00]	0.99 [0.96,1.00]	1.00 [0.98,1.00]	0.99 [0.96,1.00]	0.99 [0.96,1.00]	0.99 [0.95,1.00]	0.99 [0.95,1.00]	1.00 [1.00,1.00]	1.00 [0.98,1.00]	0.98 [0.95,1.00]	0.95 [0.84,1.00]
lung_nodule_malignancy	0.79 [0.70,0.87]	0.74 [0.64,0.82]	0.68 [0.58,0.78]	0.74 [0.63,0.84]	0.77 [0.66,0.86]	0.70 [0.60,0.80]	0.70 [0.60,0.81]	0.77 [0.67,0.86]	0.69 [0.58,0.80]	0.76 [0.66,0.85]	0.67 [0.57,0.77]

FDG-PET/CT

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ABMIL	DoRA	rsLoRA	Lin→LoRA	Attn→LoRA	Attn-only FT	Full FT
TAP-CT, per-task											
LUNG_CANCER	0.93 [0.87,0.98]	0.93 [0.87,0.97]	0.92 [0.86,0.97]	0.93 [0.88,0.97]	0.86 [0.76,0.93]	0.90 [0.82,0.96]	0.90 [0.83,0.96]	0.92 [0.86,0.97]	0.91 [0.85,0.96]	0.94 [0.88,0.98]	0.86 [0.77,0.93]
LYMPHOMA	0.93 [0.86,0.99]	0.89 [0.78,0.96]	0.93 [0.86,0.98]	0.89 [0.80,0.96]	0.92 [0.84,0.99]	0.93 [0.86,0.99]	0.94 [0.86,0.99]	0.93 [0.85,0.99]	0.96 [0.91,0.99]	0.94 [0.85,1.00]	0.80 [0.67,0.90]
MELANOMA	0.81 [0.72,0.89]	0.76 [0.64,0.86]	0.79 [0.68,0.90]	0.83 [0.74,0.90]	0.81 [0.71,0.90]	0.82 [0.73,0.89]	0.76 [0.63,0.88]	0.84 [0.75,0.90]	0.81 [0.72,0.90]	0.77 [0.65,0.88]	0.72 [0.58,0.85]
NEGATIVE	0.78 [0.69,0.86]	0.77 [0.68,0.85]	0.77 [0.68,0.85]	0.81 [0.73,0.89]	0.77 [0.68,0.85]	0.73 [0.64,0.82]	0.67 [0.57,0.76]	0.77 [0.68,0.85]	0.75 [0.65,0.83]	0.68 [0.59,0.77]	0.70 [0.60,0.79]
TAP-CT, joint											
LUNG_CANCER	0.93 [0.88,0.98]	0.92 [0.85,0.97]	0.94 [0.89,0.98]	0.94 [0.90,0.98]	0.89 [0.80,0.97]	0.93 [0.86,0.97]	0.81 [0.72,0.89]	0.93 [0.87,0.97]	0.92 [0.85,0.97]	0.93 [0.88,0.97]	0.57 [0.44,0.70]
LYMPHOMA	0.94 [0.85,1.00]	0.92 [0.82,0.99]	0.92 [0.83,0.98]	0.95 [0.87,1.00]	0.93 [0.82,1.00]	0.93 [0.84,0.99]	0.91 [0.81,0.97]	0.93 [0.85,0.99]	0.96 [0.92,1.00]	0.95 [0.88,1.00]	0.58 [0.47,0.68]
MELANOMA	0.80 [0.70,0.88]	0.79 [0.69,0.87]	0.79 [0.69,0.89]	0.80 [0.71,0.88]	0.81 [0.72,0.89]	0.81 [0.71,0.89]	0.78 [0.66,0.87]	0.81 [0.71,0.89]	0.78 [0.66,0.88]	0.80 [0.67,0.90]	0.71 [0.56,0.84]
NEGATIVE	0.76 [0.67,0.84]	0.81 [0.73,0.88]	0.81 [0.72,0.88]	0.76 [0.68,0.84]	0.76 [0.67,0.84]	0.76 [0.68,0.84]	0.65 [0.55,0.74]	0.77 [0.68,0.84]	0.79 [0.70,0.86]	0.80 [0.71,0.87]	0.58 [0.47,0.69]
SPECTRE, per-task											
LUNG_CANCER	0.93 [0.87,0.97]	0.94 [0.90,0.98]	0.95 [0.90,0.98]	0.96 [0.92,0.99]	0.96 [0.92,0.99]	0.96 [0.92,0.99]	0.95 [0.91,0.98]	0.96 [0.92,0.99]	0.96 [0.92,0.99]	0.97 [0.94,0.99]	0.80 [0.70,0.89]
LYMPHOMA	0.94 [0.85,0.99]	0.92 [0.84,0.98]	0.96 [0.89,1.00]	0.97 [0.91,1.00]	0.97 [0.91,1.00]	0.97 [0.91,1.00]	0.94 [0.86,0.99]	0.95 [0.87,0.99]	0.95 [0.88,1.00]	0.97 [0.91,1.00]	0.50 [0.50,0.50]
MELANOMA	0.72 [0.60,0.84]	0.78 [0.65,0.88]	0.79 [0.68,0.89]	0.78 [0.66,0.88]	0.74 [0.61,0.86]	0.79 [0.69,0.89]	0.75 [0.63,0.86]	0.80 [0.70,0.88]	0.77 [0.65,0.87]	0.81 [0.71,0.89]	0.63 [0.50,0.75]
NEGATIVE	0.84 [0.77,0.91]	0.81 [0.73,0.88]	0.83 [0.74,0.90]	0.81 [0.72,0.88]	0.83 [0.75,0.90]	0.84 [0.77,0.91]	0.85 [0.77,0.92]	0.84 [0.76,0.91]	0.79 [0.71,0.87]	0.65 [0.55,0.74]	0.53 [0.44,0.63]
SPECTRE, joint											
LUNG_CANCER	0.97 [0.95,0.99]	0.97 [0.93,0.99]	0.99 [0.97,1.00]	0.98 [0.95,1.00]	0.99 [0.96,1.00]	0.99 [0.97,1.00]	0.99 [0.97,1.00]	0.96 [0.92,0.99]	0.98 [0.94,1.00]	0.99 [0.97,1.00]	0.76 [0.64,0.86]
LYMPHOMA	0.93 [0.84,0.99]	0.95 [0.87,0.99]	0.96 [0.90,1.00]	0.96 [0.91,1.00]	0.95 [0.88,1.00]	0.96 [0.91,1.00]	0.96 [0.91,1.00]	0.95 [0.87,0.99]	0.94 [0.87,0.99]	0.94 [0.88,0.99]	0.69 [0.58,0.80]
MELANOMA	0.80 [0.69,0.90]	0.77 [0.65,0.88]	0.82 [0.73,0.90]	0.78 [0.68,0.88]	0.80 [0.69,0.89]	0.82 [0.73,0.90]	0.81 [0.71,0.91]	0.82 [0.72,0.90]	0.77 [0.66,0.87]	0.85 [0.75,0.93]	0.60 [0.45,0.75]
NEGATIVE	0.83 [0.75,0.90]	0.84 [0.76,0.90]	0.83 [0.75,0.90]	0.82 [0.74,0.90]	0.82 [0.73,0.90]	0.83 [0.75,0.90]	0.84 [0.75,0.91]	0.82 [0.74,0.89]	0.84 [0.76,0.91]	0.85 [0.77,0.91]	0.64 [0.53,0.74]

RSNA 2022 C-spine

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ABMIL	DoRA	rsLoRA	Lin→LoRA	Attn→LoRA	Attn-only FT	Full FT
TAP-CT, per-task											
C1	0.67 [0.54,0.79]	0.71 [0.59,0.83]	0.68 [0.57,0.79]	0.72 [0.60,0.83]	0.70 [0.58,0.82]	0.72 [0.59,0.84]	0.74 [0.62,0.83]	0.69 [0.56,0.79]	0.71 [0.60,0.83]	0.77 [0.65,0.88]	0.70 [0.57,0.79]
C2	0.71 [0.60,0.80]	0.76 [0.67,0.85]	0.73 [0.62,0.83]	0.76 [0.66,0.84]	0.75 [0.64,0.84]	0.74 [0.64,0.84]	0.72 [0.62,0.82]	0.70 [0.59,0.80]	0.73 [0.62,0.82]	0.77 [0.66,0.86]	0.70 [0.58,0.79]
C3	0.64 [0.42,0.84]	0.55 [0.24,0.84]	0.40 [0.17,0.63]	0.40 [0.19,0.59]	0.46 [0.20,0.71]	0.55 [0.25,0.85]	0.54 [0.23,0.88]	0.51 [0.29,0.76]	0.53 [0.22,0.82]	0.59 [0.30,0.89]	0.53 [0.24,0.84]
C4	0.64 [0.40,0.84]	0.58 [0.38,0.75]	0.56 [0.34,0.78]	0.62 [0.41,0.79]	0.51 [0.31,0.71]	0.53 [0.33,0.73]	0.51 [0.32,0.72]	0.64 [0.41,0.85]	0.56 [0.39,0.75]	0.52 [0.35,0.70]	0.66 [0.48,0.82]
C5	0.51 [0.36,0.67]	0.53 [0.38,0.68]	0.59 [0.46,0.72]	0.50 [0.34,0.66]	0.44 [0.31,0.58]	0.51 [0.36,0.66]	0.60 [0.45,0.74]	0.49 [0.34,0.66]	0.60 [0.46,0.74]	0.54 [0.41,0.66]	0.57 [0.43,0.68]
C6	0.50 [0.37,0.61]	0.62 [0.51,0.73]	0.66 [0.55,0.76]	0.59 [0.48,0.69]	0.66 [0.55,0.77]	0.65 [0.56,0.75]	0.64 [0.53,0.74]	0.50 [0.38,0.63]	0.64 [0.54,0.74]	0.65 [0.54,0.77]	0.68 [0.56,0.78]
C7	0.63 [0.53,0.73]	0.57 [0.46,0.67]	0.68 [0.58,0.77]	0.61 [0.50,0.72]	0.63 [0.53,0.72]	0.66 [0.57,0.75]	0.65 [0.55,0.74]	0.63 [0.53,0.73]	0.59 [0.48,0.68]	0.66 [0.55,0.75]	0.65 [0.54,0.74]
patient_overall	0.65 [0.57,0.73]	0.68 [0.60,0.75]	0.68 [0.61,0.75]	0.67 [0.59,0.74]	0.75 [0.68,0.82]	0.76 [0.70,0.83]	0.68 [0.61,0.75]	0.68 [0.61,0.75]	0.67 [0.59,0.74]	0.78 [0.72,0.84]	0.62 [0.55,0.69]
TAP-CT, joint											
C1	0.64 [0.51,0.76]	0.71 [0.59,0.82]	0.72 [0.60,0.84]	0.75 [0.63,0.85]	0.72 [0.60,0.83]	0.71 [0.58,0.83]	0.68 [0.56,0.80]	0.62 [0.49,0.75]	0.71 [0.59,0.83]	0.70 [0.57,0.82]	0.76 [0.64,0.86]
C2	0.76 [0.66,0.85]	0.73 [0.63,0.83]	0.76 [0.67,0.85]	0.75 [0.65,0.85]	0.77 [0.67,0.85]	0.74 [0.62,0.83]	0.70 [0.60,0.80]	0.74 [0.65,0.85]	0.72 [0.61,0.82]	0.79 [0.71,0.87]	0.81 [0.72,0.89]
C3	0.55 [0.28,0.84]	0.45 [0.21,0.72]	0.63 [0.35,0.89]	0.69 [0.42,0.91]	0.69 [0.43,0.92]	0.59 [0.32,0.85]	0.47 [0.16,0.80]	0.59 [0.34,0.84]	0.43 [0.20,0.67]	0.63 [0.28,0.97]	0.62 [0.28,0.95]
C4	0.69 [0.46,0.89]	0.65 [0.44,0.84]	0.74 [0.53,0.91]	0.71 [0.51,0.88]	0.70 [0.51,0.86]	0.71 [0.53,0.88]	0.53 [0.32,0.73]	0.67 [0.46,0.85]	0.65 [0.46,0.82]	0.73 [0.50,0.91]	0.78 [0.59,0.92]
C5	0.52 [0.36,0.68]	0.56 [0.41,0.70]	0.69 [0.57,0.79]	0.65 [0.53,0.76]	0.68 [0.57,0.79]	0.63 [0.51,0.75]	0.56 [0.45,0.68]	0.51 [0.36,0.66]	0.57 [0.41,0.70]	0.69 [0.57,0.80]	0.69 [0.57,0.81]
C6	0.54 [0.42,0.65]	0.58 [0.46,0.69]	0.71 [0.61,0.80]	0.68 [0.58,0.78]	0.71 [0.62,0.81]	0.68 [0.58,0.79]	0.69 [0.57,0.80]	0.53 [0.41,0.66]	0.57 [0.47,0.68]	0.68 [0.57,0.78]	0.69 [0.58,0.80]
C7	0.64 [0.53,0.74]	0.66 [0.57,0.75]	0.71 [0.62,0.80]	0.70 [0.61,0.78]	0.70 [0.62,0.77]	0.68 [0.59,0.77]	0.58 [0.49,0.68]	0.65 [0.54,0.74]	0.64 [0.55,0.73]	0.72 [0.64,0.79]	0.73 [0.64,0.81]
patient_overall	0.67 [0.60,0.75]	0.65 [0.58,0.73]	0.75 [0.68,0.81]	0.72 [0.65,0.79]	0.73 [0.66,0.80]	0.70 [0.62,0.77]	0.59 [0.52,0.67]	0.68 [0.61,0.75]	0.67 [0.60,0.74]	0.75 [0.69,0.81]	0.77 [0.70,0.83]
SPECTRE, per-task											
C1	0.55 [0.42,0.67]	0.56 [0.45,0.69]	0.68 [0.56,0.79]	0.64 [0.53,0.75]	0.61 [0.48,0.73]	0.59 [0.47,0.71]	0.59 [0.46,0.72]	0.66 [0.54,0.76]	0.61 [0.48,0.73]	0.54 [0.41,0.66]	0.58 [0.43,0.72]
C2	0.67 [0.57,0.77]	0.62 [0.50,0.72]	0.69 [0.60,0.78]	0.73 [0.62,0.82]	0.77 [0.67,0.86]	0.67 [0.57,0.77]	0.77 [0.68,0.86]	0.74 [0.65,0.83]	0.64 [0.53,0.74]	0.66 [0.56,0.75]	0.68 [0.57,0.78]
C3	0.51 [0.39,0.64]	0.72 [0.54,0.89]	0.77 [0.61,0.89]	0.82 [0.61,0.95]	0.60 [0.29,0.88]	0.71 [0.54,0.84]	0.63 [0.46,0.79]	0.57 [0.35,0.73]	0.43 [0.11,0.78]	0.71 [0.46,0.88]	0.56 [0.35,0.77]
C4	0.65 [0.44,0.84]	0.35 [0.16,0.54]	0.45 [0.25,0.64]	0.54 [0.41,0.69]	0.61 [0.48,0.74]	0.44 [0.26,0.64]	0.64 [0.47,0.78]	0.47 [0.33,0.60]	0.58 [0.38,0.76]	0.47 [0.29,0.64]	0.48 [0.35,0.64]
C5	0.56 [0.43,0.68]	0.60 [0.48,0.72]	0.46 [0.34,0.58]	0.59 [0.48,0.69]	0.51 [0.39,0.64]	0.54 [0.40,0.67]	0.49 [0.35,0.63]	0.53 [0.43,0.64]	0.49 [0.38,0.61]	0.56 [0.42,0.68]	0.55 [0.42,0.67]
C6	0.55 [0.44,0.66]	0.61 [0.50,0.72]	0.67 [0.56,0.76]	0.58 [0.47,0.67]	0.67 [0.56,0.77]	0.68 [0.57,0.79]	0.61 [0.50,0.72]	0.63 [0.51,0.74]	0.61 [0.50,0.72]	0.70 [0.59,0.81]	0.58 [0.47,0.69]
C7	0.55 [0.44,0.65]	0.62 [0.52,0.71]	0.65 [0.56,0.74]	0.66 [0.55,0.75]	0.66 [0.56,0.76]	0.67 [0.57,0.77]	0.68 [0.58,0.78]	0.64 [0.54,0.75]	0.67 [0.58,0.76]	0.66 [0.56,0.75]	0.52 [0.50,0.54]
patient_overall	0.60 [0.52,0.68]	0.64 [0.56,0.72]	0.72 [0.65,0.78]	0.68 [0.60,0.75]	0.72 [0.64,0.79]	0.66 [0.58,0.73]	0.70 [0.62,0.77]	0.68 [0.62,0.76]	0.71 [0.63,0.78]	0.78 [0.72,0.85]	0.58 [0.50,0.66]
SPECTRE, joint											
C1	0.65 [0.52,0.77]	0.53 [0.41,0.65]	0.73 [0.61,0.85]	0.70 [0.58,0.81]	0.72 [0.58,0.83]	0.72 [0.59,0.83]	0.72 [0.59,0.83]	0.70 [0.59,0.79]	0.69 [0.56,0.79]	0.73 [0.62,0.83]	0.64 [0.50,0.76]
C2	0.67 [0.56,0.77]	0.67 [0.56,0.76]	0.77 [0.69,0.86]	0.75 [0.66,0.84]	0.78 [0.70,0.86]	0.75 [0.66,0.83]	0.76 [0.67,0.85]	0.77 [0.67,0.85]	0.74 [0.64,0.83]	0.75 [0.66,0.85]	0.68 [0.58,0.78]
C3	0.66 [0.46,0.86]	0.56 [0.30,0.81]	0.65 [0.41,0.89]	0.51 [0.38,0.65]	0.68 [0.44,0.91]	0.63 [0.42,0.82]	0.63 [0.35,0.91]	0.72 [0.54,0.87]	0.52 [0.40,0.65]	0.59 [0.31,0.86]	0.46 [0.18,0.75]
C4	0.63 [0.50,0.75]	0.61 [0.39,0.79]	0.75 [0.58,0.87]	0.68 [0.57,0.78]	0.81 [0.68,0.92]	0.74 [0.58,0.86]	0.70 [0.54,0.86]	0.63 [0.49,0.75]	0.58 [0.46,0.70]	0.75 [0.56,0.90]	0.61 [0.46,0.75]
C5	0.60 [0.48,0.72]	0.57 [0.47,0.67]	0.62 [0.51,0.71]	0.55 [0.45,0.65]	0.69 [0.59,0.77]	0.63 [0.54,0.72]	0.59 [0.47,0.71]	0.59 [0.48,0.69]	0.47 [0.38,0.56]	0.70 [0.58,0.83]	0.46 [0.33,0.62]
C6	0.64 [0.54,0.74]	0.59 [0.48,0.70]	0.68 [0.56,0.79]	0.63 [0.52,0.74]	0.68 [0.58,0.78]	0.67 [0.55,0.79]	0.69 [0.57,0.79]	0.63 [0.52,0.74]	0.61 [0.51,0.71]	0.74 [0.63,0.83]	0.48 [0.36,0.61]
C7	0.60 [0.51,0.70]	0.65 [0.56,0.74]	0.71 [0.63,0.80]	0.66 [0.55,0.75]	0.74 [0.64,0.82]	0.69 [0.58,0.79]	0.71 [0.61,0.80]	0.64 [0.54,0.74]	0.62 [0.53,0.72]	0.72 [0.64,0.81]	0.52 [0.42,0.62]
patient_overall	0.58 [0.50,0.66]	0.63 [0.56,0.70]	0.74 [0.67,0.80]	0.67 [0.60,0.74]	0.74 [0.68,0.81]	0.71 [0.64,0.78]	0.72 [0.65,0.79]	0.68 [0.61,0.75]	0.66 [0.59,0.73]	0.75 [0.68,0.82]	0.59 [0.51,0.66]

RSNA 2023 trauma

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ABMIL	DoRA	rsLoRA	Lin→LoRA	Attn→LoRA	Attn-only FT	Full FT
TAP-CT, per-task											
any_injury	0.75 [0.69,0.81]	0.78 [0.72,0.83]	0.77 [0.71,0.82]	0.76 [0.71,0.81]	0.78 [0.73,0.83]	0.77 [0.71,0.82]	0.71 [0.65,0.77]	0.76 [0.70,0.81]	0.75 [0.69,0.80]	0.88 [0.84,0.92]	0.87 [0.82,0.91]
kidney_injury	0.70 [0.59,0.81]	0.64 [0.52,0.75]	0.71 [0.59,0.81]	0.69 [0.55,0.82]	0.65 [0.52,0.78]	0.80 [0.66,0.90]	0.86 [0.76,0.95]	0.67 [0.55,0.78]	0.69 [0.57,0.80]	0.88 [0.76,0.96]	0.79 [0.69,0.89]
liver_injury	0.78 [0.69,0.85]	0.77 [0.70,0.83]	0.76 [0.67,0.83]	0.80 [0.73,0.86]	0.78 [0.71,0.84]	0.78 [0.72,0.85]	0.74 [0.66,0.81]	0.77 [0.69,0.85]	0.79 [0.72,0.85]	0.79 [0.72,0.85]	0.69 [0.60,0.76]
spleen_injury	0.72 [0.64,0.79]	0.69 [0.60,0.77]	0.71 [0.64,0.79]	0.77 [0.69,0.85]	0.60 [0.52,0.69]	0.66 [0.57,0.74]	0.59 [0.50,0.68]	0.75 [0.67,0.83]	0.64 [0.54,0.74]	0.89 [0.83,0.94]	0.75 [0.67,0.83]
TAP-CT, joint											
any_injury	0.75 [0.70,0.81]	0.76 [0.70,0.81]	0.77 [0.72,0.82]	0.75 [0.70,0.81]	0.76 [0.70,0.82]	0.79 [0.73,0.84]	0.69 [0.63,0.75]	0.76 [0.71,0.81]	0.77 [0.71,0.82]	0.84 [0.80,0.89]	0.87 [0.83,0.91]
kidney_injury	0.72 [0.59,0.82]	0.70 [0.57,0.81]	0.74 [0.61,0.85]	0.71 [0.58,0.84]	0.78 [0.67,0.86]	0.75 [0.64,0.85]	0.71 [0.61,0.80]	0.70 [0.59,0.80]	0.76 [0.63,0.86]	0.84 [0.76,0.92]	0.96 [0.93,0.99]
liver_injury	0.80 [0.73,0.86]	0.79 [0.71,0.86]	0.81 [0.75,0.87]	0.81 [0.74,0.86]	0.81 [0.74,0.88]	0.79 [0.72,0.86]	0.72 [0.65,0.79]	0.80 [0.74,0.87]	0.80 [0.73,0.86]	0.84 [0.79,0.90]	0.83 [0.76,0.89]
spleen_injury	0.68 [0.60,0.76]	0.71 [0.64,0.79]	0.71 [0.62,0.79]	0.71 [0.62,0.78]	0.68 [0.60,0.76]	0.74 [0.66,0.82]	0.62 [0.53,0.69]	0.72 [0.64,0.80]	0.71 [0.63,0.80]	0.81 [0.73,0.88]	0.85 [0.80,0.90]
SPECTRE, per-task											
any_injury	0.80 [0.75,0.86]	0.79 [0.74,0.84]	0.86 [0.81,0.90]	0.86 [0.82,0.90]	0.87 [0.84,0.91]	0.85 [0.80,0.89]	0.86 [0.81,0.90]	0.84 [0.79,0.89]	0.85 [0.80,0.89]	0.85 [0.81,0.90]	0.57 [0.51,0.64]
kidney_injury	0.85 [0.74,0.93]	0.83 [0.73,0.90]	0.85 [0.74,0.93]	0.90 [0.82,0.96]	0.91 [0.82,0.97]	0.90 [0.80,0.96]	0.86 [0.76,0.93]	0.87 [0.78,0.93]	0.79 [0.68,0.88]	0.74 [0.63,0.83]	0.63 [0.60,0.65]
liver_injury	0.73 [0.65,0.80]	0.73 [0.65,0.81]	0.77 [0.70,0.83]	0.75 [0.68,0.83]	0.77 [0.68,0.86]	0.77 [0.69,0.85]	0.78 [0.69,0.85]	0.77 [0.70,0.83]	0.77 [0.69,0.84]	0.70 [0.59,0.79]	0.74 [0.66,0.80]
spleen_injury	0.78 [0.70,0.85]	0.76 [0.67,0.84]	0.82 [0.75,0.89]	0.81 [0.73,0.89]	0.85 [0.77,0.91]	0.85 [0.79,0.91]	0.85 [0.79,0.91]	0.83 [0.75,0.89]	0.78 [0.69,0.86]	0.80 [0.73,0.87]	0.56 [0.47,0.63]
SPECTRE, joint											
any_injury	0.80 [0.74,0.85]	0.79 [0.74,0.85]	0.87 [0.82,0.91]	0.85 [0.81,0.90]	0.87 [0.83,0.91]	0.86 [0.81,0.90]	0.87 [0.83,0.91]	0.85 [0.80,0.89]	0.83 [0.77,0.87]	0.82 [0.76,0.87]	0.65 [0.58,0.71]
kidney_injury	0.84 [0.71,0.92]	0.83 [0.72,0.91]	0.88 [0.81,0.94]	0.88 [0.79,0.94]	0.91 [0.83,0.96]	0.86 [0.78,0.92]	0.86 [0.79,0.92]	0.87 [0.79,0.94]	0.85 [0.77,0.93]	0.87 [0.80,0.93]	0.64 [0.53,0.77]
liver_injury	0.73 [0.65,0.80]	0.74 [0.66,0.82]	0.80 [0.71,0.88]	0.79 [0.72,0.85]	0.81 [0.73,0.88]	0.80 [0.72,0.87]	0.82 [0.75,0.88]	0.78 [0.70,0.85]	0.78 [0.70,0.85]	0.74 [0.65,0.82]	0.74 [0.65,0.81]
spleen_injury	0.79 [0.72,0.87]	0.79 [0.71,0.86]	0.84 [0.77,0.90]	0.82 [0.74,0.88]	0.84 [0.78,0.90]	0.84 [0.77,0.90]	0.85 [0.79,0.91]	0.81 [0.73,0.88]	0.79 [0.69,0.86]	0.78 [0.69,0.86]	0.59 [0.50,0.67]

RSNA 2019 ICH

Label	Lin	Attn	LoRA	$\text{LoRA}_{\text{late}}$	LoRA+ABMIL	DoRA	rsLoRA	Lin→LoRA	Attn→LoRA	Attn-only FT	Full FT
TAP-CT, per-task											
any	0.92 [0.91,0.93]	0.92 [0.91,0.93]	0.95 [0.94,0.95]	0.94 [0.93,0.95]	0.93 [0.92,0.94]	0.94 [0.93,0.95]	0.92 [0.91,0.93]	0.92 [0.91,0.93]	0.92 [0.91,0.93]	0.95 [0.94,0.95]	0.94 [0.93,0.95]
epidural	0.81 [0.76,0.86]	0.80 [0.75,0.85]	0.78 [0.73,0.83]	0.79 [0.74,0.83]	0.78 [0.73,0.82]	0.82 [0.77,0.86]	0.76 [0.71,0.82]	0.82 [0.77,0.87]	0.79 [0.74,0.84]	0.84 [0.79,0.88]	0.87 [0.83,0.90]
intraparenchymal	0.87 [0.86,0.89]	0.88 [0.87,0.89]	0.93 [0.92,0.94]	0.93 [0.91,0.94]	0.92 [0.90,0.93]	0.92 [0.91,0.93]	0.92 [0.91,0.93]	0.89 [0.88,0.90]	0.90 [0.89,0.91]	0.94 [0.93,0.95]	0.88 [0.87,0.90]
intraventricular	0.93 [0.92,0.94]	0.94 [0.93,0.95]	0.94 [0.94,0.95]	0.95 [0.94,0.96]	0.95 [0.94,0.96]	0.95 [0.94,0.96]	0.93 [0.92,0.94]	0.93 [0.92,0.94]	0.94 [0.93,0.95]	0.96 [0.96,0.97]	0.95 [0.94,0.96]
subarachnoid	0.85 [0.84,0.87]	0.86 [0.84,0.87]	0.88 [0.86,0.89]	0.88 [0.87,0.90]	0.88 [0.87,0.90]	0.88 [0.86,0.89]	0.86 [0.84,0.87]	0.86 [0.84,0.87]	0.86 [0.85,0.88]	0.87 [0.85,0.89]	0.88 [0.86,0.89]
subdural	0.84 [0.82,0.86]	0.84 [0.82,0.86]	0.89 [0.87,0.90]	0.87 [0.86,0.89]	0.88 [0.87,0.90]	0.87 [0.85,0.89]	0.83 [0.81,0.85]	0.84 [0.83,0.86]	0.85 [0.83,0.86]	0.91 [0.89,0.92]	0.81 [0.79,0.83]
TAP-CT, joint											
any	0.91 [0.90,0.92]	0.92 [0.91,0.93]	0.95 [0.94,0.95]	0.94 [0.93,0.95]	0.94 [0.93,0.95]	0.94 [0.93,0.95]	0.94 [0.93,0.95]	0.93 [0.92,0.94]	0.91 [0.90,0.92]	0.92 [0.91,0.93]	0.95 [0.95,0.96]	0.91 [0.90,0.92]
epidural	0.81 [0.76,0.86]	0.83 [0.78,0.87]	0.88 [0.84,0.91]	0.86 [0.82,0.90]	0.82 [0.78,0.87]	0.87 [0.83,0.91]	0.79 [0.75,0.83]	0.80 [0.75,0.85]	0.82 [0.77,0.86]	0.89 [0.85,0.92]	0.75 [0.70,0.79]
intraparenchymal	0.87 [0.86,0.89]	0.88 [0.87,0.89]	0.92 [0.91,0.93]	0.91 [0.90,0.92]	0.91 [0.90,0.92]	0.91 [0.90,0.92]	0.90 [0.89,0.91]	0.88 [0.86,0.89]	0.88 [0.87,0.90]	0.92 [0.91,0.93]	0.91 [0.89,0.92]
intraventricular	0.93 [0.91,0.94]	0.93 [0.92,0.94]	0.95 [0.94,0.96]	0.94 [0.94,0.95]	0.93 [0.92,0.94]	0.94 [0.93,0.95]	0.92 [0.91,0.93]	0.92 [0.91,0.93]	0.93 [0.92,0.94]	0.95 [0.94,0.96]	0.93 [0.92,0.94]
subarachnoid	0.85 [0.84,0.87]	0.86 [0.85,0.88]	0.88 [0.86,0.89]	0.87 [0.85,0.88]	0.86 [0.84,0.87]	0.87 [0.85,0.88]	0.87 [0.85,0.88]	0.85 [0.84,0.87]	0.86 [0.85,0.88]	0.88 [0.87,0.89]	0.86 [0.84,0.87]
subdural	0.84 [0.82,0.86]	0.84 [0.82,0.86]	0.87 [0.85,0.88]	0.85 [0.83,0.87]	0.85 [0.83,0.86]	0.85 [0.83,0.86]	0.81 [0.79,0.83]	0.84 [0.82,0.85]	0.83 [0.82,0.85]	0.89 [0.88,0.91]	0.78 [0.76,0.80]
Pillar-0, per-task											
any	0.91 [0.90,0.92]	0.93 [0.92,0.94]	0.96 [0.95,0.96]	0.94 [0.94,0.95]	0.95 [0.94,0.96]	0.95 [0.95,0.96]	0.96 [0.95,0.96]	0.94 [0.94,0.95]	0.94 [0.93,0.95]	0.96 [0.95,0.96]	0.96 [0.95,0.96]
epidural	0.82 [0.77,0.86]	0.77 [0.72,0.82]	0.88 [0.84,0.91]	0.86 [0.81,0.90]	0.81 [0.76,0.85]	0.89 [0.85,0.92]	0.82 [0.77,0.86]	0.86 [0.81,0.90]	0.83 [0.78,0.87]	0.89 [0.85,0.92]	0.82 [0.77,0.87]
intraparenchymal	0.91 [0.90,0.92]	0.92 [0.91,0.93]	0.95 [0.94,0.96]	0.93 [0.92,0.94]	0.93 [0.92,0.94]	0.95 [0.94,0.95]	0.95 [0.94,0.96]	0.93 [0.92,0.94]	0.92 [0.91,0.93]	0.95 [0.94,0.96]	0.94 [0.93,0.95]
intraventricular	0.95 [0.94,0.95]	0.94 [0.93,0.95]	0.96 [0.95,0.97]	0.95 [0.95,0.96]	0.96 [0.95,0.97]	0.96 [0.95,0.96]	0.96 [0.96,0.97]	0.95 [0.94,0.96]	0.95 [0.94,0.96]	0.96 [0.96,0.97]	0.96 [0.95,0.97]
subarachnoid	0.86 [0.85,0.88]	0.86 [0.84,0.88]	0.91 [0.90,0.92]	0.89 [0.87,0.90]	0.84 [0.83,0.86]	0.92 [0.91,0.93]	0.92 [0.90,0.93]	0.88 [0.87,0.90]	0.88 [0.86,0.89]	0.92 [0.90,0.93]	0.85 [0.84,0.87]
subdural	0.87 [0.86,0.89]	0.88 [0.87,0.90]	0.92 [0.90,0.93]	0.91 [0.90,0.92]	0.92 [0.90,0.93]	0.92 [0.91,0.93]	0.92 [0.90,0.93]	0.91 [0.89,0.92]	0.90 [0.89,0.91]	0.91 [0.90,0.93]	0.89 [0.87,0.90]
Pillar-0, joint											
any	0.91 [0.90,0.92]	0.92 [0.91,0.93]	0.95 [0.94,0.96]	0.94 [0.93,0.95]	0.95 [0.95,0.96]	0.95 [0.95,0.96]	0.95 [0.95,0.96]	0.94 [0.93,0.95]	0.93 [0.92,0.94]	0.96 [0.95,0.96]	0.94 [0.93,0.95]
epidural	0.83 [0.78,0.86]	0.82 [0.77,0.86]	0.86 [0.82,0.89]	0.82 [0.77,0.85]	0.88 [0.85,0.92]	0.86 [0.82,0.89]	0.87 [0.84,0.90]	0.85 [0.81,0.89]	0.84 [0.80,0.88]	0.85 [0.81,0.88]	0.78 [0.74,0.82]
intraparenchymal	0.91 [0.90,0.92]	0.91 [0.90,0.92]	0.93 [0.92,0.94]	0.92 [0.91,0.93]	0.94 [0.93,0.95]	0.93 [0.92,0.94]	0.93 [0.92,0.94]	0.92 [0.91,0.93]	0.92 [0.90,0.93]	0.94 [0.93,0.95]	0.92 [0.91,0.93]
intraventricular	0.95 [0.94,0.95]	0.94 [0.93,0.95]	0.94 [0.93,0.95]	0.94 [0.93,0.95]	0.95 [0.94,0.96]	0.94 [0.93,0.95]	0.95 [0.94,0.96]	0.95 [0.94,0.95]	0.94 [0.93,0.95]	0.94 [0.93,0.95]	0.93 [0.92,0.94]
subarachnoid	0.86 [0.85,0.88]	0.86 [0.85,0.88]	0.88 [0.87,0.90]	0.87 [0.85,0.88]	0.90 [0.89,0.92]	0.89 [0.88,0.90]	0.90 [0.88,0.91]	0.88 [0.86,0.89]	0.87 [0.85,0.88]	0.89 [0.87,0.90]	0.87 [0.86,0.89]
subdural	0.87 [0.86,0.89]	0.89 [0.87,0.90]	0.91 [0.89,0.92]	0.87 [0.85,0.88]	0.92 [0.90,0.93]	0.90 [0.89,0.92]	0.91 [0.89,0.92]	0.90 [0.88,0.91]	0.89 [0.87,0.90]	0.91 [0.89,0.92]	0.82 [0.80,0.84]

AMOS22

Structure	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→LoRA
TAP-CT, per-structure							
aorta	0.79 [0.77,0.81]	0.92 [0.91,0.92]	0.89 [0.88,0.90]	0.89 [0.88,0.90]	0.89 [0.88,0.90]	0.92 [0.91,0.93]	0.91 [0.91,0.92]
bladder	0.74 [0.69,0.78]	0.85 [0.82,0.88]	0.86 [0.83,0.88]	0.86 [0.82,0.88]	0.86 [0.83,0.88]	0.86 [0.83,0.89]	0.85 [0.81,0.88]
duodenum	0.53 [0.50,0.56]	0.68 [0.64,0.71]	0.68 [0.65,0.70]	0.67 [0.64,0.70]	0.65 [0.63,0.68]	0.70 [0.66,0.73]	0.67 [0.64,0.70]
esophagus	0.55 [0.51,0.58]	0.74 [0.71,0.76]	0.65 [0.62,0.67]	0.64 [0.62,0.67]	0.65 [0.62,0.68]	0.77 [0.75,0.79]	0.73 [0.70,0.76]
gallbladder	0.55 [0.48,0.61]	0.66 [0.60,0.73]	0.67 [0.60,0.74]	0.66 [0.58,0.73]	0.65 [0.58,0.72]	0.69 [0.61,0.76]	0.68 [0.61,0.73]
left_adrenal	0.38 [0.34,0.41]	0.58 [0.54,0.61]	0.42 [0.39,0.45]	0.41 [0.38,0.44]	0.41 [0.37,0.44]	0.59 [0.55,0.62]	0.57 [0.54,0.60]
left_kidney	0.72 [0.69,0.75]	0.88 [0.84,0.91]	0.91 [0.90,0.93]	0.91 [0.90,0.93]	0.91 [0.89,0.93]	0.92 [0.90,0.93]	0.89 [0.85,0.91]
liver	0.89 [0.87,0.91]	0.95 [0.94,0.96]	0.95 [0.94,0.96]	0.95 [0.94,0.96]	0.95 [0.94,0.96]	0.95 [0.94,0.96]	0.95 [0.93,0.96]
pancreas	0.68 [0.66,0.70]	0.78 [0.76,0.80]	0.78 [0.75,0.80]	0.78 [0.75,0.80]	0.77 [0.75,0.79]	0.78 [0.76,0.80]	0.78 [0.76,0.79]
postcava	0.70 [0.68,0.72]	0.82 [0.81,0.84]	0.77 [0.75,0.79]	0.77 [0.75,0.79]	0.77 [0.75,0.78]	0.84 [0.82,0.85]	0.82 [0.81,0.83]
prostate_uterus	0.73 [0.67,0.77]	0.80 [0.74,0.85]	0.78 [0.72,0.83]	0.76 [0.70,0.82]	0.77 [0.70,0.82]	0.78 [0.73,0.83]	0.79 [0.74,0.84]
right_adrenal	0.33 [0.29,0.37]	0.56 [0.52,0.60]	0.42 [0.39,0.44]	0.45 [0.42,0.47]	0.42 [0.39,0.44]	0.57 [0.54,0.61]	0.56 [0.52,0.60]
right_kidney	0.70 [0.67,0.74]	0.90 [0.87,0.92]	0.92 [0.90,0.94]	0.91 [0.88,0.92]	0.92 [0.91,0.93]	0.92 [0.91,0.93]	0.89 [0.87,0.92]
spleen	0.82 [0.77,0.85]	0.90 [0.87,0.93]	0.92 [0.90,0.94]	0.92 [0.88,0.94]	0.92 [0.90,0.94]	0.91 [0.88,0.93]	0.90 [0.87,0.92]
stomach	0.78 [0.74,0.81]	0.87 [0.84,0.89]	0.88 [0.86,0.90]	0.88 [0.86,0.90]	0.87 [0.85,0.90]	0.87 [0.84,0.90]	0.87 [0.84,0.89]
TAP-CT, joint							
aorta	0.78 [0.76,0.80]	0.90 [0.89,0.91]	0.91 [0.90,0.92]	0.90 [0.89,0.91]	0.91 [0.90,0.91]	0.92 [0.92,0.93]	0.90 [0.89,0.91]
bladder	0.73 [0.68,0.78]	0.81 [0.77,0.85]	0.68 [0.64,0.71]	0.76 [0.70,0.81]	0.75 [0.70,0.79]	0.86 [0.83,0.88]	0.78 [0.72,0.82]
duodenum	0.52 [0.48,0.55]	0.68 [0.65,0.71]	0.68 [0.65,0.71]	0.67 [0.64,0.69]	0.69 [0.65,0.72]	0.72 [0.68,0.75]	0.68 [0.66,0.71]
esophagus	0.52 [0.49,0.56]	0.71 [0.69,0.74]	0.73 [0.70,0.75]	0.66 [0.63,0.68]	0.71 [0.69,0.74]	0.00 [0.00,0.00]	0.00 [0.00,0.00]
gallbladder	0.50 [0.42,0.57]	0.67 [0.60,0.73]	0.65 [0.58,0.72]	0.66 [0.58,0.73]	0.69 [0.62,0.75]	0.70 [0.62,0.77]	0.65 [0.58,0.72]
left_adrenal	0.36 [0.32,0.40]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]
left_kidney	0.71 [0.67,0.74]	0.88 [0.84,0.91]	0.91 [0.89,0.93]	0.91 [0.90,0.92]	0.91 [0.89,0.93]	0.93 [0.91,0.94]	0.88 [0.84,0.91]
liver	0.89 [0.87,0.91]	0.94 [0.93,0.95]	0.93 [0.92,0.94]	0.95 [0.94,0.95]	0.94 [0.93,0.95]	0.95 [0.94,0.96]	0.94 [0.93,0.95]
pancreas	0.67 [0.64,0.69]	0.77 [0.75,0.79]	0.77 [0.75,0.79]	0.76 [0.74,0.77]	0.77 [0.75,0.79]	0.79 [0.76,0.81]	0.77 [0.75,0.78]
postcava	0.70 [0.68,0.71]	0.82 [0.81,0.83]	0.83 [0.81,0.84]	0.76 [0.74,0.77]	0.82 [0.81,0.84]	0.84 [0.83,0.86]	0.82 [0.80,0.83]
prostate_uterus	0.72 [0.67,0.76]	0.77 [0.70,0.81]	0.71 [0.65,0.76]	0.77 [0.73,0.81]	0.72 [0.66,0.77]	0.79 [0.73,0.84]	0.76 [0.71,0.81]
right_adrenal	0.34 [0.30,0.37]	0.00 [0.00,0.00]	0.57 [0.53,0.60]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]
right_kidney	0.68 [0.64,0.71]	0.89 [0.86,0.91]	0.92 [0.90,0.93]	0.91 [0.90,0.92]	0.91 [0.89,0.93]	0.93 [0.91,0.94]	0.89 [0.87,0.91]
spleen	0.81 [0.76,0.84]	0.91 [0.87,0.93]	0.92 [0.89,0.94]	0.92 [0.90,0.93]	0.92 [0.90,0.94]	0.93 [0.90,0.94]	0.90 [0.87,0.93]
stomach	0.78 [0.74,0.81]	0.87 [0.85,0.89]	0.87 [0.85,0.89]	0.85 [0.83,0.88]	0.87 [0.84,0.89]	0.88 [0.83,0.91]	0.87 [0.84,0.89]
SPECTRE, per-structure							
aorta	0.78 [0.77,0.80]	0.92 [0.91,0.92]	0.92 [0.91,0.92]	0.92 [0.91,0.92]	0.92 [0.91,0.92]	0.92 [0.91,0.93]	0.92 [0.91,0.92]
bladder	0.81 [0.78,0.84]	0.88 [0.86,0.90]	0.88 [0.86,0.91]	0.88 [0.85,0.90]	0.88 [0.86,0.91]	0.86 [0.84,0.89]	0.88 [0.86,0.90]
duodenum	0.55 [0.51,0.59]	0.71 [0.68,0.75]	0.74 [0.71,0.77]	0.73 [0.70,0.76]	0.75 [0.71,0.77]	0.72 [0.68,0.76]	0.73 [0.69,0.76]
esophagus	0.54 [0.51,0.57]	0.77 [0.75,0.79]	0.76 [0.75,0.78]	0.75 [0.73,0.77]	0.76 [0.74,0.78]	0.78 [0.76,0.80]	0.77 [0.75,0.79]
gallbladder	0.63 [0.56,0.69]	0.73 [0.66,0.79]	0.74 [0.66,0.81]	0.72 [0.65,0.79]	0.76 [0.69,0.82]	0.73 [0.65,0.79]	0.73 [0.67,0.79]
left_adrenal	0.37 [0.33,0.40]	0.57 [0.54,0.60]	0.53 [0.51,0.56]	0.53 [0.50,0.56]	0.54 [0.51,0.56]	0.52 [0.48,0.56]	0.58 [0.54,0.60]
left_kidney	0.76 [0.74,0.77]	0.87 [0.83,0.90]	0.92 [0.90,0.93]	0.91 [0.89,0.92]	0.91 [0.90,0.93]	0.78 [0.76,0.80]	0.90 [0.87,0.92]
liver	0.92 [0.91,0.93]	0.95 [0.94,0.96]	0.96 [0.95,0.96]	0.95 [0.94,0.96]	0.95 [0.94,0.96]	0.95 [0.94,0.96]	0.96 [0.95,0.96]
pancreas	0.70 [0.68,0.72]	0.79 [0.77,0.82]	0.81 [0.79,0.83]	0.80 [0.78,0.82]	0.81 [0.79,0.83]	0.80 [0.78,0.82]	0.81 [0.79,0.83]
postcava	0.65 [0.63,0.68]	0.82 [0.81,0.84]	0.84 [0.82,0.85]	0.83 [0.82,0.85]	0.84 [0.83,0.85]	0.84 [0.83,0.85]	0.83 [0.82,0.84]
prostate_uterus	0.74 [0.69,0.78]	0.80 [0.74,0.84]	0.80 [0.74,0.85]	0.78 [0.71,0.84]	0.81 [0.76,0.85]	0.78 [0.73,0.82]	0.80 [0.75,0.85]
right_adrenal	0.37 [0.33,0.40]	0.57 [0.52,0.61]	0.52 [0.50,0.54]	0.55 [0.51,0.58]	0.53 [0.50,0.56]	0.53 [0.49,0.57]	0.56 [0.52,0.60]
right_kidney	0.70 [0.69,0.72]	0.88 [0.86,0.90]	0.91 [0.89,0.93]	0.90 [0.88,0.92]	0.90 [0.88,0.92]	0.83 [0.80,0.86]	0.90 [0.88,0.92]
spleen	0.87 [0.85,0.88]	0.93 [0.91,0.94]	0.93 [0.92,0.94]	0.93 [0.92,0.95]	0.94 [0.92,0.95]	0.93 [0.91,0.94]	0.93 [0.92,0.94]
stomach	0.85 [0.82,0.87]	0.91 [0.89,0.92]	0.91 [0.89,0.92]	0.91 [0.89,0.92]	0.91 [0.89,0.92]	0.90 [0.87,0.92]	0.91 [0.88,0.92]
SPECTRE, joint							
aorta	0.77 [0.76,0.79]	0.89 [0.88,0.89]	0.90 [0.89,0.90]	0.89 [0.88,0.89]	0.90 [0.89,0.90]	0.91 [0.90,0.91]	0.89 [0.88,0.90]
bladder	0.80 [0.76,0.83]	0.83 [0.80,0.86]	0.86 [0.83,0.88]	0.83 [0.80,0.86]	0.85 [0.82,0.88]	0.84 [0.80,0.87]	0.84 [0.81,0.87]
duodenum	0.55 [0.52,0.59]	0.68 [0.64,0.71]	0.71 [0.69,0.74]	0.70 [0.67,0.72]	0.71 [0.68,0.74]	0.71 [0.67,0.74]	0.69 [0.65,0.72]
esophagus	0.53 [0.50,0.56]	0.00 [0.00,0.00]	0.70 [0.67,0.72]	0.52 [0.48,0.57]	0.70 [0.67,0.72]	0.00 [0.00,0.00]	0.00 [0.00,0.00]
gallbladder	0.65 [0.58,0.70]	0.70 [0.63,0.76]	0.74 [0.67,0.80]	0.72 [0.64,0.78]	0.55 [0.49,0.62]	0.75 [0.70,0.80]	0.70 [0.62,0.76]
left_adrenal	0.35 [0.31,0.39]	0.00 [0.00,0.00]	0.53 [0.49,0.55]	0.43 [0.39,0.47]	0.53 [0.49,0.55]	0.00 [0.00,0.00]	0.00 [0.00,0.00]
left_kidney	0.76 [0.72,0.79]	0.84 [0.80,0.88]	0.80 [0.76,0.83]	0.87 [0.83,0.90]	0.84 [0.83,0.86]	0.88 [0.85,0.90]	0.88 [0.86,0.89]
liver	0.92 [0.91,0.93]	0.95 [0.94,0.95]	0.95 [0.94,0.95]	0.94 [0.94,0.95]	0.95 [0.94,0.95]	0.95 [0.94,0.96]	0.95 [0.95,0.95]
pancreas	0.70 [0.68,0.72]	0.77 [0.75,0.79]	0.79 [0.77,0.81]	0.78 [0.75,0.79]	0.78 [0.76,0.80]	0.79 [0.77,0.81]	0.77 [0.75,0.79]
postcava	0.66 [0.64,0.68]	0.80 [0.79,0.81]	0.83 [0.81,0.84]	0.82 [0.81,0.83]	0.83 [0.81,0.84]	0.83 [0.81,0.84]	0.81 [0.79,0.82]
prostate_uterus	0.70 [0.64,0.76]	0.75 [0.70,0.79]	0.75 [0.69,0.79]	0.77 [0.72,0.82]	0.78 [0.74,0.83]	0.78 [0.72,0.82]	0.76 [0.72,0.80]
right_adrenal	0.37 [0.33,0.40]	0.00 [0.00,0.00]	0.52 [0.49,0.56]	0.50 [0.46,0.53]	0.53 [0.49,0.57]	0.00 [0.00,0.00]	0.00 [0.00,0.00]
right_kidney	0.77 [0.73,0.80]	0.87 [0.84,0.89]	0.89 [0.87,0.91]	0.88 [0.85,0.90]	0.86 [0.84,0.88]	0.84 [0.81,0.87]	0.88 [0.85,0.90]
spleen	0.86 [0.84,0.87]	0.91 [0.90,0.92]	0.90 [0.89,0.92]	0.92 [0.90,0.93]	0.92 [0.91,0.93]	0.90 [0.88,0.92]	0.91 [0.90,0.92]
stomach	0.84 [0.81,0.86]	0.89 [0.86,0.91]	0.89 [0.87,0.91]	0.89 [0.87,0.91]	0.90 [0.88,0.92]	0.87 [0.83,0.91]	0.89 [0.87,0.91]

KiTS23

Structure	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→LoRA
TAP-CT, per-structure							
cyst	0.17 [0.09,0.26]	0.19 [0.10,0.29]	0.18 [0.11,0.26]	0.17 [0.10,0.24]	0.17 [0.10,0.24]	0.20 [0.11,0.30]	0.19 [0.10,0.29]
kidney	0.89 [0.87,0.90]	0.94 [0.93,0.94]	0.94 [0.93,0.95]	0.94 [0.93,0.95]	0.94 [0.93,0.95]	0.94 [0.92,0.95]	0.94 [0.93,0.94]
tumor	0.54 [0.48,0.59]	0.62 [0.56,0.68]	0.57 [0.50,0.64]	0.59 [0.53,0.66]	0.58 [0.52,0.65]	0.63 [0.57,0.69]	0.63 [0.58,0.69]
TAP-CT, joint							
cyst	0.04 [0.01,0.08]	0.09 [0.05,0.14]	0.18 [0.11,0.27]	0.14 [0.08,0.21]	0.15 [0.08,0.21]	0.08 [0.04,0.13]	0.00 [0.00,0.00]
kidney	0.89 [0.88,0.90]	0.91 [0.90,0.92]	0.94 [0.93,0.94]	0.94 [0.93,0.95]	0.93 [0.92,0.94]	0.91 [0.90,0.92]	0.93 [0.91,0.94]
tumor	0.36 [0.28,0.44]	0.54 [0.47,0.61]	0.62 [0.56,0.68]	0.59 [0.53,0.66]	0.60 [0.53,0.66]	0.56 [0.49,0.62]	0.57 [0.50,0.65]
SPECTRE, per-structure							
cyst	0.22 [0.14,0.31]	0.22 [0.14,0.31]	0.19 [0.12,0.27]	0.21 [0.13,0.29]	0.20 [0.13,0.27]	0.05 [0.02,0.08]	0.21 [0.13,0.29]
kidney	0.90 [0.89,0.91]	0.94 [0.93,0.95]	0.95 [0.94,0.95]	0.94 [0.93,0.95]	0.95 [0.94,0.95]	0.93 [0.91,0.95]	0.94 [0.94,0.95]
tumor	0.62 [0.56,0.68]	0.63 [0.56,0.70]	0.70 [0.64,0.75]	0.70 [0.64,0.75]	0.72 [0.66,0.77]	0.60 [0.52,0.67]	0.66 [0.59,0.72]
SPECTRE, joint							
cyst	0.16 [0.07,0.27]	0.15 [0.07,0.23]	0.09 [0.04,0.15]	0.10 [0.06,0.16]	0.17 [0.09,0.24]	0.11 [0.07,0.16]	0.13 [0.06,0.21]
kidney	0.90 [0.89,0.91]	0.93 [0.92,0.94]	0.94 [0.94,0.95]	0.94 [0.93,0.95]	0.92 [0.91,0.93]	0.89 [0.87,0.91]	0.94 [0.93,0.94]
tumor	0.53 [0.45,0.61]	0.63 [0.56,0.70]	0.73 [0.68,0.78]	0.72 [0.66,0.77]	0.69 [0.63,0.75]	0.56 [0.49,0.62]	0.64 [0.57,0.71]

LiTS

Structure	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→LoRA
TAP-CT, per-structure							
liver	0.92 [0.90,0.93]	0.95 [0.93,0.97]	0.96 [0.95,0.97]	0.96 [0.95,0.97]	0.96 [0.95,0.97]	0.96 [0.96,0.97]	0.96 [0.95,0.97]
tumor	0.40 [0.26,0.53]	0.47 [0.33,0.59]	0.47 [0.32,0.62]	0.46 [0.32,0.59]	0.47 [0.31,0.61]	0.46 [0.31,0.61]	0.47 [0.34,0.60]
TAP-CT, joint							
liver	0.91 [0.89,0.93]	0.95 [0.94,0.96]	0.96 [0.94,0.97]	0.96 [0.94,0.97]	0.96 [0.94,0.97]	0.96 [0.95,0.97]	0.95 [0.94,0.96]
tumor	0.40 [0.25,0.53]	0.45 [0.30,0.60]	0.47 [0.33,0.62]	0.48 [0.35,0.63]	0.43 [0.28,0.59]	0.47 [0.33,0.60]	0.45 [0.30,0.60]
SPECTRE, per-structure							
liver	0.94 [0.94,0.95]	0.97 [0.96,0.97]	0.97 [0.96,0.97]	0.97 [0.96,0.97]	0.97 [0.96,0.97]	0.97 [0.96,0.97]	0.97 [0.96,0.97]
tumor	0.43 [0.28,0.57]	0.52 [0.36,0.66]	0.55 [0.41,0.67]	0.48 [0.33,0.62]	0.55 [0.40,0.67]	0.51 [0.36,0.66]	0.49 [0.33,0.63]
SPECTRE, joint							
liver	0.94 [0.93,0.94]	0.96 [0.96,0.96]	0.96 [0.95,0.97]	0.96 [0.96,0.97]	0.96 [0.96,0.97]	0.96 [0.94,0.97]	0.96 [0.96,0.96]
tumor	0.46 [0.31,0.61]	0.52 [0.35,0.66]	0.55 [0.40,0.67]	0.51 [0.37,0.65]	0.52 [0.37,0.65]	0.47 [0.31,0.63]	0.53 [0.37,0.67]

MSD Pancreas

Structure	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→LoRA
TAP-CT, per-structure							
pancreas	0.66 [0.62,0.70]	0.75 [0.71,0.78]	0.77 [0.73,0.79]	0.76 [0.73,0.79]	0.76 [0.72,0.79]	0.76 [0.72,0.79]	0.75 [0.71,0.78]
tumor	0.19 [0.14,0.25]	0.23 [0.16,0.31]	0.24 [0.18,0.31]	0.24 [0.18,0.30]	0.25 [0.18,0.31]	0.26 [0.19,0.34]	0.24 [0.18,0.31]
TAP-CT, joint							
pancreas	0.66 [0.62,0.69]	0.72 [0.69,0.75]	0.75 [0.72,0.78]	0.75 [0.71,0.78]	0.75 [0.72,0.78]	0.75 [0.72,0.78]	0.73 [0.69,0.77]
tumor	0.01 [0.00,0.02]	0.24 [0.17,0.31]	0.26 [0.19,0.32]	0.23 [0.17,0.30]	0.23 [0.18,0.29]	0.22 [0.15,0.30]	0.20 [0.12,0.29]
SPECTRE, per-structure							
pancreas	0.71 [0.68,0.74]	0.78 [0.74,0.80]	0.79 [0.76,0.82]	0.78 [0.74,0.81]	0.79 [0.76,0.82]	0.79 [0.75,0.81]	0.78 [0.75,0.81]
tumor	0.24 [0.18,0.31]	0.26 [0.18,0.32]	0.34 [0.26,0.41]	0.30 [0.22,0.38]	0.32 [0.24,0.40]	0.25 [0.17,0.33]	0.26 [0.18,0.33]
SPECTRE, joint							
pancreas	0.70 [0.67,0.73]	0.76 [0.72,0.78]	0.79 [0.75,0.81]	0.77 [0.73,0.79]	0.79 [0.76,0.81]	0.77 [0.74,0.80]	0.76 [0.73,0.78]
tumor	0.22 [0.15,0.29]	0.27 [0.19,0.35]	0.32 [0.24,0.40]	0.29 [0.21,0.38]	0.31 [0.24,0.40]	0.29 [0.21,0.38]	0.28 [0.20,0.36]

TotalSegmentator

Structure	Lin	Dec	LoRA	$\text{LoRA}_{\text{late}}$	DoRA	Full FT	Dec→LoRA
TAP-CT, per-structure							
arteries	0.66 [0.63,0.70]	0.78 [0.74,0.81]	0.73 [0.68,0.77]	–	–	0.78 [0.74,0.81]	–
bladder_prostate	0.83 [0.79,0.86]	0.84 [0.81,0.88]	0.83 [0.78,0.87]	–	–	0.85 [0.81,0.88]	–
brain_spinal_cord	0.78 [0.76,0.81]	0.91 [0.90,0.92]	0.84 [0.82,0.86]	–	–	0.91 [0.91,0.92]	–
gi_tract	0.76 [0.73,0.79]	0.86 [0.84,0.88]	0.84 [0.81,0.87]	–	–	0.86 [0.84,0.88]	–
glands	0.47 [0.44,0.51]	0.63 [0.59,0.66]	0.52 [0.48,0.56]	–	–	0.41 [0.38,0.44]	–
heart	0.85 [0.80,0.88]	0.89 [0.85,0.93]	0.88 [0.83,0.93]	–	–	0.90 [0.86,0.93]	–
kidneys	0.79 [0.74,0.83]	0.85 [0.80,0.89]	0.85 [0.80,0.89]	–	–	0.84 [0.80,0.89]	–
liver	0.87 [0.83,0.90]	0.92 [0.89,0.95]	0.93 [0.90,0.96]	–	–	0.93 [0.90,0.96]	–
lungs_airway	0.90 [0.88,0.92]	0.95 [0.94,0.96]	0.94 [0.91,0.96]	–	–	0.95 [0.92,0.97]	–
muscles	0.80 [0.78,0.81]	0.90 [0.89,0.91]	0.90 [0.87,0.91]	–	–	0.91 [0.90,0.92]	–
other_bones	0.73 [0.72,0.75]	0.88 [0.87,0.89]	0.88 [0.87,0.89]	–	–	0.89 [0.88,0.90]	–
pancreas	0.64 [0.60,0.68]	0.72 [0.68,0.76]	0.70 [0.64,0.75]	–	–	0.73 [0.68,0.77]	–
spine	0.73 [0.71,0.75]	0.87 [0.86,0.88]	0.87 [0.85,0.88]	–	–	0.88 [0.86,0.89]	–
spleen	0.79 [0.76,0.82]	0.88 [0.85,0.91]	0.89 [0.86,0.92]	–	–	0.91 [0.89,0.93]	–
veins	0.59 [0.56,0.61]	0.74 [0.71,0.76]	0.70 [0.67,0.72]	–	–	0.72 [0.70,0.75]	–
TAP-CT, joint							
arteries	0.67 [0.63,0.70]	0.79 [0.75,0.82]	0.80 [0.76,0.83]	0.77 [0.74,0.80]	0.80 [0.77,0.83]	0.81 [0.77,0.85]	0.79 [0.75,0.82]
bladder_prostate	0.81 [0.76,0.85]	0.85 [0.82,0.88]	0.84 [0.80,0.88]	0.82 [0.78,0.85]	0.83 [0.79,0.86]	0.87 [0.84,0.90]	0.84 [0.79,0.87]
brain_spinal_cord	0.79 [0.76,0.81]	0.91 [0.90,0.92]	0.89 [0.87,0.91]	0.89 [0.87,0.91]	0.89 [0.87,0.90]	0.92 [0.91,0.92]	0.91 [0.90,0.92]
gi_tract	0.76 [0.73,0.78]	0.86 [0.84,0.87]	0.84 [0.81,0.86]	0.85 [0.84,0.87]	0.85 [0.83,0.87]	0.88 [0.86,0.89]	0.85 [0.83,0.87]
glands	0.47 [0.43,0.50]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]
heart	0.84 [0.81,0.88]	0.89 [0.85,0.92]	0.89 [0.85,0.92]	0.88 [0.83,0.92]	0.89 [0.85,0.92]	0.90 [0.85,0.93]	0.88 [0.84,0.92]
kidneys	0.77 [0.72,0.82]	0.83 [0.78,0.88]	0.84 [0.79,0.88]	0.84 [0.80,0.88]	0.83 [0.79,0.87]	0.86 [0.82,0.90]	0.83 [0.78,0.87]
liver	0.86 [0.83,0.89]	0.92 [0.89,0.95]	0.80 [0.73,0.87]	0.81 [0.75,0.87]	0.80 [0.73,0.87]	0.94 [0.90,0.96]	0.92 [0.88,0.95]
lungs_airway	0.90 [0.87,0.92]	0.95 [0.93,0.96]	0.82 [0.76,0.89]	0.83 [0.77,0.89]	0.82 [0.76,0.88]	0.95 [0.94,0.96]	0.94 [0.93,0.95]
muscles	0.80 [0.79,0.82]	0.89 [0.88,0.90]	0.90 [0.89,0.91]	0.88 [0.85,0.90]	0.89 [0.88,0.90]	0.91 [0.90,0.92]	0.89 [0.88,0.90]
other_bones	0.73 [0.71,0.75]	0.86 [0.85,0.87]	0.86 [0.85,0.87]	0.86 [0.84,0.87]	0.86 [0.85,0.87]	0.88 [0.86,0.89]	0.85 [0.84,0.86]
pancreas	0.62 [0.57,0.66]	0.71 [0.67,0.75]	0.68 [0.63,0.73]	0.71 [0.67,0.75]	0.69 [0.64,0.74]	0.74 [0.68,0.78]	0.68 [0.63,0.72]
spine	0.73 [0.71,0.75]	0.85 [0.84,0.87]	0.84 [0.82,0.85]	0.85 [0.83,0.86]	0.84 [0.82,0.85]	0.86 [0.85,0.88]	0.84 [0.83,0.86]
spleen	0.77 [0.74,0.81]	0.89 [0.86,0.91]	0.89 [0.87,0.91]	0.91 [0.88,0.93]	0.89 [0.87,0.92]	0.93 [0.90,0.94]	0.87 [0.85,0.90]
veins	0.58 [0.56,0.61]	0.75 [0.73,0.78]	0.75 [0.72,0.77]	0.73 [0.70,0.75]	0.74 [0.72,0.76]	0.79 [0.76,0.80]	0.74 [0.71,0.76]
SPECTRE, per-structure							
arteries	0.63 [0.59,0.67]	0.79 [0.75,0.82]	0.77 [0.73,0.81]	–	–	0.80 [0.77,0.83]	–
bladder_prostate	0.81 [0.77,0.85]	0.86 [0.83,0.90]	0.87 [0.82,0.91]	–	–	0.87 [0.84,0.90]	–
brain_spinal_cord	0.74 [0.72,0.76]	0.91 [0.91,0.92]	0.90 [0.88,0.92]	–	–	0.91 [0.89,0.92]	–
gi_tract	0.75 [0.71,0.78]	0.85 [0.83,0.88]	0.87 [0.84,0.89]	–	–	0.87 [0.84,0.89]	–
glands	0.41 [0.37,0.44]	0.63 [0.59,0.66]	0.61 [0.57,0.64]	–	–	0.57 [0.53,0.60]	–
heart	0.86 [0.83,0.89]	0.91 [0.87,0.94]	0.91 [0.88,0.94]	–	–	0.91 [0.87,0.94]	–
kidneys	0.79 [0.74,0.83]	0.85 [0.81,0.89]	0.86 [0.80,0.90]	–	–	0.80 [0.75,0.85]	–
liver	0.91 [0.87,0.93]	0.94 [0.91,0.96]	0.93 [0.89,0.96]	–	–	0.92 [0.89,0.95]	–
lungs_airway	0.90 [0.88,0.92]	0.95 [0.94,0.96]	0.95 [0.94,0.96]	–	–	0.95 [0.94,0.96]	–
muscles	0.75 [0.72,0.77]	0.90 [0.88,0.91]	0.91 [0.90,0.92]	–	–	0.91 [0.90,0.92]	–
other_bones	0.68 [0.66,0.70]	0.87 [0.85,0.88]	0.87 [0.86,0.88]	–	–	0.88 [0.87,0.89]	–
pancreas	0.66 [0.62,0.69]	0.75 [0.71,0.79]	0.73 [0.68,0.78]	–	–	0.74 [0.69,0.78]	–
spine	0.67 [0.65,0.69]	0.86 [0.84,0.87]	0.87 [0.85,0.88]	–	–	0.87 [0.86,0.88]	–
spleen	0.87 [0.84,0.89]	0.91 [0.88,0.93]	0.91 [0.87,0.93]	–	–	0.80 [0.77,0.84]	–
veins	0.55 [0.52,0.57]	0.74 [0.71,0.76]	0.77 [0.74,0.79]	–	–	0.77 [0.75,0.79]	–
SPECTRE, joint							
arteries	0.63 [0.60,0.66]	0.74 [0.70,0.77]	0.77 [0.73,0.80]	0.75 [0.71,0.79]	0.76 [0.72,0.80]	0.78 [0.75,0.82]	0.74 [0.70,0.77]
bladder_prostate	0.80 [0.75,0.84]	0.80 [0.76,0.83]	0.69 [0.63,0.74]	0.74 [0.70,0.78]	0.60 [0.52,0.67]	0.85 [0.82,0.88]	0.81 [0.79,0.84]
brain_spinal_cord	0.74 [0.71,0.76]	0.90 [0.89,0.91]	0.90 [0.88,0.92]	0.90 [0.88,0.92]	0.90 [0.88,0.91]	0.91 [0.91,0.92]	0.90 [0.89,0.91]
gi_tract	0.74 [0.71,0.78]	0.83 [0.81,0.86]	0.85 [0.83,0.87]	0.85 [0.83,0.87]	0.85 [0.83,0.87]	0.86 [0.84,0.88]	0.83 [0.81,0.86]
glands	0.38 [0.34,0.42]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]	0.00 [0.00,0.00]
heart	0.86 [0.82,0.89]	0.89 [0.86,0.92]	0.88 [0.83,0.92]	0.88 [0.84,0.92]	0.88 [0.83,0.92]	0.91 [0.87,0.93]	0.89 [0.86,0.92]
kidneys	0.77 [0.72,0.81]	0.81 [0.77,0.85]	0.79 [0.73,0.84]	0.81 [0.77,0.85]	0.79 [0.74,0.84]	0.81 [0.76,0.86]	0.80 [0.76,0.84]
liver	0.90 [0.87,0.93]	0.92 [0.89,0.95]	0.75 [0.67,0.82]	0.83 [0.76,0.89]	0.75 [0.68,0.82]	0.93 [0.89,0.95]	0.92 [0.89,0.95]
lungs_airway	0.90 [0.87,0.92]	0.93 [0.90,0.95]	0.83 [0.77,0.88]	0.88 [0.83,0.92]	0.85 [0.80,0.90]	0.94 [0.93,0.95]	0.93 [0.90,0.94]
muscles	0.74 [0.71,0.76]	0.87 [0.86,0.88]	0.88 [0.86,0.90]	0.89 [0.88,0.90]	0.89 [0.86,0.90]	0.90 [0.89,0.91]	0.88 [0.86,0.89]
other_bones	0.67 [0.65,0.69]	0.82 [0.81,0.84]	0.85 [0.83,0.86]	0.84 [0.83,0.85]	0.85 [0.84,0.86]	0.85 [0.84,0.86]	0.82 [0.81,0.84]
pancreas	0.64 [0.60,0.68]	0.67 [0.63,0.72]	0.63 [0.59,0.67]	0.64 [0.59,0.68]	0.66 [0.61,0.70]	0.70 [0.66,0.74]	0.69 [0.64,0.73]
spine	0.66 [0.64,0.68]	0.81 [0.79,0.82]	0.82 [0.81,0.84]	0.82 [0.81,0.84]	0.83 [0.81,0.84]	0.83 [0.81,0.84]	0.81 [0.79,0.82]
spleen	0.86 [0.84,0.88]	0.86 [0.84,0.89]	0.65 [0.58,0.71]	0.72 [0.66,0.77]	0.71 [0.64,0.76]	0.88 [0.85,0.91]	0.87 [0.84,0.89]
veins	0.54 [0.52,0.56]	0.69 [0.66,0.71]	0.72 [0.70,0.74]	0.72 [0.70,0.73]	0.73 [0.71,0.75]	0.75 [0.73,0.77]	0.69 [0.67,0.71]

Appendix F. Cohorts and examples

Table 21: Cohorts and datasets. Labels = binary labels (classification) or structures (segmentation); counts are cases in the training / selection / held-out splits as read from the manifests. Split rule: "official val, halves" = the official validation cases sorted by identifier and alternately assigned to the selection and held-out split; "seeded" = one fixed seeded division of the released labelled cases (80/10/10 of the released training data for the three RSNA and FDG cohorts, 70/15/15 for KiTS23 and MSD Pancreas, 91/20/20 volumes for LiTS); TotalSegmentator holds out its official test split and takes an equally large selection split from the official training split. FDG-PET/CT cases are studies (1014 studies of 900 patients; no patient spans splits); RSNA 2019 training cases are a stratified subset of the released training patients. Backbones: T = TAP-CT, S = SPECTRE, P = Pillar-0, V = VoCo.

Cohort / dataset	Labels	Split rule	Backbones	Train	Sel.	Held-out
Classification						
CVPR26: AMOS	15 disease labels	official val, halves	T S	1463[†]	178	178
CVPR26: COVID-CTset	COVID-19	official val, halves	T S	308	32	32
CVPR26: LUNA25	nodule malignancy	official val, halves	T S	1000	100	100
FDG-PET/CT	4 diagnosis groups	seeded, patient level	T S V	776	116	122
RSNA 2022 C-spine	patient, C1–C7 (8)	seeded, per study	T S V	1615	202	202
RSNA 2023 trauma	any, kidney, liver, spleen (4)	seeded, one series per patient	T S V	2517	315	315
RSNA 2019 hemorrhage	any + 5 subtypes (6)	stratified, patient level	T S P	5000	3269	3264
Segmentation						
AMOS22	15 organs	official val, halves	T S V	200	50	50
KiTS23	kidney, tumor, cyst	seeded	T S V	343	73	73
LiTS	liver, tumor	seeded	T S V	91	20	20
MSD Pancreas	pancreas, tumor	seeded	T S V	197	42	42
TotalSegmentator	15 merged structures	official test held out	T S V	1721	109	109

[†] distinct AMOS cases over the 15 per-label files; each label is known for a subset of cases.

Table 22: TotalSegmentator: the 117 released structures merged into 15 within anatomical systems, as implemented. Where two source masks overlap, the voxel is assigned to the smaller structure.

Structure	Source structures
lungs and airway	five lung lobes, trachea
heart	heart, left atrial appendage
arteries	aorta, brachiocephalic trunk, left and right subclavian, common carotid and iliac arteries
veins	superior and inferior vena cava, left and right brachiocephalic and iliac veins, portal and splenic vein, pulmonary vein
liver	liver, gallbladder
spleen	spleen
pancreas	pancreas
kidneys	left and right kidney, left and right kidney cysts
GI tract	esophagus, stomach, duodenum, small bowel, colon
bladder and prostate	urinary bladder, prostate
glands	thyroid, left and right adrenal gland
brain and spinal cord	brain, spinal cord
spine	25 vertebrae (C1–L6), sacrum
other bones	24 ribs, sternum, costal cartilages, clavicles, scapulae, humeri, hip bones, femurs, skull
muscles	gluteus maximus, medius and minimus, iliopsoas, autochthon (left and right)

Table 23: Evaluation data that appear in a backbone's pretraining data, as listed in the backbones' papers (images only; no labels of our tasks were used in pretraining). TAP-CT was pretrained on in-house oncological CT (Veenboer et al., 2026) and Pillar-0 on CT and breast MRI of one academic center (Agrawal et al., 2025), neither listing a public dataset. SPECTRE lists, among others, AMOS, NLST, AbdomenAtlas 1.0 Mini and AbdomenCT-1K (Claessens et al., 2026); the last two aggregate public datasets whose members we did not check. VoCo's self-supervised checkpoint follows the PreCT-160K recipe (Wu et al., 2026), which lists AMOS22, TotalSegmentator, NLST, LUNA16 and TCIA-Covid19 and cites the MSD and KiTS challenges among its tumor datasets. Possibly: only through an aggregated dataset.

Evaluation data	TAP-CT	SPECTRE	Pillar-0	VoCo
AMOS (CVPR26) and AMOS22	no	yes	no	yes
LUNA25	no	yes[a]	no	yes[a]
TotalSegmentator	no	possibly	no	yes
KiTS23, MSD Pancreas	no	possibly	no	likely[b]
LiTS	no	possibly	no	not listed
FDG-PET/CT	no	possibly	no	no[c]
COVID-CTset, RSNA 2019/2022/2023	no	not listed	no	not listed

[a] through NLST, which is in the pretraining data; the public LUNA25 training and development scans are NLST scans (Peeters et al., 2025). [b] cited among the tumor datasets without a version. [c] AutoPET is marked as unseen in pretraining.

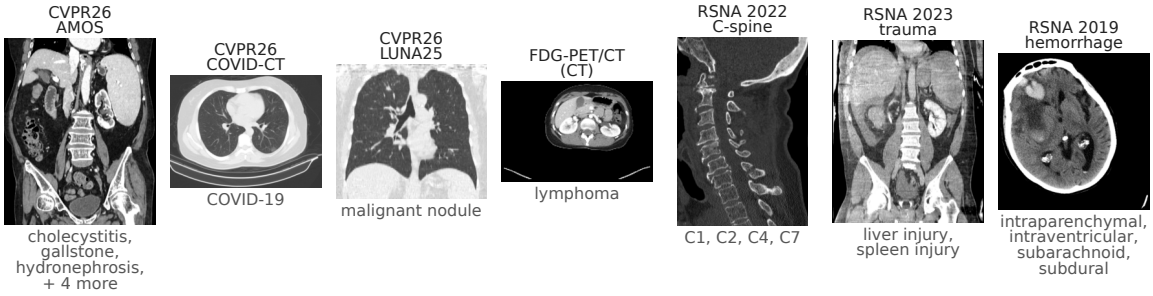


Figure 16: One held-out case per classification source: the case with the most positive labels, listed below it. Views are fixed per source and pass through the body's center of mass: coronal for AMOS, LUNA25 and RSNA 2023, sagittal for the cervical spine, and axial for COVID-CTset (thick slices), FDG-PET/CT (whole body) and head CT. Windows are soft tissue, lung (COVID-CTset, LUNA25), bone (cervical spine) and brain (hemorrhage).

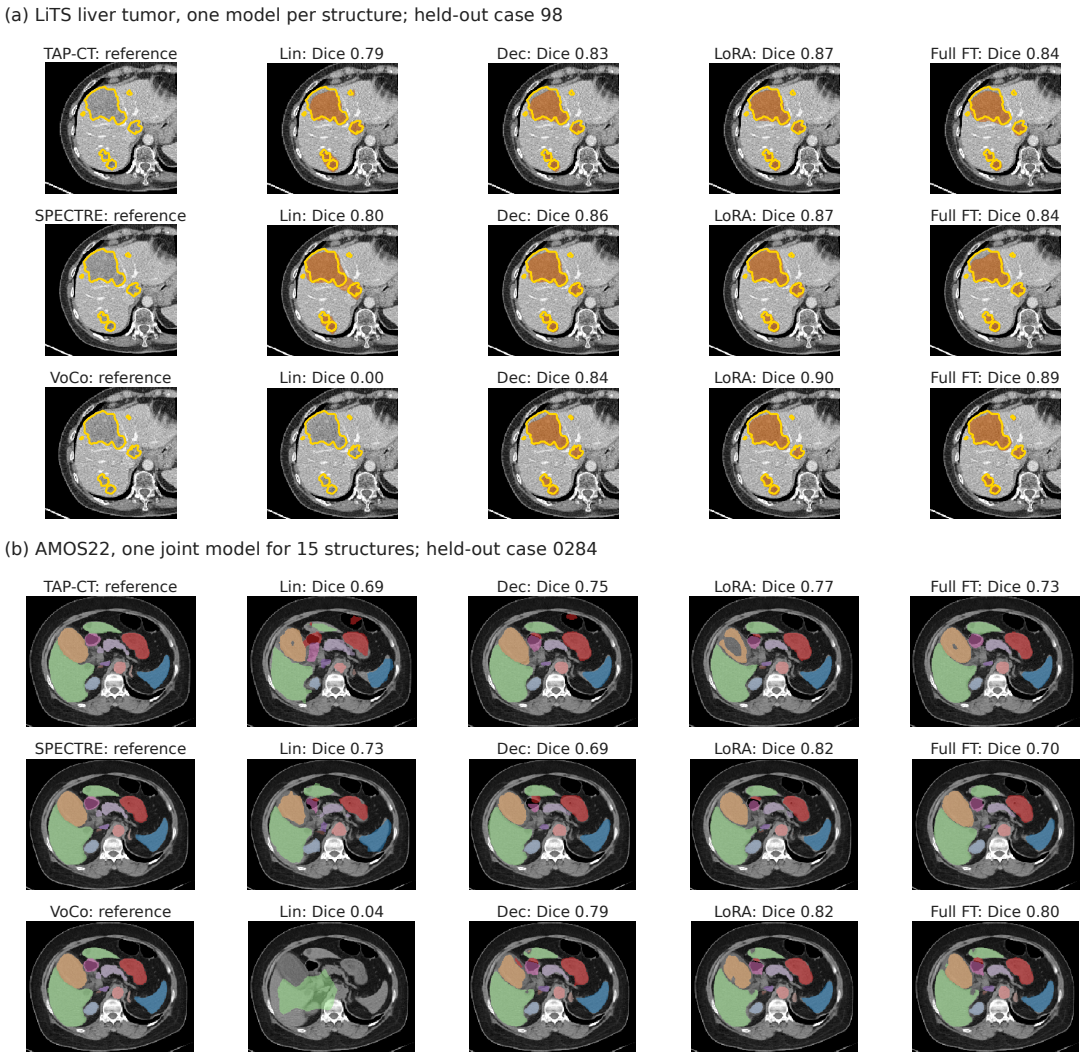


Figure 17: Segmentation examples on held-out cases, each on the backbone's own grid at the soft-tissue window; panel titles give the case's Dice (mean over structures in b). (a) LiTS liver tumor with one model per structure; yellow is the reference outline, orange the prediction. (b) AMOS22 with one joint model for all 15 structures (colors are structures). On TAP-CT and SPECTRE the decoder improves on the linear probe, LoRA has the highest Dice in both panels, and full fine-tuning does not improve on LoRA. On VoCo the linear probe predicts no tumor in (a) and almost nothing in (b). Cases were chosen by a fixed rule on TAP-CT and SPECTRE among held-out cases with a valid Dice in every run: the case agrees with each backbone's mean held-out Dice on the best and the worst method (a) or on the full ranking (b), and among such cases its pairwise Dice differences are closest to the mean differences; 3 of 18 cases qualify for (a) and 23 of 50 for (b).

Appendix G. Success and failure cases

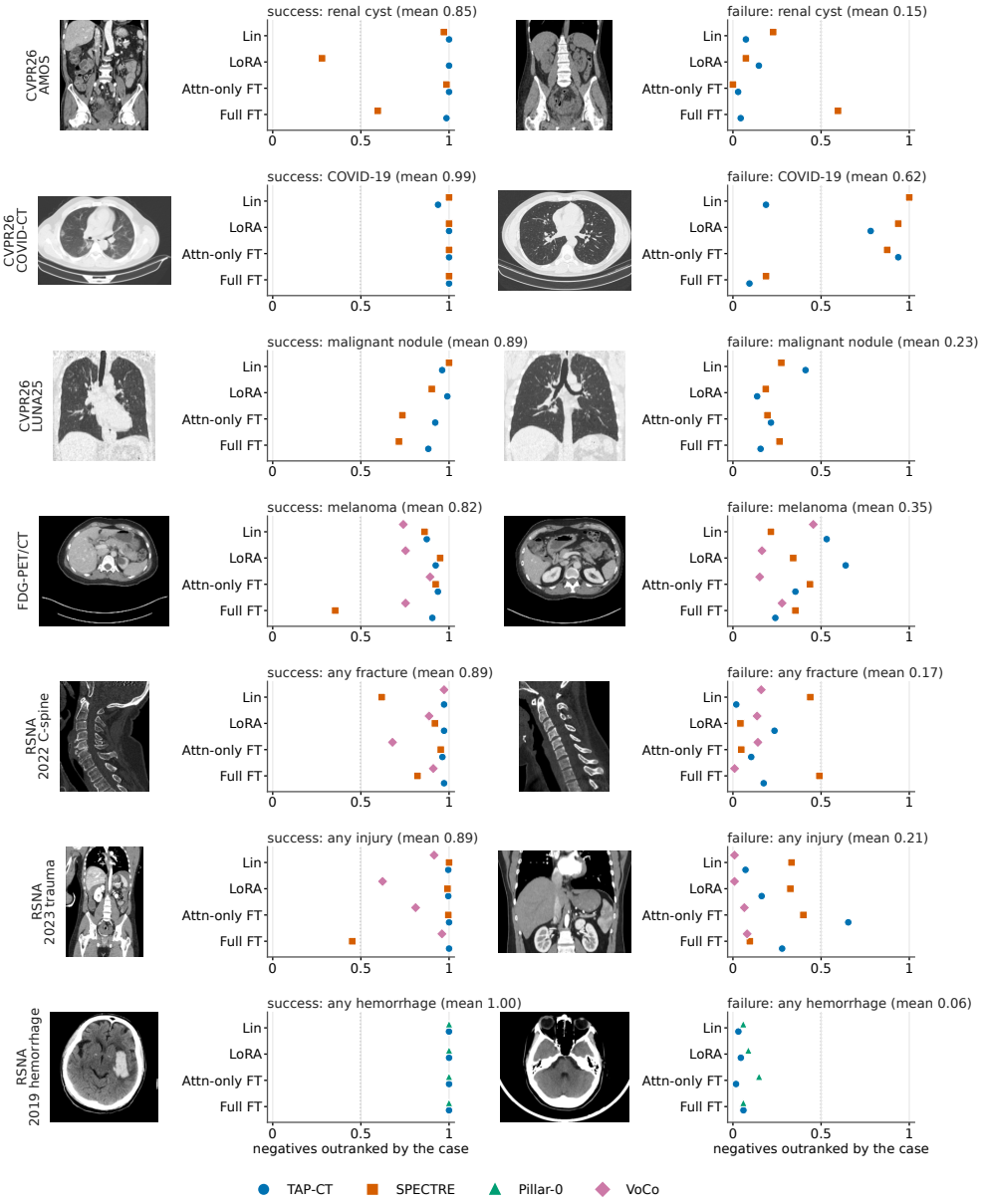


Figure 18: Classification success (left) and failure (right) cases, one positive held-out case per source. Markers give the share of held-out negatives the case outranks for each method and backbone (1: ranked above every negative; dotted line: 0.5); the title gives the label and the case's mean over methods and backbones. Rows are the three CVPR26 sources, FDG-PET/CT, the RSNA 2022 cervical spine, RSNA 2023 trauma and RSNA 2019 hemorrhage.

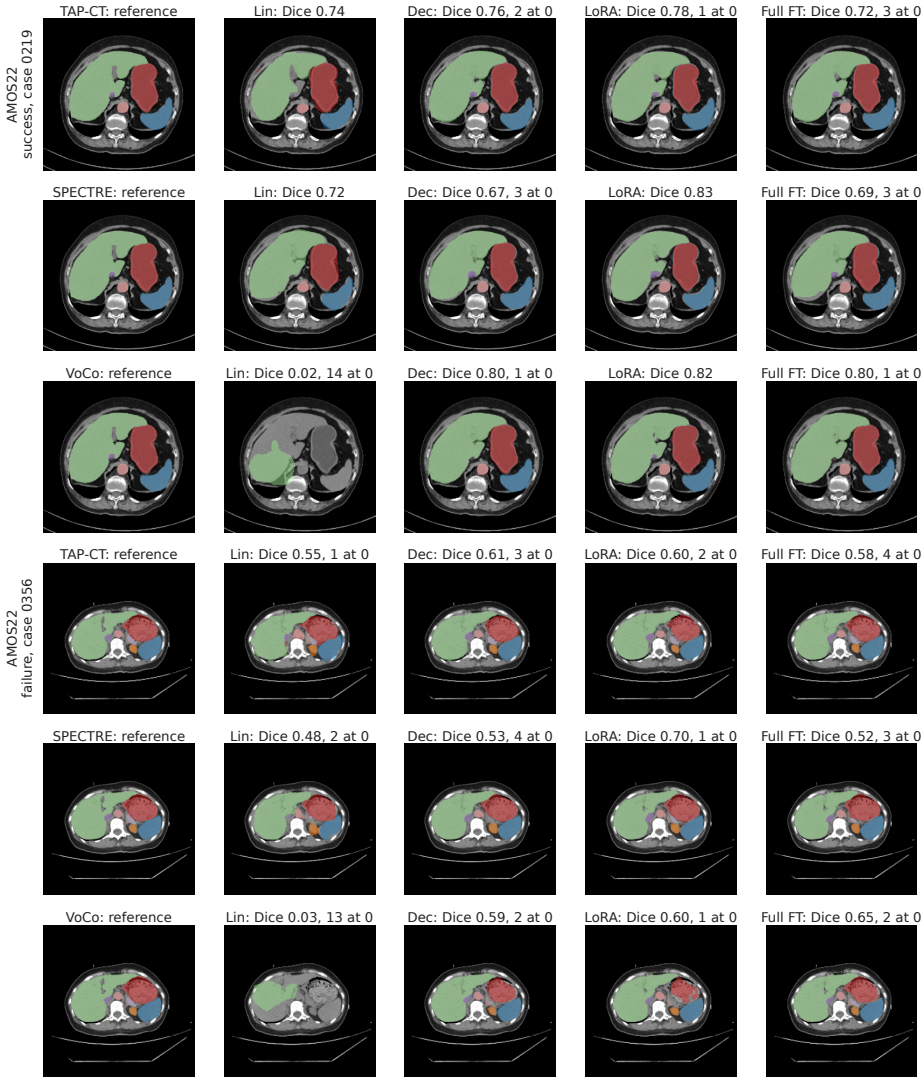


Figure 19: Segmentation success and failure cases on AMOS22 (15 structures), joint model. Rows: the success case (top three) and the failure case (bottom three), each on TAP-CT, SPECTRE and VoCo; columns: the reference and the predictions of Lin, Dec, LoRA and Full FT on the axial slice with the most reference foreground, soft-tissue window, colors per structure. Panel titles give the method's Dice for the whole case, averaged over its structures, and how many of them score 0. On TAP-CT and SPECTRE these are the adrenal glands for most adapted methods and, for some, the esophagus or the gallbladder. On VoCo the decoder and full fine-tuning score 0 on the prostate or uterus in both cases and on the right adrenal gland in the failure case, LoRA only on the right adrenal gland in the failure case, and the linear probe segments only the liver in both cases.

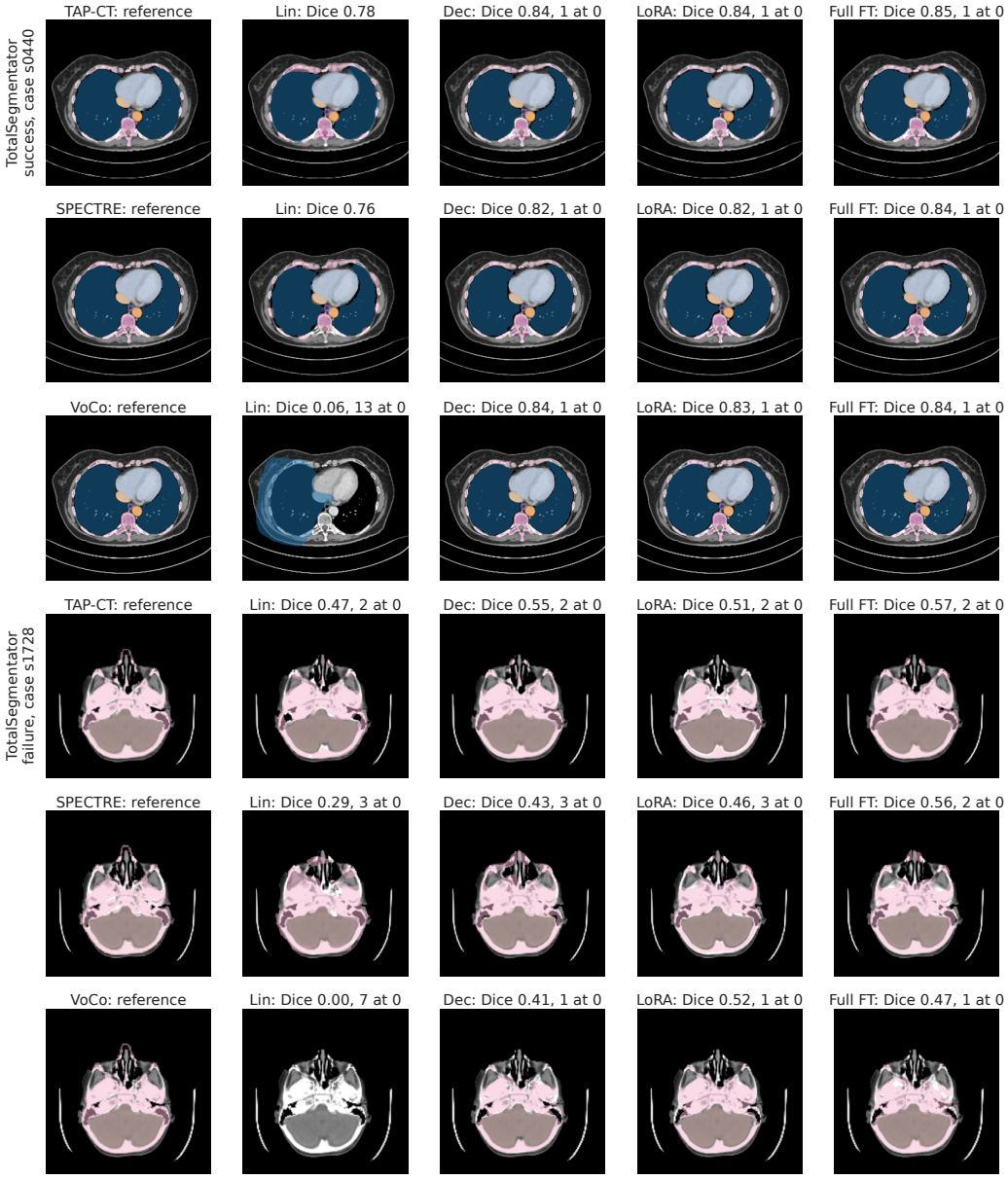


Figure 20: As Figure 19, for TotalSegmentator (15 merged structures). Every adapted joint model scores 0 on the merged glands in both cases; in the failure case some TAP-CT and SPECTRE models also score 0 on the arteries, the liver, the lungs and airway or the gastrointestinal tract. The VoCo linear probe scores 0 on 13 structures in the success case and on all 7 structures present in the failure case.

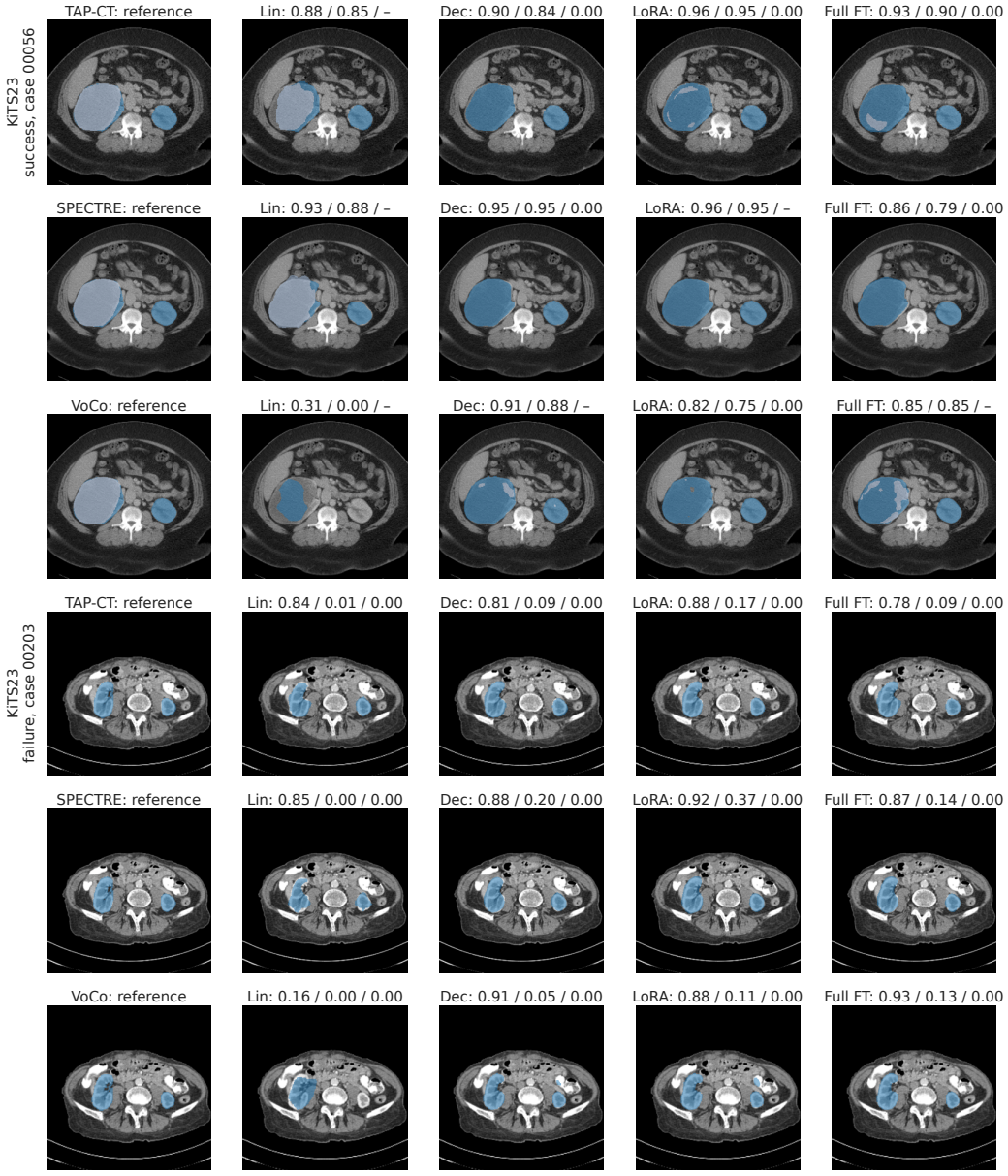


Figure 21: As Figure 19, for KiTS23; panel titles give the Dice of each structure (kidney / tumor / cyst; –: absent from both reference and prediction).

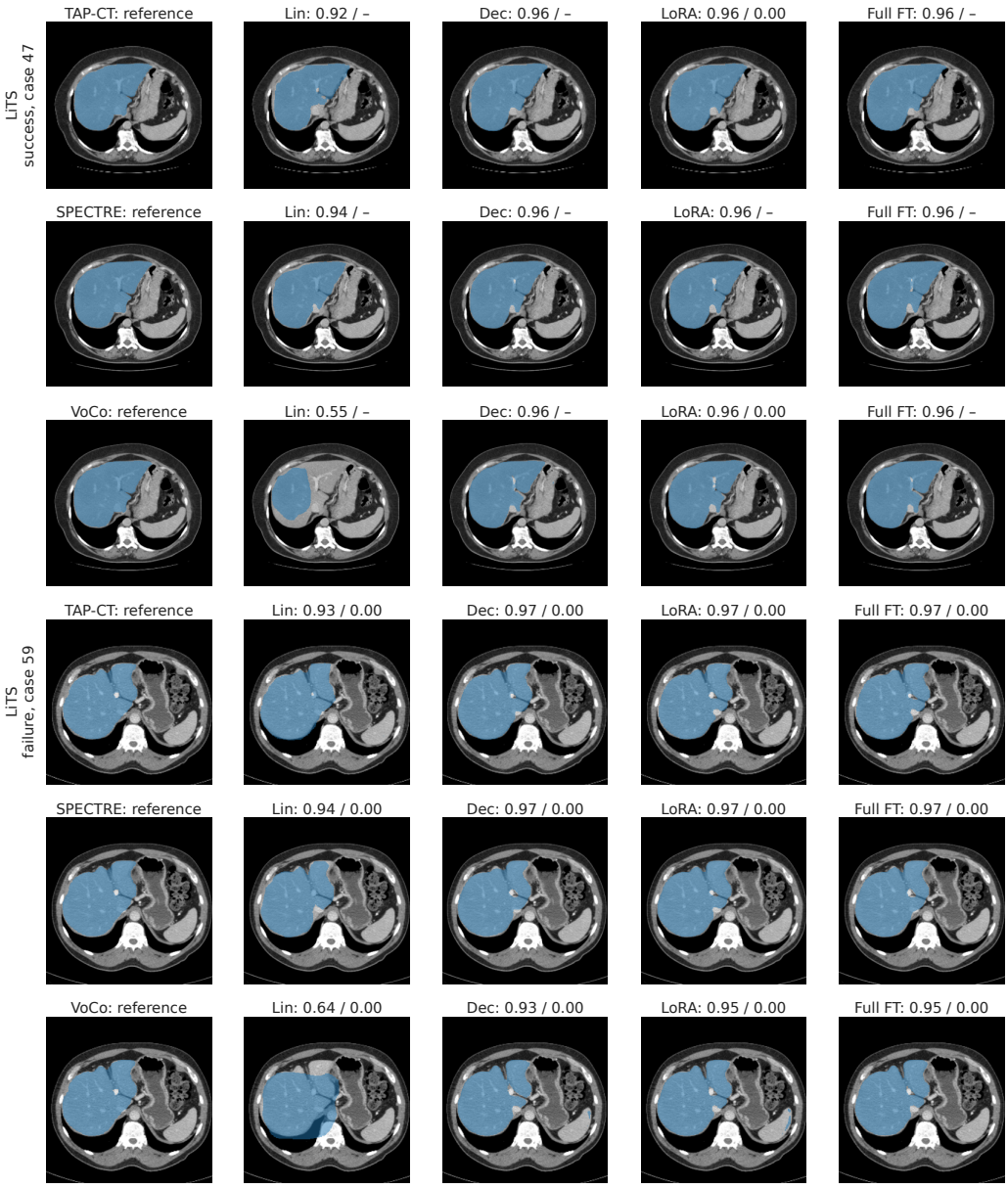


Figure 22: As Figure 19, for LiTS, except that panel titles give the Dice of each structure (liver / tumor; –: absent from both reference and prediction). Neither case has a tumor in the reference, so a method's tumor Dice is undefined when it predicts no tumor and 0 when it predicts any: in the success case only LoRA on TAP-CT and on VoCo predict a tumor somewhere in the volume, and in the failure case every method does, while all segment the liver well. The failure is therefore a false-positive tumor, which the per-structure Dice counts as 0, and not a missed one; the overlay shows the structure with the highest logit at each voxel, which hides the predicted tumor under the liver in all but one run.

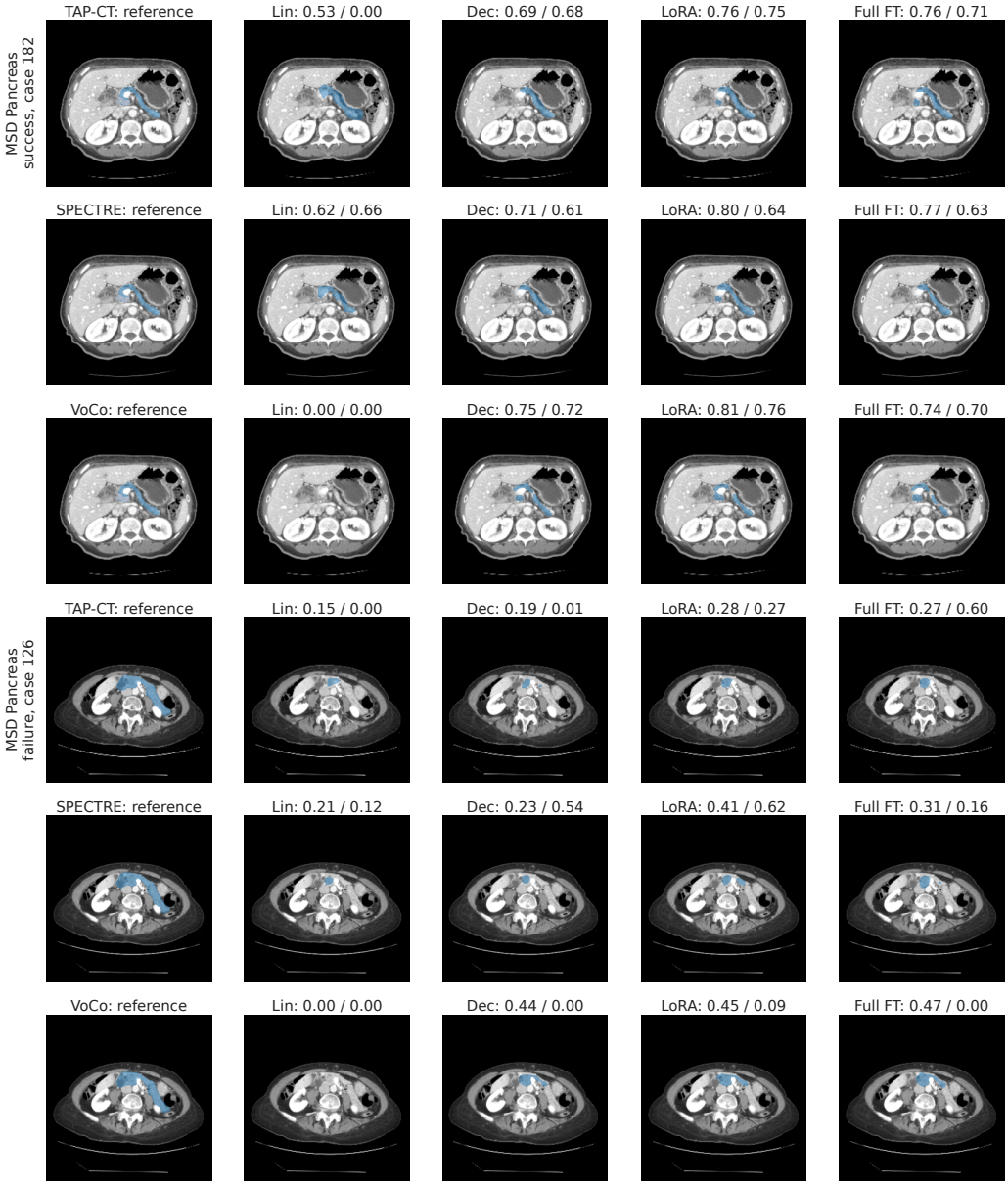


Figure 23: As Figure 19, for MSD Pancreas; panel titles give the Dice of each structure (pancreas / tumor; –: absent from both reference and prediction).